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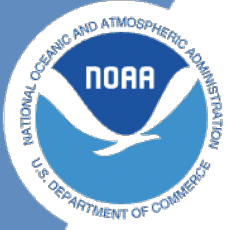
Side-by-Side Survey Catchability Studies: Cooperative Research to Estimate Survey Catchability for Tanner Crab in the Eastern Bering Sea DRAFT

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Side-by-Side Survey Catchability Studies: Cooperative Research to Estimate Survey Catchability for Tanner Crab in the Eastern Bering Sea DRAFT

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Introduction

Scientific surveys conducted to estimate current population size and changes in abundance over time for a given species provide important information to inform decisions on conservation and management, but most are not without challenges to interpret the resulting raw data (counts or biomass of individuals, perhaps classified into different population categories) in terms of the population of interest. In the eastern Bering Sea (EBS), the National Marine Fisheries Service's Alaska Fisheries Science Center (NMFS AFSC) has conducted an annual survey of the EBS shelf using a standardized survey grid [Figure 1; Wakabayashi et al. (1985); Zacher et al. (2023)] and benthic trawl gear since 1982 to quantify trends in abundance and biomass of commercially- and ecologically-important marine species. Among other uses, these data provide a critically-important resource to stock assessment scientists and fishery managers and are used to determine the status of exploited groundfish and crab stocks (e.g., CPT 2022; BSAI GPT 2023), as well as characterize general ecosystem health (e.g., Siddon 2023).

The EBS Shelf Trawl Survey is inherently a multi-species survey designed to simultaneously provide information on a variety of species and ecosystem components, and thus may not be the "optimal" approach to quantify abundance and changes in abundance for any individual species. In particular, the trawl gear used in the survey is size selective for many species (Somerton et al. 2013), meaning that the probability of capturing an individual in the path of the trawl gear depends on its size. For a given species, smaller individuals in the path of the gear may be less likely to be caught than larger individuals because they can slip through the mesh of the trawl net. Larger, more mobile individuals in the path of the gear may be less likely to be caught than intermediate-sized individuals because they are able to outswim it. Alternatively, intermediate-sized individuals may engage in behaviors such as ducking under the trawl footrope that enable some fraction to escape capture. In addition, the probability of capture (or, conversely, escape) may vary with local environmental factors such as temperature and, for benthic species and benthic trawl gear, sediment characteristics. Determining the size-dependent capture probability of a survey gear for a given species can be important for assessing population size because it allows one to estimate the size-specific density of the species across the area in which hauls were conducted. This size-dependent probability of capture is referred to here as "haul-level selectivity."

Side-by-side haul studies

Side-by-side haul (SBS) studies are one approach to characterizing the selectivity of one trawl gear relative to another (Somerton et al. 2013; Kotwicki et al. 2017). At its simplest, a SBS study uses paired hauls to simultaneously sample the size distribution of a targeted species at a given location with the target and reference gears. Conceptually, the ratio of the observed size distribution from the target gear to that from the reference gear yields the size selectivity of the target gear for the target species *relative* to the reference gear. If the size selectivity of the reference gear is known (or assumed), then the *absolute* size selectivity of the target gear can subsequently be determined.

Haul-level equations

For each paired haul h , the number of crab caught within a size bin z using gear type g , $n_{h,z}^g$, is a random variable with some sampling distribution which we represent as $Pr(\bar{n}_{h,z}^g, \theta_{h,z})$, where $\bar{n}_{h,z}^g$ is the *expected* number and $\theta_{h,z}$ represents parameters that characterize other aspects of the distribution (e.g., the variance). Further, the expected number $\bar{n}_{h,z}^g$ is related to the gear-specific selectivity $S_{h,z}^g$ for crab in size bin z at the location of the paired haul h , the gear/haul-specific area swept $A_{h,z}^g$, and the local density of crab (e.g., abundance/km²) at the location of the paired haul location in the size bin, $D_{h,z}$, by

$$\bar{n}_{h,z}^g = S_{h,z}^g \cdot A_{h,z}^g \cdot D_{h,z} \quad (1)$$

The gear-specific selectivity $S_{h,z}^g$ is characterized by haul to reflect potential environmental factors such as depth or temperature that may influence selectivity and vary among paired hauls. The nature of the sampling distribution $Pr(\dots)$ will be determined by the spatial process determining how crab are distributed across the local benthic seascape.

In the case that individuals are distributed completely randomly across the seascape, the spatial process is a homogeneous spatial Poisson process with intensity $D_{h,z}$ and the sampling distributions are Poisson distributions with intensities $S_{h,z}^g \cdot A_{h,z}^g \cdot D_{h,z}$. Furthermore, in this (special) case, it can be shown that the target gear catch, conditional on the total catch (target + reference) $n_{h,z}$, follows a binomial distribution $B(N, p)$ with $N = n_{h,z}$ and

$$p = p_{h,z} = \frac{\bar{n}_{h,z}^T}{\bar{n}_{h,z}^T + \bar{n}_{h,z}^R} \quad (2)$$

where the superscripts T and R indicate the target and reference gears, respectively. Substituting eq. 1 for the target and reference gears into eq. 2 yields

$$\begin{aligned} p_{h,z} &= \frac{S_{h,z}^T \cdot A_{h,z}^T \cdot D_{h,z}}{S_{h,z}^R \cdot A_{h,z}^R \cdot D_{h,z} + S_{h,z}^T \cdot A_{h,z}^T \cdot D_{h,z}} \\ &= \frac{S_{h,z}^T \cdot A_{h,z}^T}{S_{h,z}^R \cdot A_{h,z}^R + S_{h,z}^T \cdot A_{h,z}^T} \\ &= \frac{r_{h,z} \cdot q_h}{1 + r_{h,z} \cdot q_h} \end{aligned} \quad (3)$$

where $r_{h,z}$ represents the ratio of the gear-specific selectivities $\frac{S_{h,z}^T}{S_{h,z}^R}$ and q_h is the ratio of areas swept by the two gear types (i.e., $\frac{A_{h,z}^T}{A_{h,z}^R}$). Eq. 3 can be recast slightly to yield

$$\text{logit}(p_{h,z}) = \ln\left(\frac{p_{h,z}}{1 - p_{h,z}}\right) = \ln\left(\frac{\frac{r_{h,z} \cdot q_h}{1 + r_{h,z} \cdot q_h}}{1 - \frac{r_{h,z} \cdot q_h}{1 + r_{h,z} \cdot q_h}}\right) = \ln(r_{h,z} \cdot q_h) = \ln(r_{h,z}) + \ln(q_h) \quad (4)$$

This allows one to fit statistical models of the form

$$\text{logit}(\hat{p}_{h,z}) = \ln R(z, e_{1_h}, e_{2_h}, \dots) + \ln(q_h) + \epsilon_{h,z} \quad (5)$$

where $\hat{p}_{h,z}$ is the observed proportion, q_h is a known constant, $e_{1_h}, e_{2_h}, \dots$ represent environmental variables associated with the paired haul, and $\epsilon_{h,z}$ represents sampling variability. Under the assumption that the reference gear captures all crab in its swept area (i.e., the reference gear is not size-selective and has catchability equal to 1),

$$\tilde{S}_{h,z}^T = \exp(\ln R(z, e_1(h), e_2(h), \dots)) \quad (6)$$

is the estimated haul-level selectivity of the target gear.

Survey-level equations

An alternative to the above haul-level analysis is based on aggregating the haul-level area-swept abundance-at-size to the entire SBS study area (A^{SBS}) such that

$$N_z^g = A^{SBS} \cdot \sum_h \frac{n_{h,z}^g}{A_h^g} \quad (7)$$

is a gear-specific estimate of area-swept abundance-at-size in the SBS study area. By analogy with the haul-level equation (Eq. 1), N_z^g is related to total abundance-at-size in the SBS study area, N_z by

$$N_z^g = \tilde{S}_z^g \cdot N_z \quad (8)$$

where $\tilde{S}_z^g$ represents “survey-level” selectivity for gear g .

Taking the ratio of N_z^g from the target gear ($g = T$) to that for the reference gear ($g = R$) yields

$$\frac{N_z^T}{N_z^R} = \frac{\tilde{S}_z^T}{\tilde{S}_z^R}$$

which, under the assumption that the reference gear is non-size selective and has catchability equal to 1, yields

$$\tilde{S}_z^T = \frac{N_z^T}{N_z^R} \quad (9)$$

If haul-level selectivity is independent of local environmental factors, then $\tilde{S}_z^T$ is also an estimate of haul-level selectivity and characterizes “survey-level” selectivity across the entire area covered by the survey using the target gear. However, if haul-level selectivity depends strongly on local environmental factors, then $\tilde{S}_z^T$ represents a weighted average

of individual haul-level values appropriate for the SBS study area. If the SBS study area does not coincide with the area covered by the survey that uses the target gear, $\tilde{S}_z^T$ is not necessarily applicable to the entire survey area (and thus may not represent true “survey-level” selectivity for the target gear).

Application

SBS studies in the eastern Bering Sea (EBS) were first conducted jointly by the National Marine Fisheries’ Alaska Science Center (NMFS AFSC) and the Bering Sea Fisheries Research Foundation (BSFRF) in 2009 and 2010 at standard NMFS EBS Shelf Survey stations in the northwest of the EBS shelf survey grid to characterize the selectivity of the NMFS bottom trawl survey gear for snow crab, *Chionoecetes opilio* (Somerton et al. 2013). Similar studies were conducted in 2013, 2014, 2015, and 2016 in the southeastern EBS (Bristol Bay) to characterize the selectivity of the NMFS bottom trawl survey gear for red king crab (*Paralithodes camtschaticus*) and Tanner crab (*Chionoecetes bairdi*), as well as further west on the EBS shelf in 2017 and 2018 to focus more specifically on Tanner and snow crab.

This document provides a detailed report on the data and selectivity analyses for Tanner crab (similar analyses are in progress for red king crab and snow crab). While the NMFS bottom trawl gear (an 83-112 otter trawl) is considered size-selective for Tanner crab because escapement under the trawl footrope is size-dependent, the BSFRF nephrops trawl (the reference gear in these studies) has better bottom-tending abilities than the NMFS gear such that the BSFRF gear is considered to be non-size selective for Tanner crab across the range of sizes of interest to the stock assessment (25-200 mm carapace width [CW]; Stockhausen (2023)). Estimates of size-specific selectivity, by sex, for Tanner crab in the NMFS EBS survey are presented based on both the survey-level and haul-level approaches described above (i.e., eq.s 9 and 6, respectively).

Methods

SBS data collection

The National Marine Fisheries’ Alaska Science Center (NMFS AFSC) has conducted an annual design-based summer bottom trawl survey on the eastern Bering Sea continental shelf using standardized gear since 1982 (Lauth and Nichol 2013), with the exception of 2020. A series of studies were conducted jointly by AFSC and the Bering Sea Fisheries Research Foundation (BSFRF) in 2013, 2014, 2015, and 2016 at standard survey stations in Bristol Bay, Alaska to characterize the selectivity of the NMFS bottom trawl survey gear for red king crab, *Paralithodes camtschaticus*, and Tanner crab, *Chionoecetes bairdi* (Figure 3). Similar studies focused more specifically on Tanner crab were also conducted in 2017 and 2018 further west on the EBS shelf (Figure 3). These “side-by-side” (SBS) studies were similar to earlier studies conducted to characterize NMFS survey selectivity for snow crab, *C. opilio*, using the same gear but conducted on the northwest EBS shelf (Somerton et al. 2013). The NMFS bottom trawl is a 83-112 otter trawl, which is deployed at standard survey stations located on a fixed grid with (generally) 20 x 20 nmi² cells across the EBS shelf and towed for ~30 minutes on-bottom (Figure 2). The complete NMFS survey includes additional stations located at grid corners near the Pribilof Islands and St. Matthew Island to

provide higher sampling densities in those areas for blue king crab (e.g., [Lauth and Nichol 2013](#)); some of these were included in the 2017 and 2018 studies.

For the SBS studies, a BSFRF-chartered vessel towed a nephrops bottom trawl, Figure 2 starting at the same time and in the direction as the associated NMFS tow, but offset by ~0.5 nmi to create a “paired haul”. Tow durations were limited to ~5 minutes on bottom because the BSFRF gear was somewhat smaller than the NMFS gear and had better bottom-tending performance ([Conan et al. 1994](#)).

Haul sampling protocols for both survey groups were based on the standard NMFS EBS survey protocols ([Lauth and Nichol 2013](#); [Zacher et al. 2020](#)) For each tow, bottom depth and bottom temperature were based on sensor readings. Area swept was determined from net mensuration data recording net width and distance towed on-bottom. All crab were removed from the catch, sorted by species and sex, and a total catch weight was obtained for each species. Biological sampling for various crab species during survey generally consists of whole haul sampling, in which all crab are enumerated and various biological data such as sex, maturity state, shell condition, and carapace width (or length, depending on species) were recorded for each individual (e.g., [Zacher et al. 2020](#)). However, when the total catch was exceptionally large (approximately > 300 individuals), a random subsample of the catch was taken for enumeration and biological data collection. When subsampling occurred, the subsample could be taken at the level of the total catch, or at the level of a particular size and sex category. The weights for the entire catch component to be subsampled and the total weight of all individually-measured crab in the subsample were recorded and a “sampling factor”, θ , was calculated from the size composition of the measured crab to “expand” the measured crab abundance to the entire catch component:

$$\theta = \frac{W_T}{W_S} \quad (10)$$

where W_T was the total weight of crab in the catch component and W_S was the total weight of measured crab in the subsample. A sampling factor was associated with each measured crab (measured crab which were not part of a subsample were assigned a sampling factor of 1).

Gear-specific catch abundance was standardized by sex (x) and 5-mm carapace width (CW) size bin (z) to catch-per-unit-effort (CPUE; numbers/sq nmi) for each haul by: 1) summing the sampling factors associated with each measured crab by sex/size category and 2) dividing by the area swept by the gear g during the haul, $a_{g,h}$, as in:

$$CPUE_{g,h,x,z} = \frac{\sum_{i_{g,h} \in \{x,z\}} \theta_{i_{g,h}}}{a_{g,h}} \quad (11)$$

where h denotes the paired haul, $i_{g,h}$ indicates the i -th measured crab caught by gear g in haul h , θ_i is the sampling factor associated with the i -th crab, and $i_{g,h} \in \{x,z\}$ indicates that the sum is only over measured crab falling within sex/maturity state/size bin $\{x,z\}$. The area swept $a_{g,h}$ was estimated using bottom contact sensor data to determine time/distance on bottom and net mensuration data to determine net width.

Gear-specific, haul-level CPUE was subsequently expanded to total abundance size compositions by sex and 5-mm CW size bin within each year-specific study area using standard

design-based calculations for area expansion (Wakabayashi et al. 1985), where each study area was considered a single stratum and the NMFS EBS survey grid provided the associated area expansion factors (~400 sq. nmi) for the paired hauls. Specifically,

$$N_{g,x,z} = \sum_h CPUE_{g,h,x,z} \cdot A_h \quad (12)$$

The estimated uncertainties associated with gear-specific, haul-level CPUEs were found to be large (i.e., CV's > 1, see Section), with a substantial fraction of hauls (> 25%) catching no crab within a size bin. Consequently, we conducted a 2-stage bootstrapping analysis, by crab sex, to better characterize the uncertainties associated with the area-level size compositions obtained using Eq. Equation 12 and subsequent estimates of survey-level selectivity using Eq. Equation 9. For each year, a bootstrap replicate consisted of first selecting n_y paired hauls from that year at random with replacement, where n_y was the original number of paired hauls conducted in year y . For each randomly-selected paired haul, n_g crab of the relevant sex were then selected from the haul data for gear type g at random with replacement, where n_g was the original number of crab sampled by gear type g . A bootstrap replicate for the gear-specific total abundance size composition was subsequently calculated using Eq. Equation 12, and a corresponding replicate for the survey-level selectivity was calculated using Eq. Equation 9. This was repeated 1,000 times for each study year, resulting in 1,000 sex-specific bootstrap replicates of gear-specific size composition and survey-level selectivity.

Survey-level selectivity

“Survey-level” selectivity for the NMFS gear was estimated for each SBS study year y by sex x using Eq. 9 from the corresponding area-expanded total abundance size compositions (Eq. 12) as

$$\tilde{S}_{y,z|x}^{NMFS} = \frac{N_{g=NMFS,y,x,z}}{N_{g=BSFRF,y,x,z}} \quad (13)$$

where the $\tilde{S}_{y,z|x}^{NMFS}$ were regarded as noisy estimates of the true, smoothly-varying, SBS area-specific survey-level NMFS selectivity $\tilde{S}_{y,z|x}^{NMFS}$. Subsequent analysis considered each sex separately, and hence we drop the x indices from subsequent notation, although this remains implied.

The $\tilde{S}_{y,z}^{NMFS}$ were fit using generalized additive models (GAMs; Wood et al. (2016)) to estimate the $\tilde{S}_{y,z}^{NMFS}$ as smoothly-varying functions of size. GAMs have the following form

$$g(\mu_i = E[y_i]) = \eta_i = A_i + \mathbf{x}_i\gamma + \sum_{j=1}^J f_j(\mathbf{x}_i), \quad y_i \sim \text{EF}(\mu_i, \phi)$$

where $\mu_i = E[y_i]$ is the expected value (mean) for the i th observation y_i , $g(\cdot)$ is a “link” function, η_i is the value of the link function for the i th observation, A_i is a fixed offset, $\mathbf{x}_i$

is the i th row of a parametric model matrix, γ is a vector of parameters for the parametric terms, f_j is a smooth function of the covariates $\mathbf{x}$ associated with the i th observation. $y_i \sim \text{EF}(\mu_i, \phi)$ denotes that the observations y_i are randomly distributed in accordance with a member of the exponential family of distributions, EF , with mean μ_i and scale parameter ϕ . In general, the choice of link function g is limited by the selected exponential family.

The smooth functions f_j are represented in the model via penalized spline basis expansions of the covariates, with each a weighted sum of basis functions $b_{jk}(\dots)$

$$f_j(x_{ij}) = \sum_{k=1}^K \beta_{jk} b_{jk}(x_{ij})$$

where β_{jk} is the weight (coefficient) associated with the k th basis function $b_{jk}(\dots)$ evaluated at the covariate value x_{ij} for the j th smooth function f_j . "Wiggleness" penalties $\sum_j \lambda_j \beta^T \mathbf{S}_j \beta$ control the degree of smoothing applied to the f_j through the smoothing parameters λ_j .

We wanted to estimate a smoothly-varying function of size $\tilde{S}_{y,z}^{NMFS}$ common to all years, and to test whether there was substantial variation in the $\tilde{S}_{y,z}^{NMFS}$ among years. If the latter were true, this would be strong evidence that NMFS haul-level selectivity varied spatially and/or temporally, possibly due to environmental covariates. If the latter were false, this would suggest that the estimated function could represent NMFS selectivity at the scale of the entire EBS shelf survey. To this end, we fit three GAMs of the form

$$\eta_i = M_{y,z} = \begin{cases} a) \alpha + s(z) \\ b) \alpha + s(z) + t_y(z) \\ c) \alpha + s(z) + \dot{t}(z, y) \end{cases} \quad (14)$$

to the $y_i = \tilde{S}_{y,z}^{NMFS}$, where i maps appropriately to y, z . $M_{y,z}$ is the estimated link-scale mean, α is an estimated intercept, $s(z)$ represents a smooth function of z , and $t_y(z)$ and $\dot{t}(z, y)$ represent sets of smooth functions which are constrained to sum to 0 across the y 's for each z . We selected the Tweedie family of exponential distributions (Tweedie 1984) to represent the statistical error structure of the observations, i.e.,

$$\tilde{S}_{y,z}^T \sim Tw(\mu_{y,z}, p_{tw}) \quad (15)$$

where $\mu_{y,z} = g^{-1}[M_{y,z}]$, the variance of $\tilde{S}_{y,z}^T$ is given by $\sigma^2 \mu_{y,z}^{p_{tw}}$, σ^2 is the positive dispersion parameter, and p_{tw} is the Tweedie power parameter. Tweedie distributions are a family of probability distributions which include, depending on the value of p_{tw} , the normal, gamma and inverse Gaussian distributions, the Poisson distribution, and the class of compound Poisson-gamma distributions which have positive mass at zero, but are otherwise continuous.

We implemented the three models (eq. 14) in R using the *mgcv* package Wood (2017) and `tw`, the *mgcv* version of the Tweedie family of distributions with p_{tw} estimable but

limited to values between 1 and 2 (the limits representing the Poisson and gamma distributions, respectively). The natural logarithm was taken as the link function g , the smooth function $s(z)$ was implemented as thin-plate splines using *mgcv*'s main spline function $s(z, bs="tp")$, the $t_y(z)$'s were implemented with y as a factor smooth variable using *mgcv*'s tensor interaction spline function ($ti(z, by=y, bs="sz")$), and $\hat{t}(z, y)$ was implemented as a random factor smooth ($ti(z, y, bs="fs")$), with y represented as a factor (rather than a continuous covariate) in the latter two cases. The three models were fit for each sex/maturity category using residual maximum likelihood ("REML") to estimate the smooths. Individual observations (i.e., the $\tilde{S}_{y,z}^{*T}$) were weighted by the total number of crab sampled at size z in year y by the BSFRF gear.

Sediment characteristics

Somerton et al. (2013) found that sediment grain size was an important spatial covariate, along with bottom depth, in characterizing haul-level selectivity for snow crab in the NMFS 83-112 trawl gear. Taking sediment samples is not part of the standard NMFS EBS shelf survey protocols, so sediment samples coincident with NMFS survey haul locations are generally unavailable. Consequently, we followed Somerton et al. (2013) and used an updated version of the EBSED-2 database (Richwine et al. 2018) as the basis for estimating sediment characteristics at paired haul locations. Values for sediment mean grain size (in $-\log(\phi)$ units) and sorting coefficient (reflecting the variance in sediment grain size) at the NMFS survey haul locations were interpolated from rasters (Figures 4 and 5) created using co-kriging geostatistics function in ArcGIS (ESRI 2021) from sediment core data collected across the EBS shelf and slope contained in the EBSED-2 database.

Haul-level selectivity

"Raw" haul-level data

Paired NMFS and BSFRF hauls from the SBS studies were identified by station id, haul date and time, and distance between starting locations. In general, more crab were caught by the NMFS gear relative to the BSFRF gear except at the smallest sizes (< 25 mm CW; Figures 6 and 7) because the area swept by the NMFS gear was ~ 6 x larger than that of the BSFRF gear. When corrected for the difference in areas swept and subsampling, CPUE for the BSFRF gear was generally higher at all sizes, although many exceptions also occurred (i.e., where the circles indicating the values of $\ln(R)$ are above 0 in the righthand column of each figures).

Censored haul-level data

For SBS paired hauls, the ratio of the NMFS area swept to the BSFRF area swept was expected to be ~ 6 . Area-swept ratios outside the range $\{3, 12\}$ were regarded as indicative of a problem with one or both of the hauls and the associated paired haul were dropped from further analysis (Figure 8). Applying this criterion resulted in dropping 10 paired hauls and 222 size composition records (Figures 9 and 10). An additional 5275 size composition

records were dropped after requiring that each reflected a minimum of sampled individuals (Figure 11). This resulted in a total of 2430 used in the haul-level selectivity analyses (Figure 12).

Models

The proportion of individuals caught in paired haul h in size bin z by the target gear relative to all individuals caught in the size bin, $p_{h,z}$, is related to the ratio of the target to reference gear selectivities $r_{h,z}$ by

$$\text{logit}(p_{h,z}) = \ln\left(\frac{p_{h,z}}{1 - p_{h,z}}\right) = \ln(R_{h,z}) + \ln(q_h) \quad (16)$$

where q_h is the ratio of gear expansion factors $\frac{S_h^t \cdot A_h^t}{S_h^r \cdot A_h^r}$ (S_h^x is the sampling fraction and A_h^x is the area swept for gear x on haul h).

A number of candidate paired-haul selectivity models were evaluated by sex to select the “best” sex-specific models for NMFS Tanner crab haul selectivity under the assumption that BSFRF gear selectivity was 1 for all Tanner crab sizes (such that the selectivity ratio R is equivalent to the NMFS selectivity). These included binomial models fit to the observed proportions $p_{h,z}$ and Tweedie models fit to the observed ln-scale selectivity ratio $\ln(R_{h,z}) = \text{logit}(p_{h,z}) - \ln(q_h)$.

All candidate models were implemented as generalized additive models (GAMs) using the *gam* function from the R package *mgcv* (Wood 2011; Wood 2017) and fit with various non-parametric “smooth” functions to evaluate potential main effects and two-way interactions of size (z), gear depth (d), bottom temperature (t), and local sediment characteristics ($-\log_2(\phi)$), denoted as f in the following, and sorting coefficient (s) on NMFS gear size selectivity represented by the general form

$$\ln(R_{h,z}) = \alpha + s(z) + \sum_i \beta_i \cdot s_i(x_{i,h}) + \sum_i \gamma_i \cdot s_i(z, x_{i,h}) + \sum_i \delta_{i,j} \cdot s_{i,j}(x_{i,h}, x_{j,h}) \quad (17)$$

where α is a global intercept, the s 's represent smooth functions, and the β_i 's, γ_i 's, and $\delta_{i,j}$'s are 1 or 0 to include/exclude the associated smooth term. The first two terms (intercept α and main size effect $s(z)$) were common to all models. Candidate models of additional complexity that included environmental covariates were created by setting the β_i , γ_i , and $\delta_{i,j}$ coefficients to 0 or 1 to include or exclude the associated smooth term in the model. All combinations were evaluated. The smooth terms were implemented using the “tensor interaction” *mgcv* function *ti*, which was appropriate for evaluating the significance of individual terms with the “ts” basis. The “ts” basis is a thin plate spline with extra shrinkage for each variable in the smooth function (Wood 2011; Wood 2017). Model estimation used a Laplace approximation to restricted maximum likelihood (REML) as the optimization method for all models (Wood 2011; Wood 2017). Wood (2011) suggests this may be the most effective choice among the alternatives available in *mgcv*.

For both assumed distributions, the suite of candidate models outlined above was evaluated by sex for out-of-sample performance using k-fold generalized cross validation (GCV) with 20 “folds”. Fold selection consisted of randomizing the order of paired haul size-specific observations across all years and excluding consecutive sets of 5% of the observations from the data used to fit a particular candidate model. The candidate was then evaluated by its ability to predict the out-of-sample data and the mean GCV score over the 20 folds was used to rank the candidate model configurations to identify the “best” model. In addition to the models with environmental covariates described above, a model that treated the haul-level variability as random effects was also evaluated for both distributions.

Binomial models with haul-level environmental covariates

Candidate binomial models were fit to the sex-specific observed proportion of Tanner crab caught by 5-mm CW size bin in the NMFS gear using terms from the following full model expressed as an *mgcv* formula:

$$\begin{aligned}
 p_{h,z} \sim & s(z, bs = "ts", k = k1) + \\
 & ti(d, bs = "ts", k = k2) + ti(t, bs = "ts", k = k2) + \\
 & ti(f, bs = "ts", k = k2) + ti(s, bs = "ts", k = k2) + \\
 & ti(z, d, bs = "ts", k = c(k1, k2)) + ti(z, t, bs = "ts", k = c(k1, k2)) + \\
 & ti(z, f, bs = "ts", k = c(k1, k2)) + ti(z, s, bs = "ts", k = c(k1, k2))
 \end{aligned} \tag{18}$$

where the smoothing constants $k1$ and $k2$ were set to 10 and 8, respectively. The number of measured crab in a size bin was used as the sample size and $\ln(q_h)$ was entered as a haul-specific constant offset.

Binomial model with haul-level random effects

The binomial model with haul-level random effects was implemented in *mgcv* using

$$p_{h,z} \sim s(z, bs = "ts", k = k1) + ti(z, h, bs = "fs", k = k1) \tag{19}$$

where $k1$ was 10.

Tweedie models with haul-level environmental covariates

Candidate Tweedie models were fit to the sex-specific observed proportion of Tanner crab caught by 5-mm CW size bin in the NMFS gear using terms from the following full model expressed as an *mgcv* formula:

$$\begin{aligned}
R_{h,z} \sim & s(z, bs = "ts", k = k1) + \\
& ti(d, bs = "ts", k = k2) + ti(t, bs = "ts", k = k2) + \\
& ti(f, bs = "ts", k = k2) + ti(s, bs = "ts", k = k2) + \\
& ti(z, d, bs = "ts", k = c(k1, k2)) + ti(z, t, bs = "ts", k = c(k1, k2)) + \\
& ti(z, f, bs = "ts", k = c(k1, k2)) + ti(z, s, bs = "ts", k = c(k1, k2))
\end{aligned} \tag{20}$$

where a log link function was specified and smoothing constants $k1$ and $k2$ were set to 10 and 8. Observed R values greater than 10 were excluded.

Tweedie model with haul-level random effects

The Tweedie model with haul-level random effects was implemented in *mgcv* using

$$R_{h,z} \sim s(z, bs = "ts", k = k1) + ti(z, h, bs = "fs", k = k1) \tag{21}$$

where a log link function was specified and $k1$ was set to 10. Observed R values greater than 10 were excluded.

Results

SBS surveys results

Paired-haul characteristics

In total, 384 SBS paired hauls were conducted during the 2013-2018 study period (Table 1). For NMFS hauls, areas swept ranged from 0.00626 to 0.0169 sq. nmi. and from 0.001 to 0.00497 sq. nmi. for BSFRF hauls (Figure 13). Area swept ratios for individual paired hauls (NMFS/BSFRF) ranged from 1.28 to 13.5, with a mean of 6.24 (Figure 14). NMFS haul depths ranged from 28 to 156 m while BSFRF haul depths ranged from 24 to 157 m. On average, NMFS tows tended to be 1.4 m deeper than the paired BSFRF haul, but depth differences (NMFS-BSFRF) ranged from -5.3 to 8.7 m (Figure 15). NMFS haul bottom temperatures ranged from 0 to 8.4 °C while BSFRF haul bottom temperatures ranged from 0.03 to 10 °C. Although one paired haul exhibited a large difference between the measured bottom temperatures for the two gear types (-5.2 °C), the minimum, maximum, and mean differences excluding this paired haul were small (-0.21, 0.93, and 0.021 °C, respectively; Figure 16).

Catch data

On average, the NMFS gear tended to catch more crab per haul at all but the smallest sizes compared with the BSFRF gear, largely reflecting the larger area swept by the NMFS gear during any given haul (Figures 17 and 18). Pulses of small crab (< 25 mm CW) were apparent in the BSFRF catch in every year except 2014, but absent from the NMFS catch. The percentage of hauls for which crab were caught varied with sex, size bin, and gear type across the SBS study years, generally reflecting the patterns obtained for the number of crabs caught (Figures 19 and 20). Percentages ranged from almost 0 (NMFS gear, 2013-2015, crab size < 25 mm CW) to > 80% (BSFRF gear, 2017 and 2018, crab size < 25 mm CW). Hauls with the BSFRF gear exhibited higher percentages of catches with small crab in most years, relative to the NMFS gear (2014 and 2015 being the exceptions, with very few hauls catching small crab for either gear type). Hauls with the NMFS gear exhibited higher percentages (~65%) of catches with large males (> 80 mm CW) in all years relative to the BSFRF gear (~45%), but for large females (> 75 mm CW) only in 2013 and 2014 (30-45% vs 20-30%, respectively; otherwise, the percentages were similar for the two gear types). For both gear types, the percentage of hauls in which crab < 50 mm CW were caught were similar for both sexes (Figures 21 and 22). For larger crab, both gear types also exhibited similar patterns by sex, although these differed by year.

Changes in the abundance of crab catches or percentage of hauls with crab across years did not necessarily indicate similar changes in the EBS Tanner crab stock because the former were the result of changes in the SBS study area each year confounded with changes in abundance and distribution of the Tanner crab population itself. Thus, for example, it is unknown to what extent the dramatic increases in catch of small crab in 2016-2018 relative to 2013-2015 were associated with actual recruitment events or were the result of the shifts in the SBS study areas westward from Bristol Bay (2013-2015) towards the Pribilof Islands (2016-2018).

After scaling the numbers of sampled crab for sampling fractions and areas swept by the gear to catch-per-unit-effort (CPUE) by haul, the BSFRF gear was generally more efficient (i.e., it had a comparatively higher CPUE) than the NMFS gear across the size range for both sexes (Figures 23 and 24). The associated CVs were large (>1 at all sizes for both sexes and gear types) and exhibited similar, somewhat U-shaped, curves: relatively high for small crab decreasing, decreasing to lower (but still large) values at intermediate sizes, then increasing again at large sizes (Figures 25 and 26).

Mean-variance analysis

The number of hauls for which no crab were caught within a size bin suggests that the SBS CPUE data may be better represented as arising from non-negative compound distributions that combine a discrete probability of obtaining a 0 with a continuous distribution that describes the conditional probability of obtaining a (positive) value given that the value is non-zero. The Tweedie family of distributions (Tweedie 1984) are a subset of exponential dispersion models that include the normal, Poisson, gamma, and compound Poisson-gamma distributions, among others, as special cases. Distributions consistent with the Tweedie exponential family exhibit variance-mean relationships of the form

$$V = \sigma^2 \mu^{p_{tw}} \quad (22)$$

where V represents the variance of some random variable, μ represents its mean, σ is a positive scaling coefficient, and p_{tw} is the so-called Tweedie power parameter. The normal, Poisson ($p_{tw} = 1$), and gamma ($p_{tw} = 2$) distributions are obtained when $p_{tw} = 0, 1$, or 2 , respectively. For $1 < p_{tw} < 2$, the distributions are compound Poisson-gamma distributions, with positive mass at 0 but otherwise positive and continuous. Taylor's Law in ecology (Xiao et al. 2015), relating the variance of the number of individuals of a species per unit area of habitat to some power of the corresponding mean, can be viewed as a result of individuals undergoing dispersion consistent with a Tweedie distribution.

Equation 22 implies a linear relationship between $\ln(V)$ and $\ln(\mu)$:

$$\ln(V) = \alpha + p_{tw} \cdot \ln(\mu) \quad (23)$$

Figures 27 and 28 suggest the relationship between the ln-scale variance and mean is close to linear, although it also appears to be slightly convex when treating results across years and size bins as equivalent. Ignoring the latter detail as a simplifying assumption, we fit the following model for the annual variance in CPUE V by gear type, sex, year, and size bin (g, x, y, z) as a function of the associated mean CPUE $\mu_{g,x,y,z}$

$$\ln(V_{g,x,y,z}) = \alpha + \sum_{g,x} \delta\alpha_{g,x} + \beta \cdot \ln(\mu_{g,x,y,z}) + \sum_{g,x} \delta\beta_{g,x} \cdot \ln(\mu_{g,x,y,z}) + \epsilon_{g,x,y,z}$$

where $\epsilon_{g,x,y,z} \sim N(0, \sigma_\epsilon^2)$

(24)

where α and β are fixed effects representing the common ln-scale intercept and slope for the linear relationship between $\ln(V)$ and the covariate $\ln(\mu)$ expected for a Tweedie-distributed process, the $\delta\alpha_{g,x}$ and $\delta\beta_{g,x}$ represent factor combination-specific offsets to the common intercept and slope parameters, and $\epsilon_{g,x,y,z}$ is the error term, which was assumed to be normally-distributed with mean 0 and variance σ_ϵ^2 . We also fit a "reduced" (i.e., simplified) version of Equation 24 that included only the common slope term by fixing the $\delta\beta_{g,x}$ coefficients to 0.

Using the Bayesian Information Criterion [BIC; (?)], the reduced model was found to provide a better, more parsimonious fit to the data than the full model (Table 2; the difference in BIC was 13 units): the factor-specific interaction terms did not improve the fit substantially when offset by the increase in model complexity. A quantile-quantile plot of the residuals did not indicate any issues with the model (Figure 29), although a plot of the residuals against the model predictions showed evidence of the slight convexity previously noted (Figure 30). The estimated Tweedie power parameter (the common slope) was 1.546.

Size compositions

Because the SBS study areas differed by total area covered each year, to facilitate comparisons across years it was convenient to calculate gear/sex-specific annual size compositions $Z_{g,y,x,z}$ as the area-weighted mean CPUE for gear type g over the SBS study area in year y for sex x in size bin z using

$$Z_{g,y,x,z} = \frac{\sum_h a_{y,h} \cdot CPUE_{g,y,h,x,z}}{\sum_h a_{y,h}}$$

where $a_{y,h}$ denotes the NMFS station area associated with paired haul h in year y used to expand the CPUE to the total abundance in the area associated with the station, and the sum is over the paired hauls conducted in year y (Tables 3 and 4; Figures 31 and 32). The results are similar in size dependence and relative scale to the mean CPUE results (Figures 23 and 24), although the correspondence is not exact because the mean CPUE results were not weighted by NMFS station area. As with the mean CPUE, the BSFRF gear was generally more efficient (i.e., it had a comparatively higher estimates of total abundance) than the NMFS gear across sizes for both sexes in all years.

While it would have been possible to obtain variance estimates for the $Z_{g,y,x,z}$ based on assumptions of normality, given that the CPUE results appeared to be non-normal and better-described by a Tweedie distribution (see previous section), a two-stage bootstrap analysis was undertaken to characterize the variability of the size compositions. Briefly, the paired haul data was resampled on an annual basis 1000 times using a two-stage, hierarchical empirical bootstrap and size compositions were recalculated for each replicate. For each bootstrap repetition for a given year, stage one consisted of randomly selecting, with replacement, n_h paired hauls (where n_h matched the number of actual paired hauls). Stage two then consisted of randomly selecting $n_{g,h,x}^i$ individuals by gear type and sex with replacement, where $n_{g,x}^i$ was the number of sex x crab measured in the actual gear type g haul. The resulting dataset was then processed identically to the actual dataset to obtain a bootstrapped size composition realization.

Despite the large number of repetitions, the bootstrapped size compositions exhibit instances of quantization for both gear types and sexes (Figures 31 and 32), as exemplified for the BSFRF gear in 2013 and 2016 and the NMFS gear in 2015 in Figure 31. The quantization is a result of the fairly low percentage of hauls with non-zero catches of crab in the relevant size bins. The relative patterns between the two gear types (Figures 33 and 34) are, unsurprisingly, similar to those for mean CPUE (Figures 23 and 24), with the BSFRF gear typically having higher values relative to the NMFS gear across the entire size range for both sexes (although exceptions exist for specific size bins in specific years, e.g. several size bins in 2015 for males). The CVs for the size compositions (Figures 35 and 36) are all less than one and typically much smaller than those obtained with the haul-level CPUE results (Figures 25 and 26).

Survey-level selectivity

Males

The “observed” annual empirical survey catchability curves for males are substantially variable across the size range collected in the SBS studies (Figure 37). Taken as a whole, they exhibit an increasing, slightly concave trend from 25 mm CW to ~125 mm CW, with a suggestion of a subsequent decline at larger sizes, although sample sizes are small above 150 mm CW.

Both the intercept and the year-invariant smooth term $s(z)$ were highly significant (p values < 0.0001) for all three models (Tables 5-7). Of the six year-specific “factor” smooths ($ti(z, y)$) in Model 2, three were significant (p values < 0.001 ; for 2013, 2014, and 2018) while the remainder were not significant ($p > 0.4$; 2015, 2016, 2017; Table 6). The random effects term in Model 3, $s_{RE}(z, y)$, was also found to be highly significant ($p < 0.0001$; Table 7). The estimated Tweedie power parameter ranged from 1.436 for Model 1 to 1.537 for Model 2, with an intermediate value (1.467) for Model 3 (Table 8). Of the three, Model 3

exhibited the lowest AIC and BIC scores and the highest adjusted R^2 (Table 8). Diagnostic plots for residuals (Figures 38-40) exhibited departures from expected cumulative distributions at the upper ends of those distributions, as well as significant deviations from the KS, dispersion, and outlier tests provided by the DHARMA R package (Hartig 2022).

The time-invariant estimates of catchability, $S(z) (= \exp(\alpha + s(z)))$, where α is the estimated link-scale intercept and $s(z)$ is the estimated link-scale smooth function of size, z , were quite similar across the three models, exhibiting patterns similar to that noted for the “observed” values (Figure 41). Estimates for fully-selected catchability, or q (the maximum of $S(z)$), varied over a narrow range, from 0.575 (Model 1) to 0.603 (Model 2), while the 5-mm size bin at which full selection occurred varied from 135 mm CW (Model 3) to 145 mm CW (Model 2) (Table 9). While the estimated curves from all three models exhibited a decline in catchability after reaching their peaks, the uncertainty in the estimated curves above 150 mm CW was considerable. Below 150 mm CW, the uncertainty associated with the estimated curve from Model 3 was larger (on the order of 2x) than that of the other two models at any size because it reflects the estimated *population-level* (i.e., for any randomly-selected year) variability in catchability, not the variability associated with the specific years of the SBS studies.

Females

As with male Tanner crab, the “observed” annual empirical survey catchability curves for females are substantially variable across the size range collected in the SBS studies (Figure 45). Based on a simple unweighted smooth fit, they exhibit an initially-increasing trend from 25 mm CW to 70 mm CW, with catchability at larger sizes essentially constant at ~0.4.

Both the intercept and the year-invariant smooth term $s(z)$ were highly significant (p values < 0.0001) for all three models (Tables 10-12). Of the six year-specific “factor” smooths ($ti(z, y)$) in Model 2, four were significant (p values < 0.005 ; 2013, 2015, 2016, and 2018), while two were not significant ($p > 0.4$; 2014 and 2017; Table 11). The random effects term in Model 3, $s_{RE}(z, y)$, was also found to be highly significant ($p < 0.0001$; Table 12). The estimated Tweedie power parameter ranged from 1.642 for Model 1 to 1.753 for Model 2, with an intermediate value (1.723) for Model 3 (Table 13). Of the three, Model 3 exhibited the lowest AIC score (by 15 likelihood units over the next best model) and the highest adjusted R^2 (Table 13). The BIC scores for Model 2 was 1 likelihood unit less than that for Model 3, but both were substantially less than Model 1 (~40 likelihood units). Diagnostic plots for residuals (Figures 46-48) exhibited departures from expected cumulative distributions at both ends of those distributions, as well as significant deviations from the KS, dispersion, and outlier tests provided by the DHARMA R package (Hartig 2022).

The time-invariant estimates of catchability, $S(z) (= \exp(\alpha + s(z)))$, where α is the estimated link-scale intercept and $s(z)$ is the estimated link-scale smooth function of size, z , were reasonably similar for Models 1 and 3, although the Model 1 curve was more logistic-like (i.e., rising to an asymptote) while the Model 3 curve was more dome-shaped, exhibiting a slightly declining trend after reaching its maximum (Figure 49). In contrast, the estimated time-invariant catchability curve from Model 2 exhibited a linearly increasing trend across the 25-135 mm CW size range for females with a slope less than the initial slopes of the Model 1 and Model 3 curves such that the estimated Model 2 curve was less than those for the other models at sizes below ~80 mm CW but higher at sizes above 100 mm CW. Estimates for fully-selected catchability, or q (the maximum of $S(z)$), were similar for Models 1 and 3 (0.299 and 0.329, respectively), while q for Model 2 was 0.464.

The 5-mm size bins at which full selection occurred were similar for Models 1 and 3 (70-75 mm CW), while the curve for Model 2 reached its maximum at the upper bound of female sizes (135 mm CW; Table 14). As with males, the uncertainty in the estimated curves at large sizes (above 100 mm CW for females) was considerable. Below 100 mm CW, the uncertainty associated with the estimated curve from Model 3 was larger (as with males, on the order of 2x) than that of the other two models at any size because it reflects the estimated *population-level* (i.e., for any randomly-selected year) variability in catchability, not the variability associated with the specific years of the SBS studies.

Best models

Based on the summary model statistics (Tables 8 and 13) and the residuals plots, Model 3 is the most satisfactory of the three models for each sex to describe NMFS survey catchability for male and female Tanner crab using the “observed” empirical selectivity. In addition to fitting the data equal to or better than the other two models, Model 3 captures the population-level uncertainty in the estimated curves more appropriately than the other two models when the goal is to apply the curves to arbitrary years.

Sediment characteristics

Mean grain size and sorting coefficient varied across the EBS (Figures 4 and 5). At NMFS SBS haul locations, mean grain size and sorting coefficient were positively correlated with each other ($\rho = 0.571$) and haul depth ($\rho = 0.530$ and 0.433 , respectively) but negatively correlated with bottom temperature ($\rho = -0.530$ and -0.482 ; Figure 54). Although the correlations themselves were significantly different from 0, none approached a level that precluded the use of the individual datasets as covariates in the haul-level selectivity analysis.

Haul-level selectivity

Males

The joint distribution for the numbers of male Tanner crab caught by gear type in a paired haul in the same 5-mm CW size bin is illustrated in Figure 56. Haul/size bins where the total number of males caught was less than 5 were excluded from the data sets fit by the models described below as having inadequate sample sizes. The NMFS gear caught no crab in a size bin when the paired BSFRF gear caught at least five in 246 cases whereas the BSFRF gear caught no crab in only 150 cases when the paired NMFS gear caught at least five crab.

Results for binomial models with environmental covariates

The model including smooth functions for crab size, depth, temperature, and sorting as main effects and an interaction between size and grain size ($s(z)+ti(d)+ti(t)+ti(s)+ti(z,f)$) was considered to be the best of the candidate binomial models that included environmental covariates, based on selecting the simplest ‘equivalent’ model after a comparison of the mean and range of GCV scores across the models (Figure 57). This model also

exhibited the lowest mean AIC and predicted R-squared values and all folds passed *mgcv*'s concurvity test. The size-only model ranked 32nd out of the 256 candidate models.

The five models with the best GCV scores and the size-only model exhibited remarkably similar predictions of size-dependent catchability in the absence of environmental effects (Figure 58). The size-only effects increased in two logistic curve-like sweeps from 0 to 0.25 as size increased from 15 to 50 mm, then to ~0.5 from 50 to 100 mm CW. While the size-only model remained at ~0.5 up to ~135 mm CW, the other models increased again to ~0.6 at 125 mm CW. Above 135 mm CW the estimated curves diverged and the confidence intervals of all the models increased dramatically.

The partial effects of size, temperature, and grain size are illustrated for the best model ($s(z)+ti(d)+ti(t)+ti(s)+ti(z,f)$) in Figures 59-62. The ln-scale size effect increased rapidly between 15 and 50 mm CW, then increased much more gradually up to 150 mm CW (Figure 59) while ln-scale partial residuals exhibited relatively wide ranges below 60 mm CW and above 110 mm CW. The ln-scale effect of depth were slightly positive for depths between 40 and 80 m, slightly negative from 80 to 120 m, and slightly positive again below 120 m (Figure 60). The ln-scale temperature effect was negative at temperatures below 4 °C and close to zero above it (Figure 61). The ln-scale effect of the sediment sorting coefficient was slightly positive for values of s below 1 (moderately sorted), somewhat negative for values from 1 to 3 (poorly sorted), and slightly positive again for values above 3 (very poorly sorted). The ln-scale partial effect of the interaction between size and grain size for small crab (< 50 mm CW) was positive in coarse sand ($\phi < 2.5$) but slightly negative in medium to very fine sand ($\phi > 2.5$) (Figure 62). For larger males (> 50 mm CW), the trends with grain size were reversed, but with much smaller effect sizes.

Residuals to the best binomial model simulated using the DHARMA R package (Hartig 2022) passed the package's dispersion test ($p=0.080$) but not the KS or outlier tests ($p<0.001$ for both; Figure 64).

Haul-specific catchability curves for male Tanner crab were predicted for all hauls from 1982-2023 for the NMFS EBS trawl survey using the best binomial model, then averaged by year to obtain estimates of annual "survey-level" catchability after weighting by the size-specific CPUE for each haul (Figure 65). The majority of the annual weighted-average curves are similar to the size-only selectivity curves (Figure 58), exhibiting an increase with size similar to two "stacked" logistic curves, with an increasing trend to ~80 mm CW, a plateau from 80-100 mm CW, a second increasing trend to ~125 mm CW, followed by a plateau or a gently decreasing trend above 125 mm CW. Fully-selected catchability for these curves ranged from ~0.4 to ~0.7. In several years, however, the annual curves were substantially different: most exhibiting strong peaks at sizes < 100 mm CW with catchability above 1.

Results for the binomial model with haul-level random effects

The catchability curve estimated from the binomial model with haul-level random effects (Figure 66) exhibited a roughly logistic increase from 0 to 0.48 from 10 to 125 mm CW, reaching its asymptote near 90 mm CW, before decreasing slightly with size above 125 mm CW. The confidence intervals associated with the estimated curve were extremely large above 125 mm CW.

Residuals to the model simulated using the DHARMA R package (Hartig 2022) passed the package's dispersion ($p>0.8$) and outlier ($p > 0.05$) tests, but not the KS test ($p < 0.01$; Figure 67).

Results for Tweedie models with environmental covariates

The model including smooth functions for crab size, temperature, and grain size as main effects and an interaction between size and depth ($s(z)+t_i(t)+t_i(f)+t_i(z,d)$) was considered to be the best of the candidate Tweedie models that included environmental covariates, based on selecting the simplest 'equivalent' model after a comparison of the mean and range of GCV scores across the models (Figure 68). This model also exhibited the lowest mean AIC and predicted R-squared values and all folds passed *mgcv*'s concurvity test. The size-only model ranked 26th out of the 256 candidate models.

The five models with the best GCV scores and the size-only model exhibited remarkably similar predictions of size-dependent catchability in the absence of environmental effects (Figure 69). The size-only effects increased parabolically from 0 to ~0.4 as size increased from 15 to 45 mm, then linearly to ~0.6 from 50 to 95 mm CW. From 95 mm to 125 mm CW, the curves dipped slightly then returned ~0.6. Above 125 mm CW the estimated curves diverged and the confidence intervals for all the models increased dramatically.

The partial effects of size, temperature, grain size and the interaction between size and depth are illustrated for the best model ($s(z)+t_i(t)+t_i(f)+t_i(z,d)$) in Figures 59-62. Similar to the same curve for the best binomial model with covariates, the ln-scale size effect increased rapidly between 15 and 50 mm CW, then increased much more gradually up to 150 mm CW (Figure 70). The positive ln-scale partial residuals appear to be constrained while the negative residuals exhibit relatively wide ranges. The ln-scale effect of temperature is slightly positive below 1 °C, somewhat negative from 1.5 to 3.7 °C and slightly positive again from 3.7 to 4.5 °C (Figure 71). The ln-scale grain size effect was positive for $\phi < 2.6$ (coarse to fine sand), negative for ϕ values from 2.6 to 4.5 (fine sand to silt), and positive again at values above 4.5 (silt) (Figure 61). The ln-scale partial effect of the interaction between size and depth for small crab (< 50 mm CW) was positive for depths < 80 m, negative between 80 and 120 m, and positive again for depths > 120 m (Figure 62), although there were no observations of small crab at depths < 60 m to support the predictions. For large males (> 100 mm CW), the effects with depth were reversed but with smaller effect sizes, additionally, there were more observations at shallow depths but fewer at deeper depths.

Residuals to the best Tweedie model simulated using the DHARMA R package (Hartig 2022) passed all of the package's significance tests (p-values for the dispersion, KS, and outlier tests were all > 0.05; Figure 74).

Haul-specific catchability curves for male Tanner crab were predicted for all hauls from 1982-2023 for the NMFS EBS trawl survey using the best Tweedie model, then averaged by year to obtain estimates of annual "survey-level" catchability after weighting by the size-specific CPUE for each haul (Figure 75). The weighted-average curves were similar in overall shape across years, exhibiting a "double hump" shape similar to that for the size-only curves (Figure 69) with peaks near 90 and 125 mm CW, with most exhibiting a declining trend in catchability as size increased above 130 mm CW. Catchabilities at the peak near 125 mm CW ranged from 0.35 to 0.85.

Results for the Tweedie model with haul-level random effects

The catchability curve estimated from the Tweedie model with haul-level random effects (Figure 76) was remarkably similar to that from the binomial model with haul-level effects, exhibiting a roughly logistic increase from 0 to 0.55 from 15 to 125 mm CW, approaching

its approximate asymptote near 90 mm CW. As with the binomial model with haul-level random effects, the confidence intervals associated with the estimated curve were large.

Residuals to the model simulated using the DHARMA R package (Hartig 2022) passed both the package's dispersion test ($p=0.50$) but failed the KS and outlier tests ($p < 0.01$ and $p < 0.05$, respectively; Figure 77).

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Table 1. Summary of SBS paired hauls, by study year. Area swept is in square nautical miles, bottom depth in meters, and bottom temperature in degrees centigrade. NA: no value.

year	N	area swept				bottom depth				bottom temp.			
		BSFRF		NMFS		BSFRF		NMFS		BSFRF		NMFS	
		min	max	min	max	min	max	min	max	min	max	min	max
2013	35	0.001300	0.00285	0.006455	0.01602	41.70	91.44	44	89	0.03	3.50	0.0	3.5
2014	38	0.001000	0.00409	0.011437	0.01498	41.33	89.15	43	89	3.45	5.55	3.4	5.5
2015	47	0.001350	0.00314	0.006669	0.01469	34.02	90.34	37	88	3.24	5.78	3.3	5.8
2016	102	0.001070	0.00361	0.006261	0.01568	23.77	156.00	28	155	4.04	10.18	4.2	8.4
2017	87	0.001100	0.00449	0.007178	0.01691	44.99	156.73	47	156	NA	NA	0.4	5.5
2018	75	0.001189	0.00497	0.006382	0.01575	59.62	152.34	61	154	NA	NA	3.5	5.9

Table 2. Results for the estimated power exponent based on fitting two linear regression models for the CPUE variance-to-mean relationship (calculated by gear type, sex, year, and size bin) expected for observations described by a Tweedie statistical distribution. The “reduced” model estimates a common intercept, individual intercepts (as offsets) for the sex and gear type factors, and a common slope for the ln-scale mean covariate. The “full” model additionally estimates (offset) slopes for each sex/fleet interaction term. See text for details. Only the estimated value for the common slope (the Tweedie power coefficient) is reported here. Based on the BIC scores, the reduced model provides a better (more parsimonious) fit to the data than the full model.

	reduced		full	
	Est.	CI	Est.	CI
TW exponent	1.546	[1.513, 1.579]	1.605	[1.538, 1.673]
Num.Obs.	559		559	
R2	0.953		0.953	
AIC	13 023.7		13 023.4	
BIC	13 049.7		13 062.4	
RMSE	0.58		0.58	

Table 3. Expanded total abundance size compositions (millions) for males from the SBS studies, by 5-mm size bin and gear type.

size	2013		2014		2015		2016		2017		2018	
	BSFRF	NMFS	BSFRF	NMFS	BSFRF	NMFS	BSFRF	NMFS	BSFRF	NMFS	BSFRF	NMFS
27.5	0.29027	0.000000	0.042428	0.002049	0.006776	0.006261	0.23303	0.0133296	4.007982	0.162784	3.41452	0.330597
32.5	0.12222	0.002029	0.048808	0.002859	0.060531	0.001565	0.25427	0.0181406	3.221492	0.259193	1.74203	0.346853
37.5	0.07225	0.004509	0.060357	0.007767	0.014598	0.015585	0.13316	0.0156008	0.755892	0.053704	2.06909	0.486356
42.5	0.05234	0.010797	0.037510	0.020304	0.043882	0.014127	0.13731	0.0170978	0.622592	0.075860	2.54041	0.474827
47.5	0.02631	0.006438	0.008516	0.015321	0.054551	0.024909	0.05095	0.0108641	0.335552	0.088207	1.02702	0.229406
52.5	0.06181	0.006564	0.111459	0.008231	0.031289	0.015670	0.04988	0.0116072	0.328388	0.045343	0.53844	0.153190
57.5	0.18035	0.021457	0.061165	0.012537	0.023467	0.017236	0.06458	0.0124659	0.301203	0.040404	0.64888	0.190858
62.5	0.08653	0.043117	0.069036	0.022749	0.022421	0.011109	0.09101	0.0119462	0.158746	0.053118	0.55370	0.138588
67.5	0.05830	0.028243	0.051571	0.003960	0.000000	0.011731	0.04014	0.0153347	0.134897	0.024392	0.23892	0.069270
72.5	0.12522	0.038863	0.082201	0.008103	0.013215	0.015175	0.05189	0.0163387	0.157319	0.043548	0.27677	0.077073
77.5	0.19378	0.094228	0.028815	0.024473	0.064689	0.052530	0.09234	0.0344597	0.202515	0.049516	0.16900	0.087739
82.5	0.36718	0.148436	0.209834	0.085131	0.073928	0.075863	0.09854	0.0591257	0.128067	0.045229	0.14490	0.075880
87.5	0.35324	0.189179	0.218115	0.088358	0.070573	0.086508	0.19415	0.0510461	0.223469	0.060849	0.22817	0.096066
92.5	0.26674	0.132166	0.179372	0.145459	0.128965	0.078221	0.24313	0.0668828	0.217872	0.081246	0.27066	0.126343
97.5	0.42575	0.111322	0.221283	0.175422	0.127482	0.082915	0.25926	0.1011115	0.231044	0.119094	0.30980	0.139788
102.5	0.55379	0.182575	0.537195	0.338744	0.086501	0.098682	0.28744	0.1059500	0.265729	0.154569	0.33842	0.182856
107.5	1.05362	0.263537	0.550759	0.476440	0.295298	0.108632	0.26534	0.1460781	0.484834	0.195463	0.23481	0.137177
112.5	1.48362	0.390093	0.669488	0.540827	0.232069	0.130201	0.22544	0.1799579	0.462052	0.193336	0.29628	0.158549
117.5	1.41135	0.501264	0.622612	0.444692	0.256490	0.150429	0.37961	0.1780809	0.473415	0.216329	0.32514	0.216900
122.5	1.20406	0.468045	0.391389	0.307542	0.150865	0.120350	0.28211	0.1745637	0.408464	0.245005	0.30777	0.209911
127.5	0.62022	0.211226	0.403175	0.245684	0.136406	0.098178	0.24605	0.1020419	0.420321	0.189442	0.29463	0.190267
132.5	0.25125	0.155637	0.235499	0.187993	0.170955	0.128569	0.16266	0.0835932	0.447641	0.178320	0.24939	0.166088
137.5	0.20242	0.122192	0.313251	0.270860	0.173622	0.121854	0.10112	0.0916131	0.213114	0.109823	0.21030	0.104208
142.5	0.25437	0.095020	0.269479	0.200528	0.119891	0.114817	0.08275	0.0547008	0.180663	0.079996	0.07736	0.073474
147.5	0.10152	0.063265	0.271380	0.165449	0.120173	0.067798	0.04435	0.0205203	0.071405	0.045234	0.07909	0.051280
152.5	0.05821	0.043482	0.184928	0.081252	0.067346	0.061148	0.02518	0.0180364	0.045834	0.013516	0.05504	0.025184
157.5	0.02808	0.004322	0.033956	0.046541	0.049052	0.038857	0.02295	0.0084850	0.024141	0.008228	0.01814	0.010072
162.5	0.01305	0.008612	0.021053	0.012900	0.000000	0.011317	0.01206	0.0014990	0.005526	0.006334	0.01041	0.003792
167.5	0.00000	0.004284	0.029951	0.015679	0.000000	0.003062	0.00000	0.0006848	0.003819	0.000000	0.00000	0.002884
172.5	0.00000	0.000000	0.000000	0.007069	0.000000	0.004638	0.00000	0.0014657	0.003548	0.001953	0.00000	0.000000
177.5	0.00000	0.000000	0.000000	0.001952	0.000000	0.000000	0.00000	0.0000000	0.000000	0.000000	0.00000	0.000000
182.5	0.00000	0.000000	0.000000	0.000000	0.000000	0.000000	0.00000	0.0000000	0.000000	0.000000	0.00000	0.000000

Table 4. Expanded total abundance size compositions (millions) for females from the SBS studies, by 5-mm size bin and gear type.

size	2013		2014		2015		2016		2017		2018	
	BSFRF	NMFS	BSFRF	NMFS	BSFRF	NMFS	BSFRF	NMFS	BSFRF	NMFS	BSFRF	NMFS
27.5	0.28688	0.002060	0.01893	0.000000	0.015418	0.001565	0.276268	0.0101814	4.103006	0.286111	2.89280	0.2295022
32.5	0.09719	0.002280	0.00000	0.006376	0.060852	0.004659	0.386479	0.0173336	4.189249	0.430000	2.73376	0.3464133
37.5	0.04516	0.009161	0.00000	0.008286	0.045434	0.003131	0.100496	0.0089940	1.148431	0.154340	2.90058	0.5601088
42.5	0.02785	0.002060	0.02249	0.020570	0.045434	0.012584	0.191592	0.0167592	0.780319	0.079188	2.71717	0.7012719
47.5	0.01335	0.008093	0.06748	0.031879	0.023467	0.011080	0.053496	0.0144071	0.471133	0.091685	1.48088	0.4188451
52.5	0.01429	0.002196	0.03803	0.003188	0.049600	0.020298	0.061345	0.0085189	0.301182	0.053730	0.77689	0.2113943
57.5	0.03873	0.010816	0.09391	0.011827	0.024437	0.018907	0.034737	0.0176517	0.365028	0.028999	0.81300	0.2502250
62.5	0.07821	0.017510	0.01175	0.012089	0.007822	0.007768	0.046118	0.0185098	0.198381	0.051961	0.64041	0.1894543
67.5	0.04595	0.008852	0.03052	0.012376	0.006976	0.006337	0.059813	0.0087530	0.229603	0.070398	0.51841	0.1419531
72.5	0.09929	0.032154	0.05084	0.007147	0.021374	0.028019	0.141969	0.0265285	0.608756	0.129880	0.64125	0.2156637
77.5	0.20468	0.085031	0.14346	0.042699	0.090613	0.034430	0.392413	0.0900076	1.017936	0.269598	0.78928	0.2358192
82.5	0.37350	0.160596	0.08282	0.051767	0.116710	0.044229	0.367539	0.1021071	0.784958	0.199378	0.71251	0.2033222
87.5	0.50986	0.187188	0.17927	0.042000	0.072913	0.054322	0.239731	0.0665449	0.457285	0.121992	0.49930	0.1567783
92.5	0.51557	0.191342	0.08083	0.050611	0.061430	0.012379	0.230273	0.0432671	0.305890	0.079550	0.27707	0.0977919
97.5	0.67320	0.218290	0.11958	0.069728	0.090636	0.015882	0.085170	0.0312672	0.128558	0.042577	0.16615	0.0388109
102.5	0.53680	0.187097	0.23556	0.059100	0.091397	0.027715	0.069543	0.0150371	0.017246	0.012220	0.02998	0.0134005
107.5	0.18918	0.078749	0.14844	0.047022	0.087272	0.013031	0.039643	0.0039329	0.006457	0.000717	0.00000	0.0009771
112.5	0.05074	0.013332	0.01040	0.014692	0.000000	0.009497	0.006532	0.0007309	0.000000	0.000000	0.00000	0.0009771
117.5	0.00000	0.000000	0.00000	0.000000	0.007022	0.003510	0.004085	0.0007237	0.000000	0.000000	0.00000	0.0000000
122.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.001518	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
127.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
132.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
137.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
142.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
147.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
152.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
157.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
162.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
167.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
172.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
177.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000
182.5	0.00000	0.000000	0.00000	0.000000	0.000000	0.000000	0.000000	0.0000000	0.000000	0.000000	0.00000	0.0000000

Table 5. Summary results for Model 1 for males.

Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	-1.018	0.036	-28.410	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(z)	4.753	5.796	96.884	0.0000 ***

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Adjusted R-squared: 0.659, Deviance explained 0.755

-REML : -161.934, Scale est: 0.0698, N: 169

Table 6. Summary results for Model 2 for males.

Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	-1.076	0.032	-33.246	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(z)	5.659	9.000	79.683	0.0000 ***
	ti(z):y2013	3.217	4.000	6.828	0.0000 ***
	ti(z):y2014	1.611	4.000	2.985	0.0007 ***
	ti(z):y2015	0.000	4.000	0.000	0.3888
	ti(z):y2016	0.000	4.000	0.000	0.8288
	ti(z):y2017	0.000	4.000	0.000	0.7106
	ti(z):y2018	3.376	4.000	16.667	0.0000 ***

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Adjusted R-squared: 0.755, Deviance explained 0.855

-REML : -192.502, Scale est: 0.0506, N: 169

Table 7. Summary results for Model 3 for males.

Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	-0.981	0.130	-7.536	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(z)	5.124	9.000	11.493	0.0000 ***
	s(z,y)	16.100	59.000	3.660	0.0000 ***

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Adjusted R-squared: 0.821, Deviance explained 0.896

-REML : -213.075, Scale est: 0.0345, N: 169

Table 8. Comparison of model summary results for males.

quantity	Model 1	Model 2	Model 3
family	Tweedie p=1.436	Tweedie p=1.537	Tweedie p=1.467
AIC	-330.10	-412.35	-458.82
BIC	-302.83	-352.09	-372.12
deviance	13.42	9.13	5.95
adj. R ²	0.66	0.75	0.82
df	5.75	14.86	22.22
N(params)	10	34	80
N(obs)	169	169	169

Table 9. Results for fully-selected catchability ("q") and size at full selection for males.

model	q size (mm CW)	
Model 1	0.575	140
Model 2	0.603	145
Model 3	0.591	140

Table 10. Summary results for Model 1 for females.

Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	-1.453	0.039	-37.606	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(z)	4.703	5.733	54.244	0.0000 ***

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Adjusted R-squared: 0.545, Deviance explained 0.692

-REML : -195.293, Scale est: 0.0545, N: 104

Table 11. Summary results for Model 2 for females.

Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	-1.573	0.035	-44.666	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(z)	3.451	9.000	31.891	0.0000 ***
	ti(z):y2013	2.660	4.000	5.654	0.0000 ***
	ti(z):y2014	0.000	4.000	0.000	0.4796
	ti(z):y2015	1.775	4.000	2.657	0.0021 **
	ti(z):y2016	2.219	4.000	4.980	0.0001 ***
	ti(z):y2017	0.000	4.000	0.000	0.4332
	ti(z):y2018	3.708	4.000	19.723	0.0000 ***

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Adjusted R-squared: 0.686, Deviance explained 0.844

-REML : -222.201, Scale est: 0.0366, N: 104

Table 12. Summary results for Model 3 for females.

Component	Term	Estimate	Std Error	t-value	p-value
A. parametric coefficients	(Intercept)	-1.479	0.102	-14.519	0.0000 ***
Component	Term	edf	Ref. df	F-value	p-value
B. smooth terms	s(z)	4.801	9.000	4.815	0.0000 ***
	s(z,y)	15.059	59.000	2.738	0.0000 ***

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Adjusted R-squared: 0.714, Deviance explained 0.873

-REML : -228.485, Scale est: 0.0298, N: 104

Table 13. Comparison of model summary results for females.

quantity	Model 1	Model 2	Model 3
family	Tweedie p=1.642	Tweedie p=1.753	Tweedie p=1.725
AIC	-397.91	-469.71	-484.90
BIC	-376.57	-418.41	-417.25
deviance	7.36	4.55	3.52
adj. R ²	0.55	0.69	0.71
df	5.70	14.81	20.86
N(params)	10	34	80
N(obs)	104	104	104

Table 14. Results for fully-selected catchability ("q") and size at full selection for females.

model	q size (mm CW)	
Model 1	0.299	70
Model 2	0.464	135
Model 3	0.329	75

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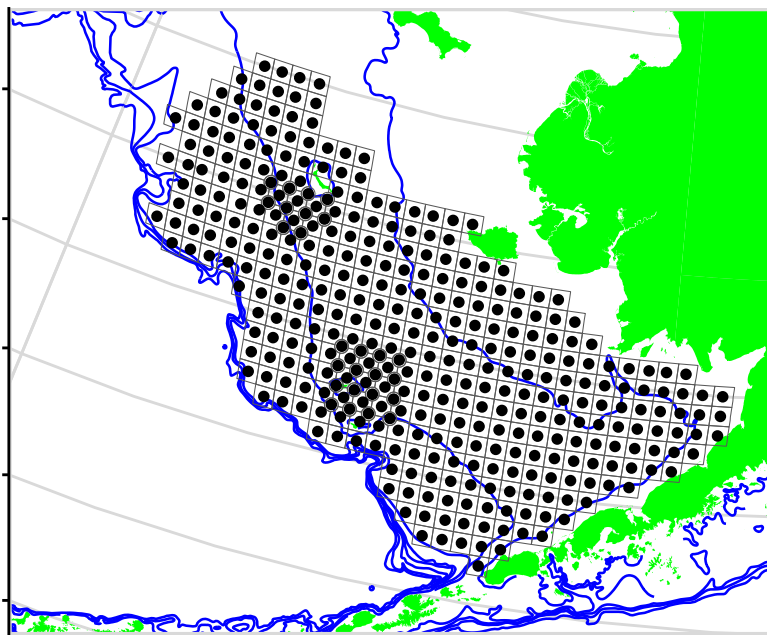


Figure 1. Standard NMFS eastern Bering Sea shelf bottom trawl survey grid. Rectangular areas are approximately 20 nm x 20 nm. Circles represent "corner" stations. Dots indicate nominal station locations. Blue lines indicate bathymetric contours (50m, 100m, 200m,...).

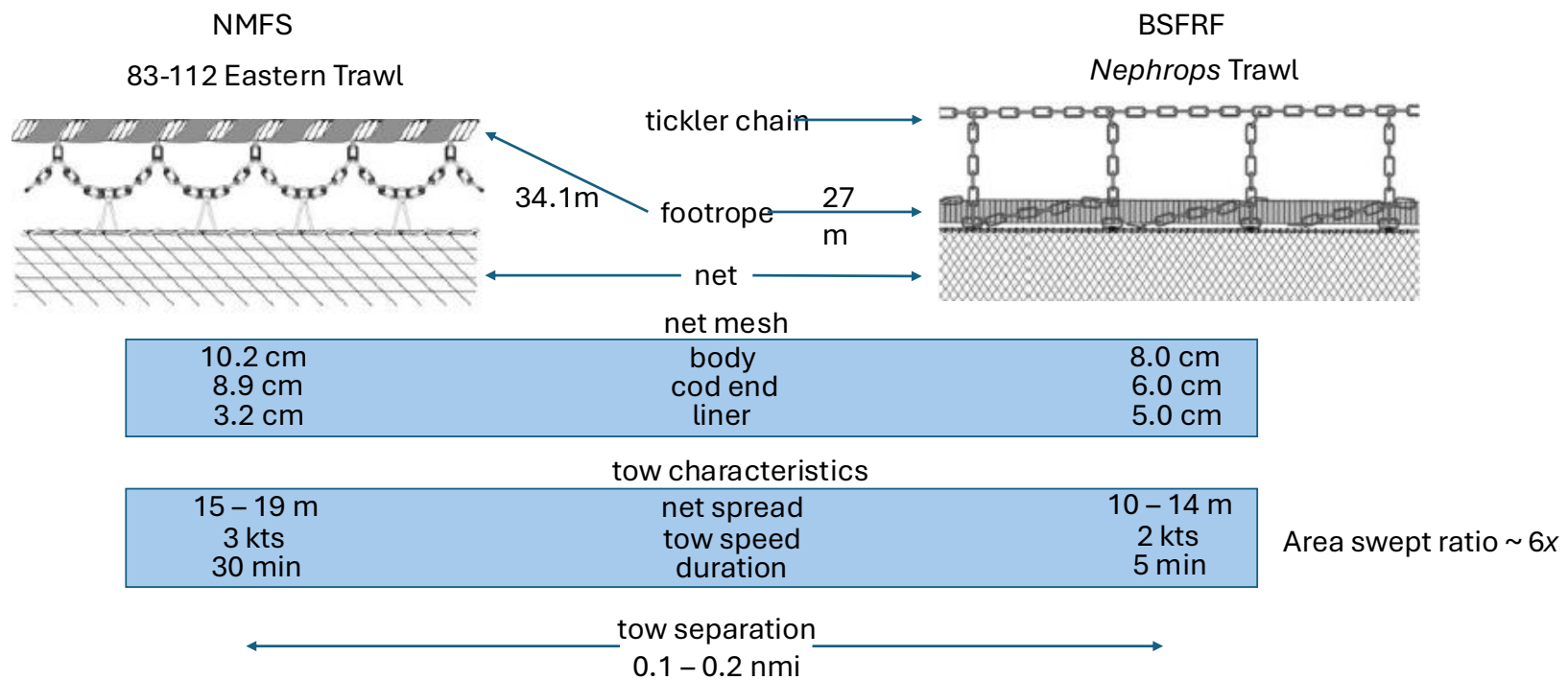


Figure 2. Comparison of the NMFS and BSFRF survey gear and tow characteristics.

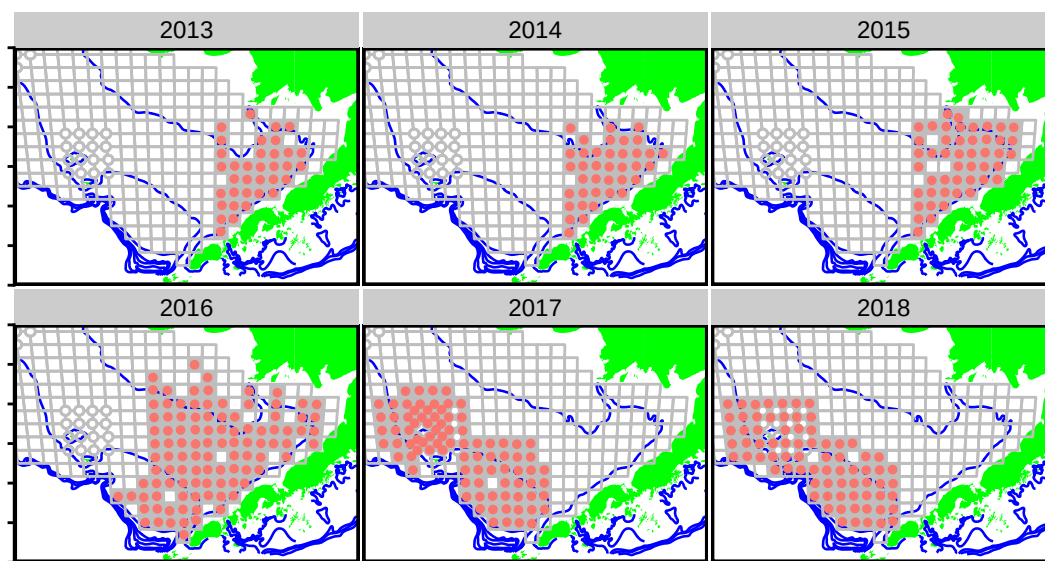


Figure 3. Side-by-side haul (SBS) study stations for Tanner crab selectivity in the NMFS Eastern Bering Sea Shelf Survey. Grey lines: NMFS survey grid stations; filled squares: SBS stations with paired hauls; blue lines: 50 m, 100 m, 200 m and deeper bathymetric contours.

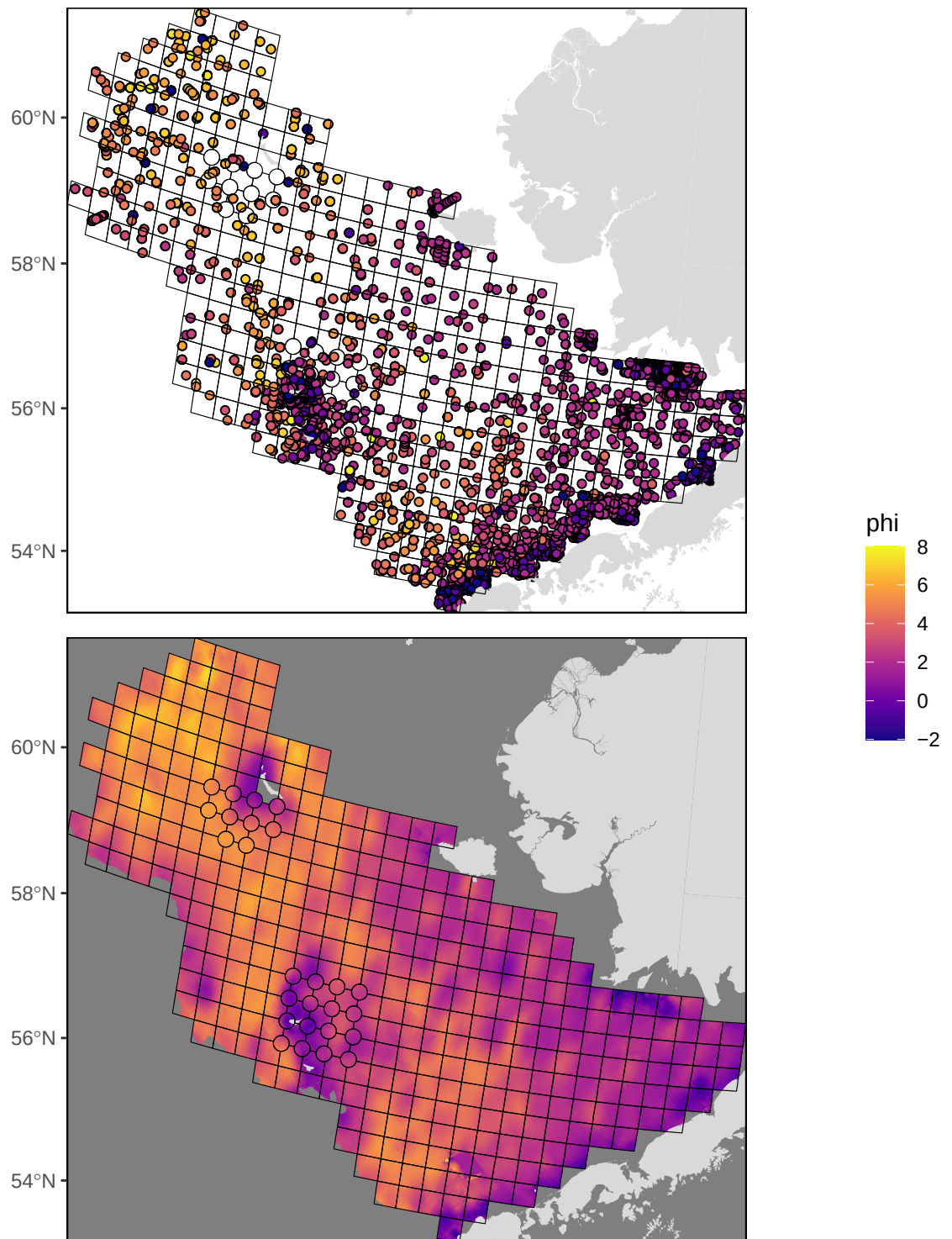


Figure 4. Upper: Mean sediment grain size (phi units) from the EBSED2 database. Lower: Distribution of mean grain size (phi units), based on interpolation of the above data by kriging.

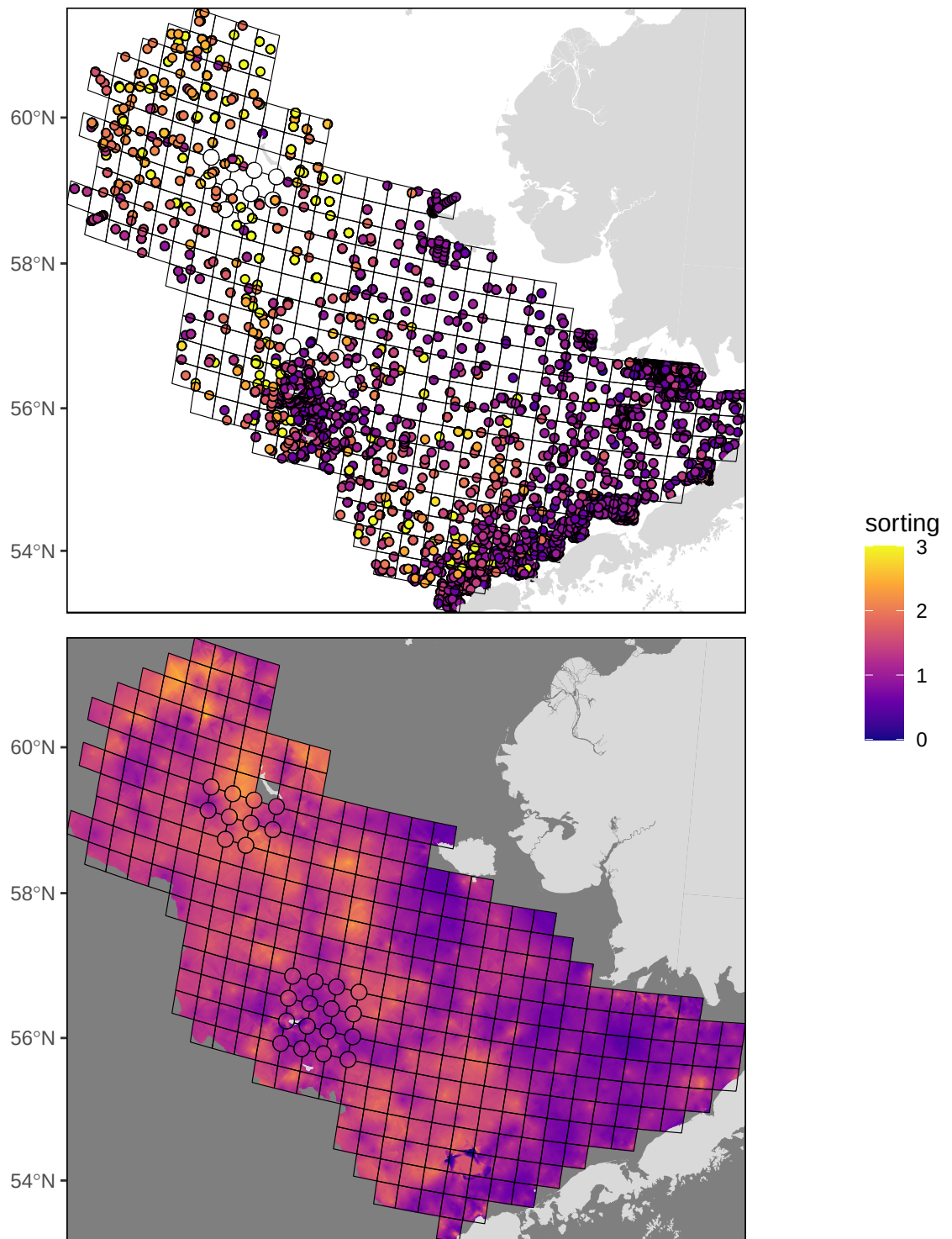


Figure 5. Upper: Sediment sorting coefficient from the EBSED2 database. Lower: Distribution of the sorting coefficient, based on interpolation of the above data by kriging.

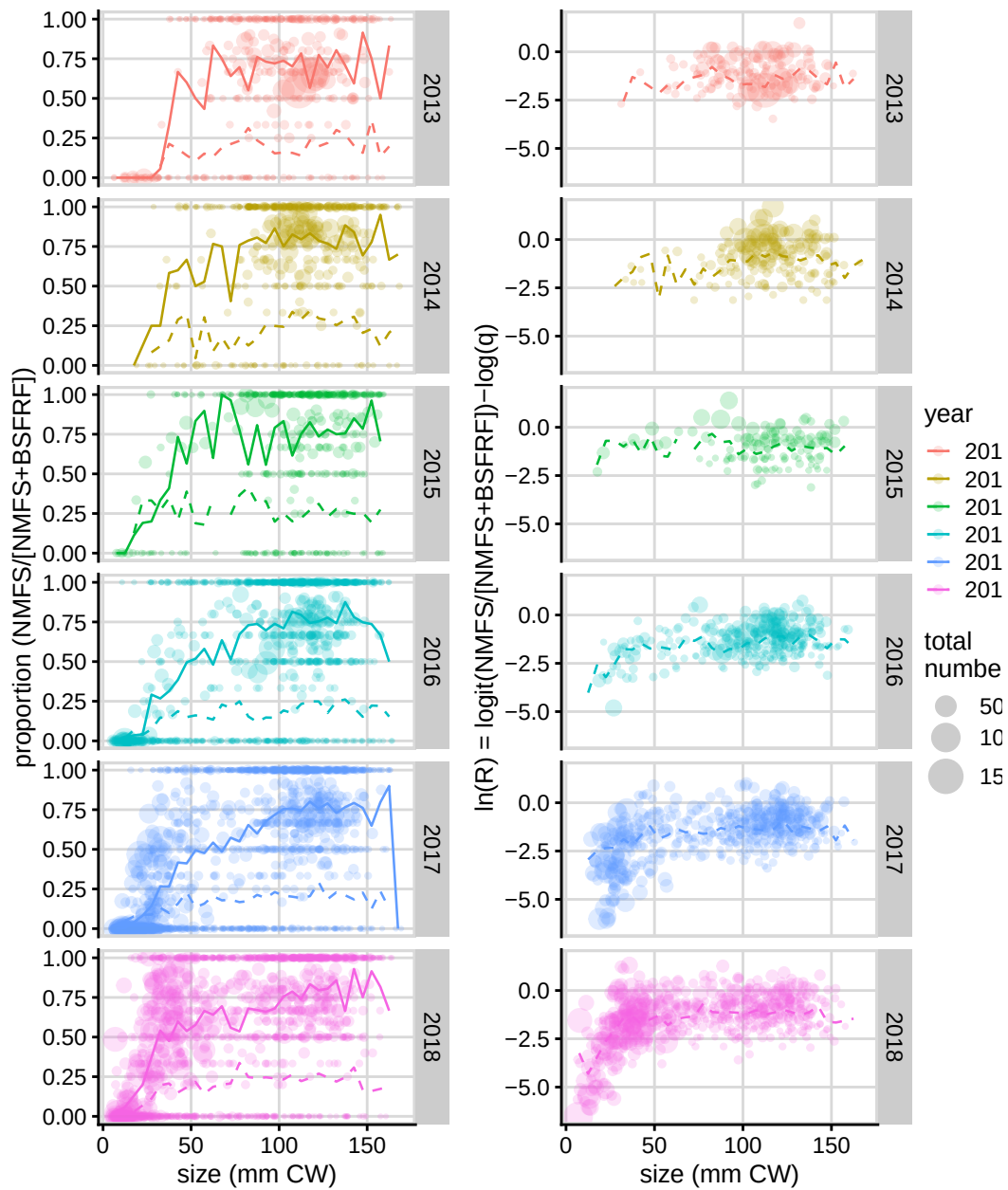


Figure 6. Left column: ratio of haul-level NMFS catch numbers to total catch numbers for males, by size bin and paired haul; solid line: haul-level mean; dashed line: survey-level proportion based on CPUE. Right column: corresponding $\ln(R)$ values; dashed line: $\ln(R)$ using survey-level CPUE proportion. Symbol size indicates the total number of males caught, by size and paired haul; color indicates SBS study year.

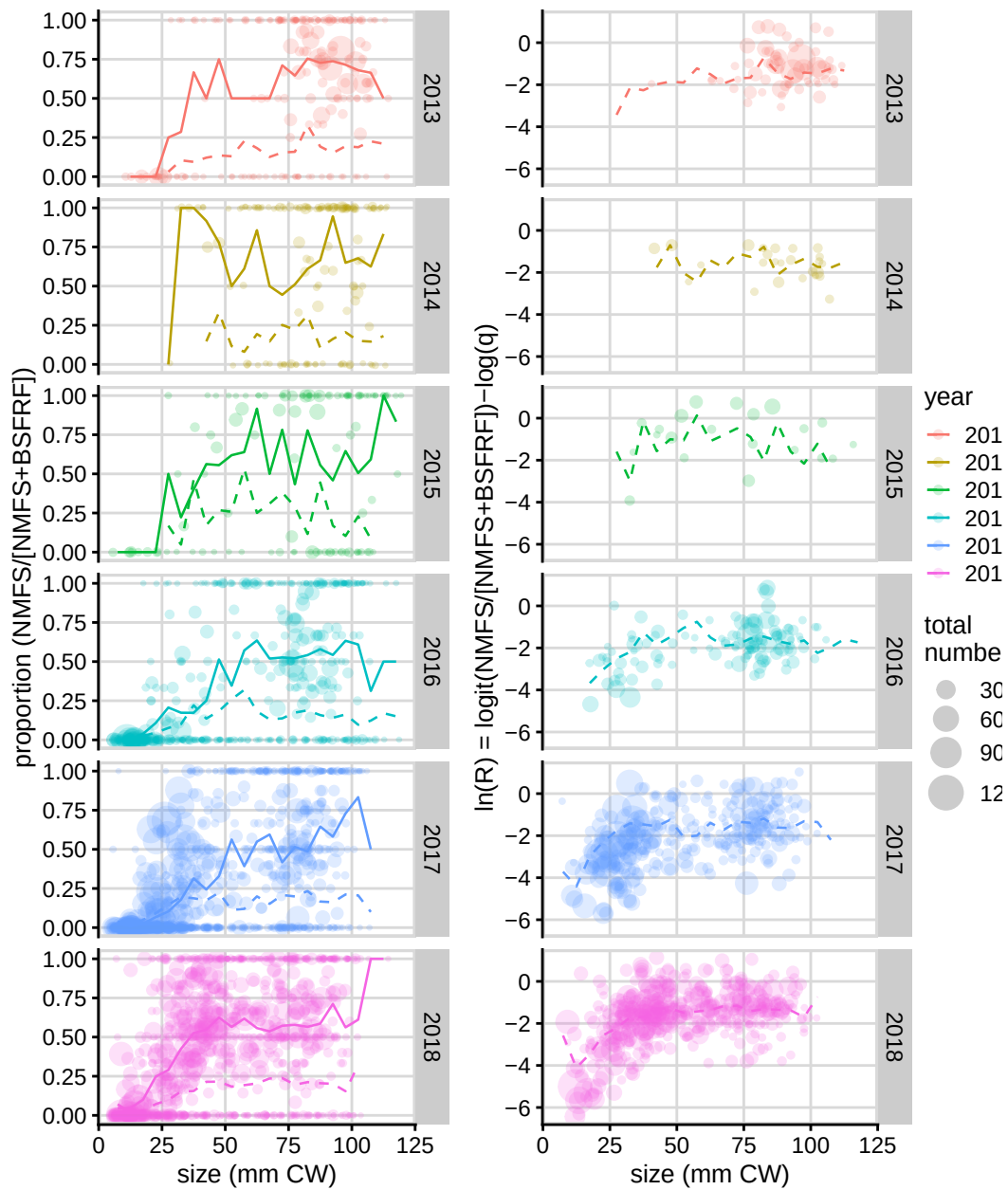


Figure 7. Left column: ratio of haul-level NMFS catch numbers to total catch numbers for females, by size bin and paired haul; solid line: haul-level mean; dashed line: survey-level proportion based on CPUE. Right column: corresponding $\ln(R)$ values; dashed line: $\ln(R)$ using survey-level CPUE proportion. Symbol size indicates the total number of females caught, by size and paired haul; color indicates SBS study year.

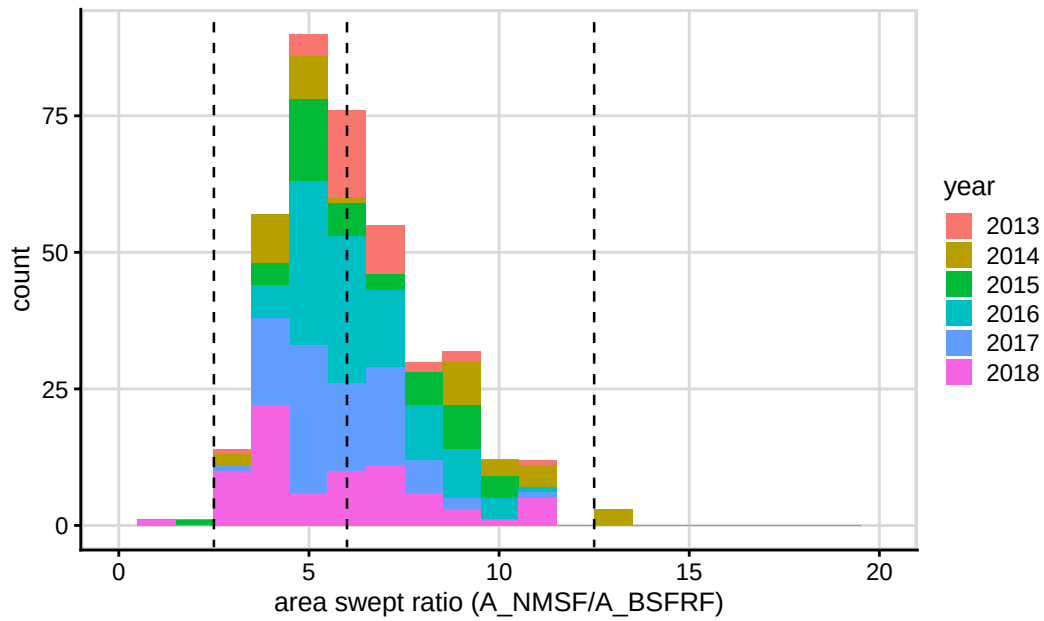


Figure 8. Histogram of paired haul area-swept ratios. Paired hauls with area swept ratios outside the outer dotted lines were dropped from further analysis. Vertical dashed lines: left: minimum acceptable ratio, center: expected ratio, right: maximum acceptable ratio.

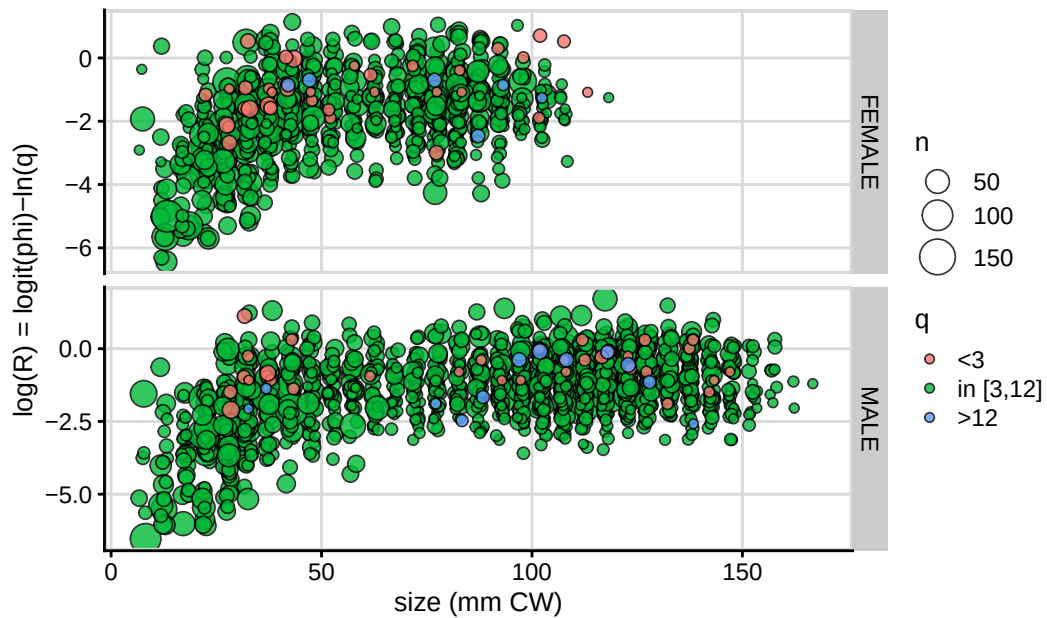


Figure 9. Haul-level $\ln(R)$ values; symbol size indicates number sampled. Colour indicates area swept ratio. Upper plot: females; lower plot: males.

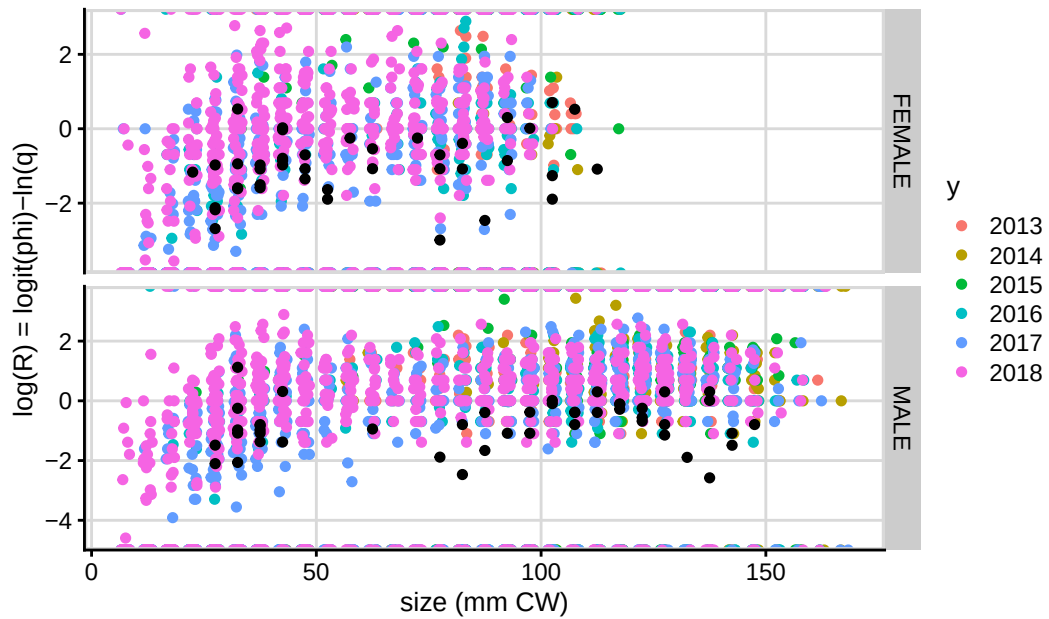


Figure 10. Haul-level $\ln(R)$ values; colours indicate SBS study years. Black symbols indicate values with area swept ratio < 3 or > 12 which will be dropped. Upper plot: females; lower plot: males.

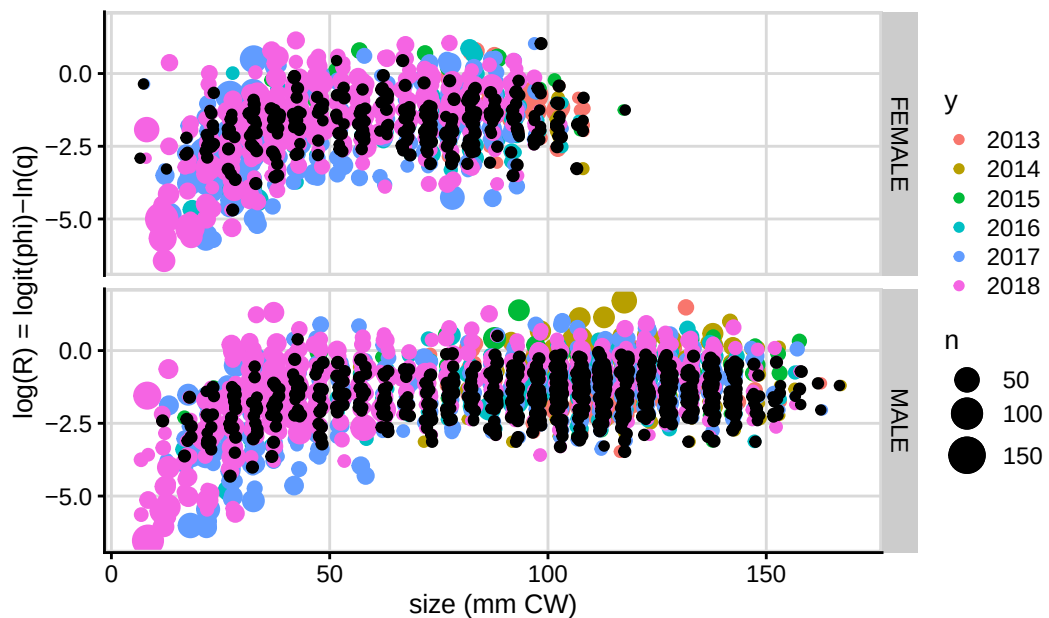


Figure 11. Haul-level $\ln(R)$ values; colours indicate SBS study years; symbol size indicates number of crab sampled. Black symbols indicate values with $<$ individuals in the size bin. Upper plot: females; lower plot: males.

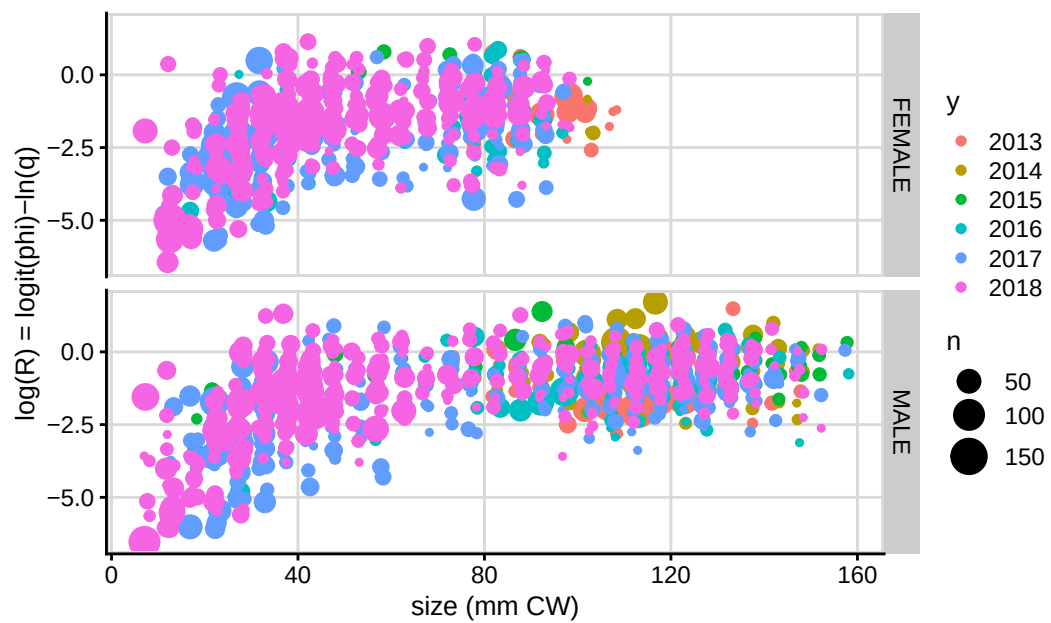


Figure 12. Haul-level $\ln(R)$ values after applying minimum catch of individuals in size bin. Upper plot: females; lower plot: males.

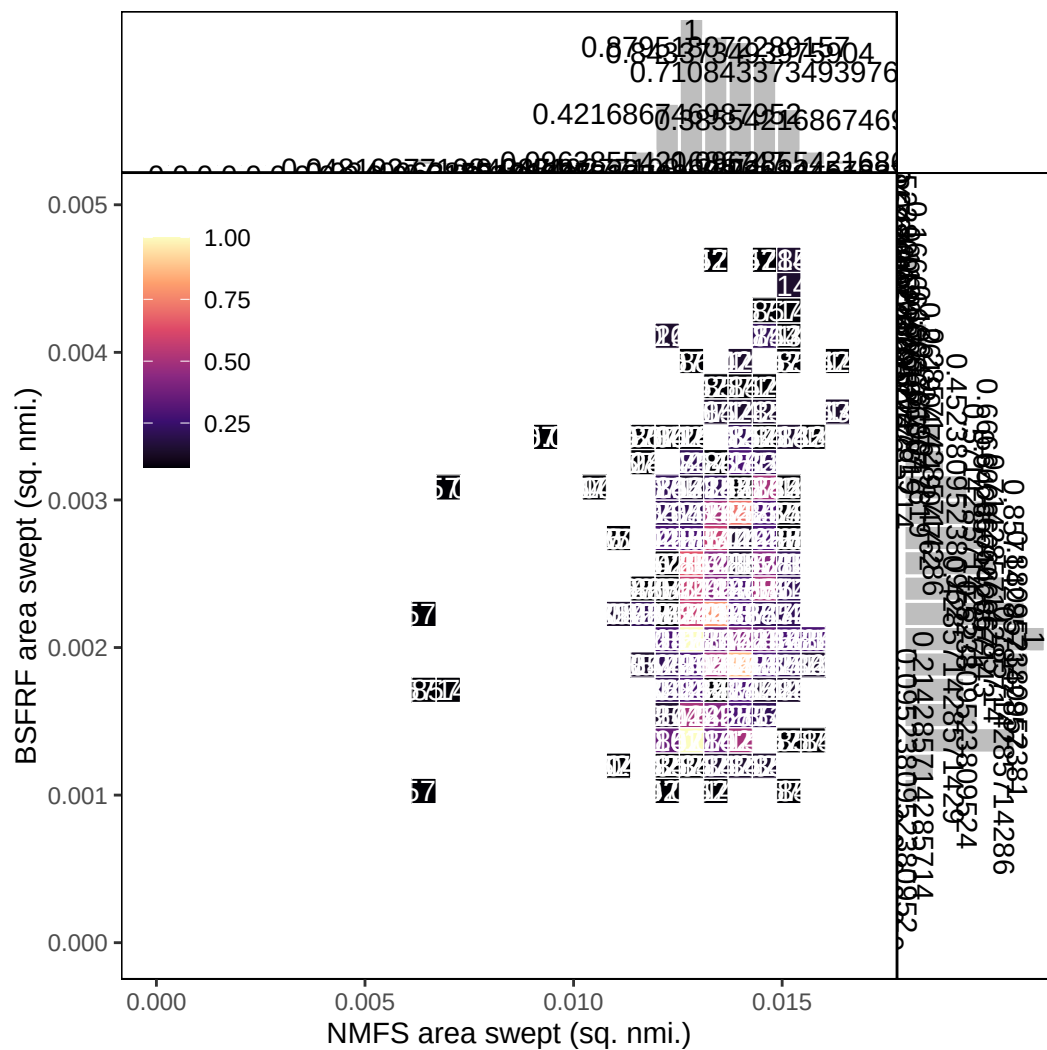


Figure 13. Comparison of areas swept by the NMFS (x-axis) and BSFRF (y-axis) gear types. Each histogram is normalized by its maximum to show relative frequencies across bins.

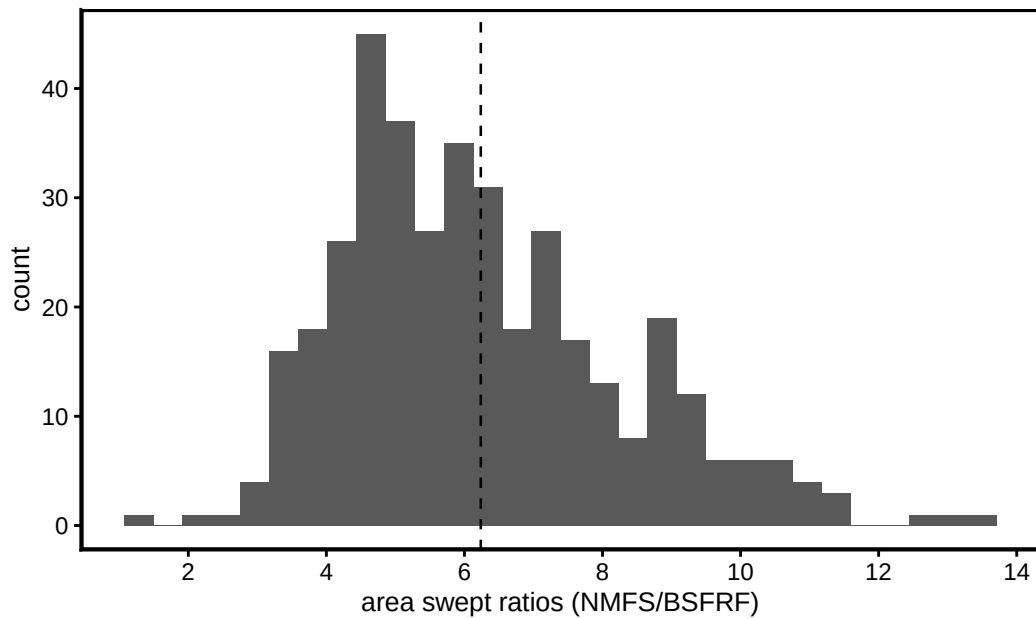


Figure 14. Ratios of haul-specific area swept by the NMFS and BSFRFs gears.

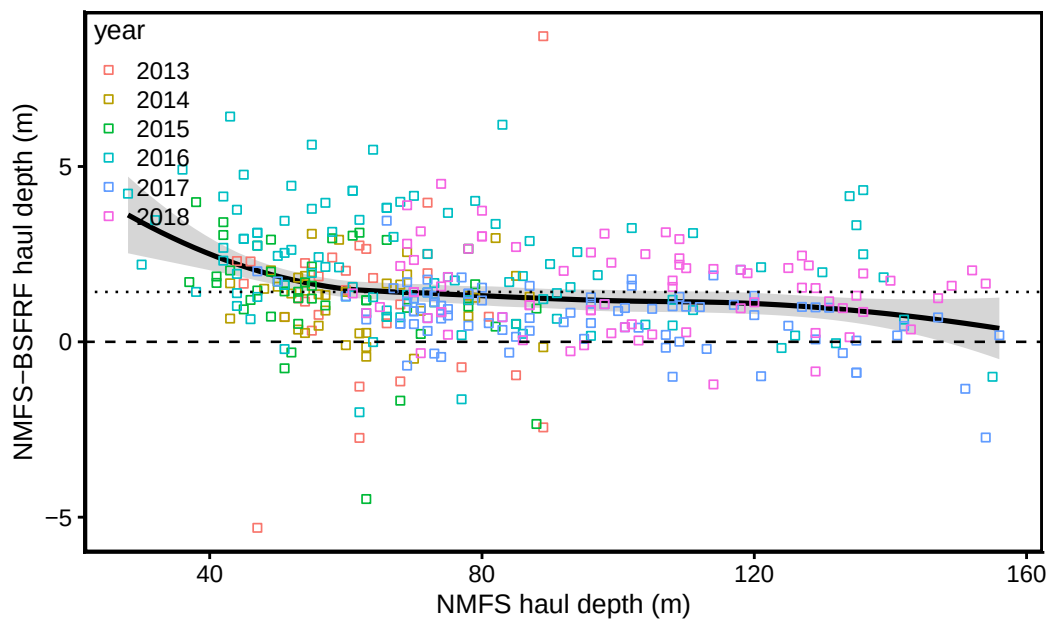


Figure 15. Comparison of the difference (NMFS-BSFRF) paired-haul depths, by NMFS haul depth. Colour indicates the year each haul was conducted. The smooth line (and shading) indicates the overall trend with depth. The dotted line indicates the mean difference.

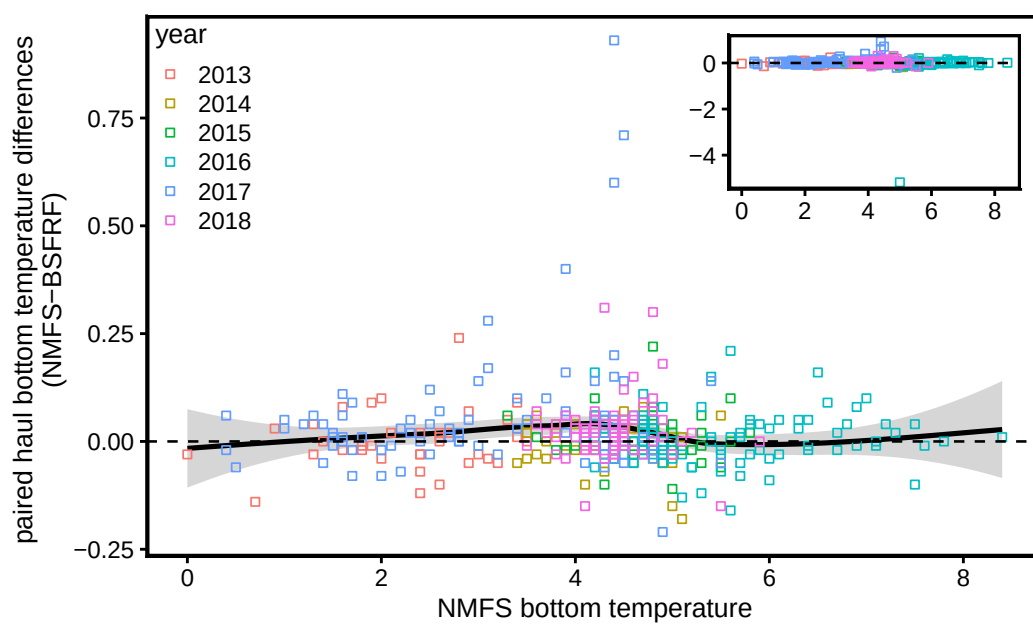


Figure 16. Comparison of measured bottom temperatures by paired haul, with the largest difference excluded to show detail. The difference (NMFS-BSFRF) between the paired haul depths is plotted against the NMFS haul depth. Colour indicates the year each haul was conducted. The smooth line (and shading) indicates the overall trend with depth. The inset shows the same, but with the largest difference included.

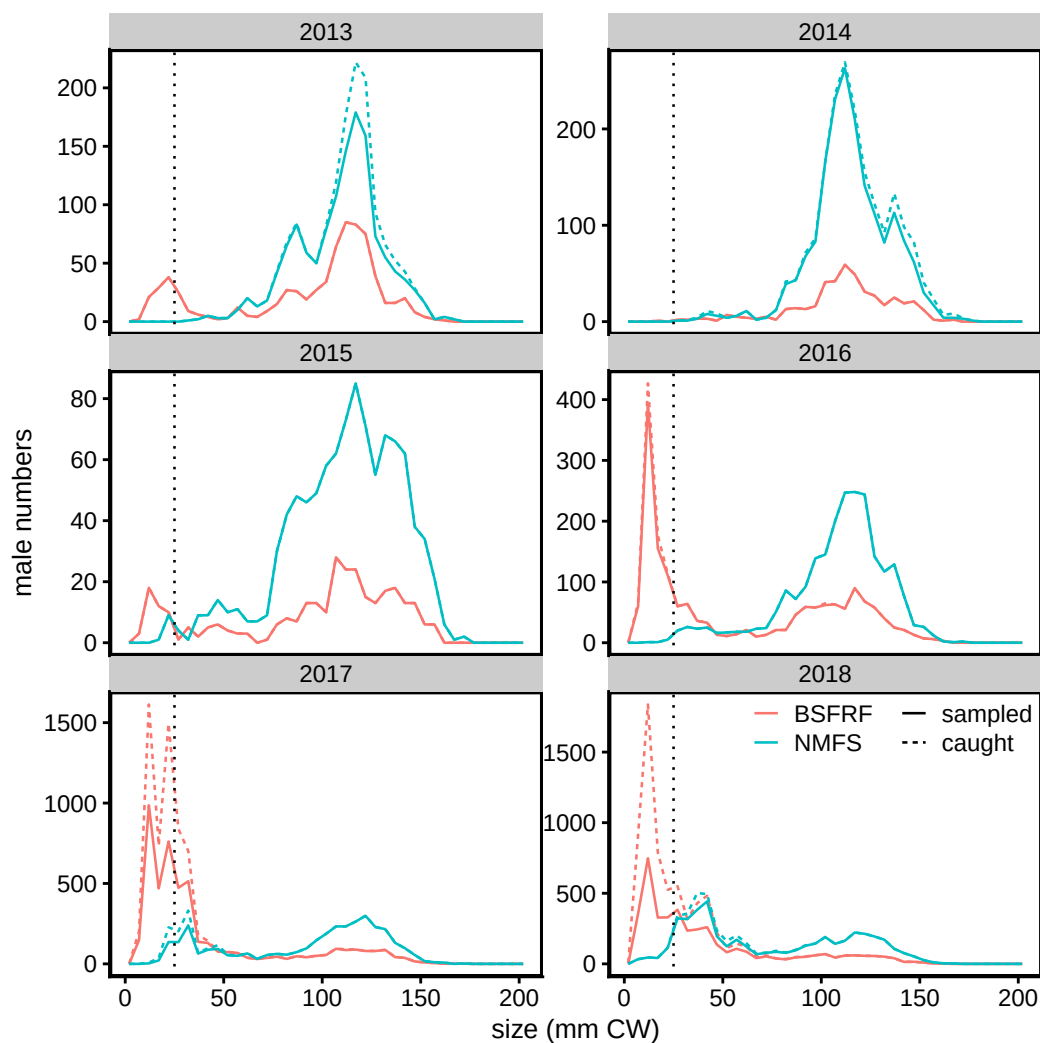


Figure 17. Total numbers of males sampled (measured) in the SBS studies, by 5-mm size bin and gear type (solid lines). Also shown are estimates of the actual numbers caught (i.e., prior to any sub-sampling; dotted lines). The dotted vertical line marks the lower limit of the size bins used in the assessment.

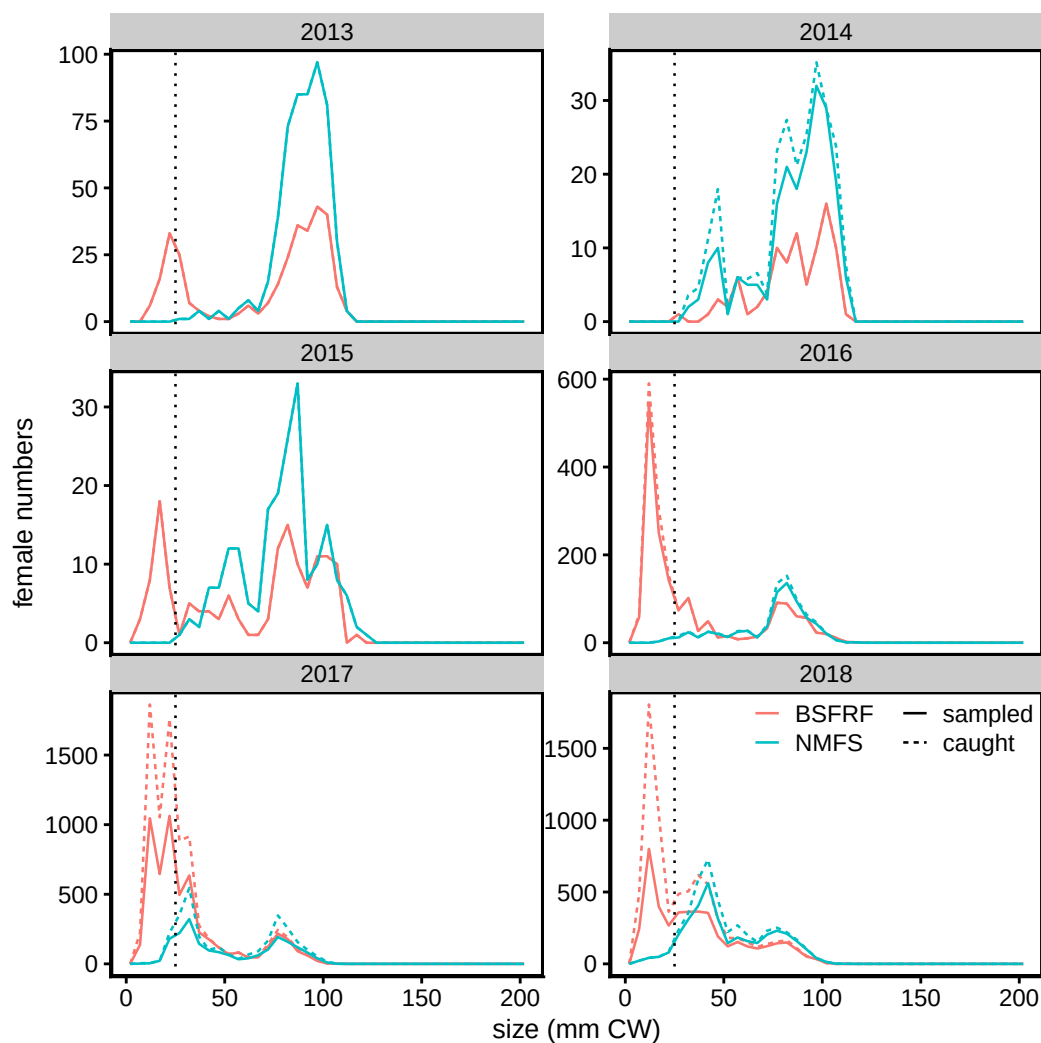


Figure 18. Total numbers of females sampled (measured) in the SBS studies, by 5-mm size bin and gear type (solid lines). Also shown are estimates of the actual numbers caught (i.e., prior to any sub-sampling; dotted lines). The dotted vertical line marks the lower limit of the size bins used in the assessment.

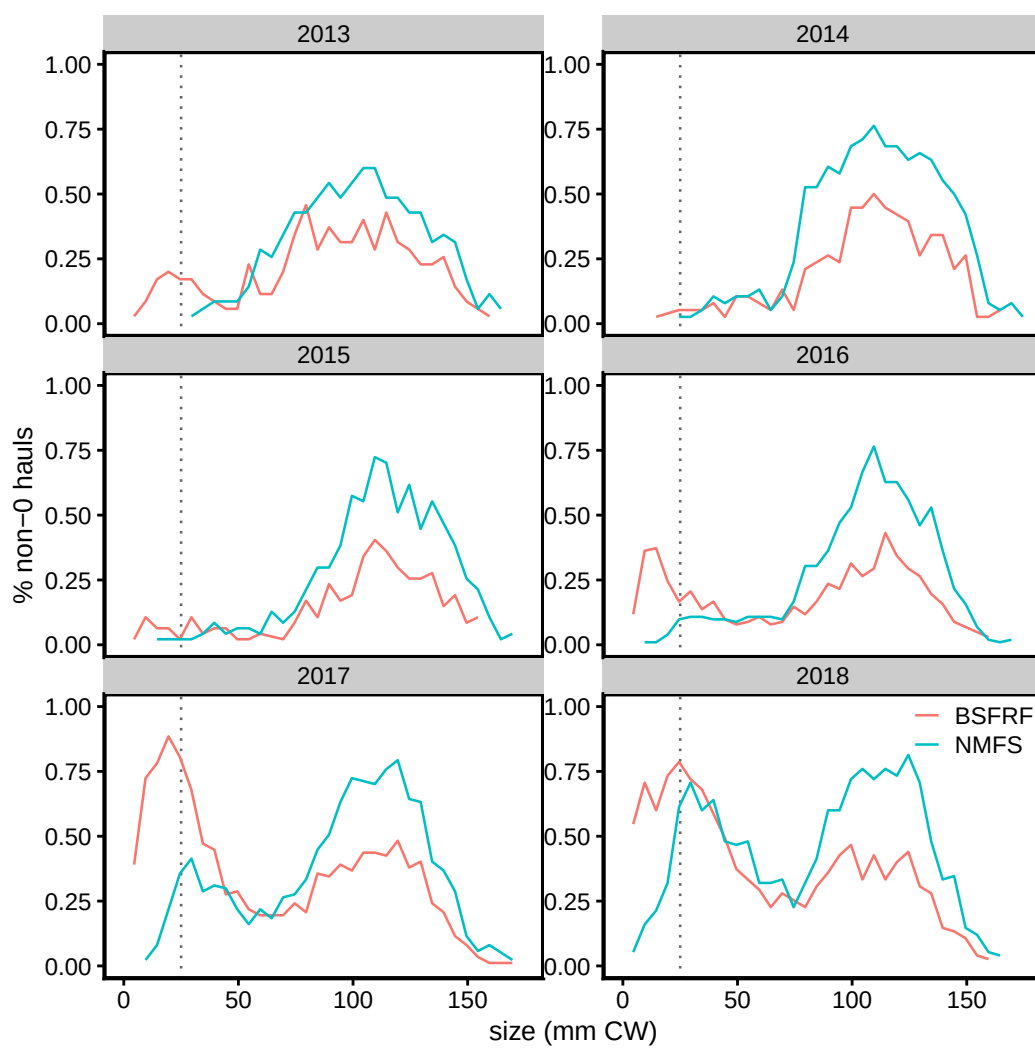


Figure 19. The percentage of non-zero catch hauls for males in the SBS studies, by 5-mm size bin and gear type. The dotted vertical line marks the lower limit of the size bins used in the assessment.

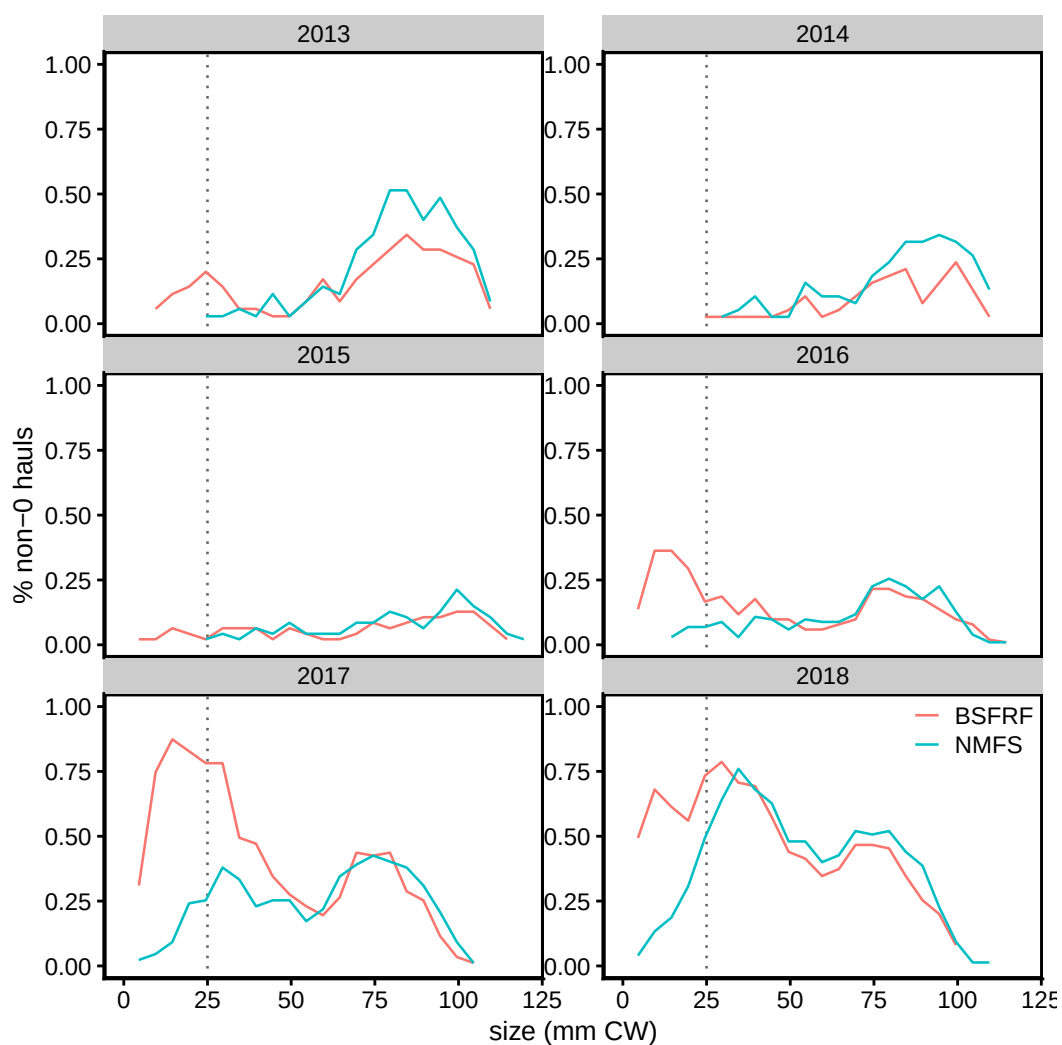


Figure 20. The percentage of non-zero catch hauls for females in the SBS studies, by 5-mm size bin and gear type. The dotted vertical line marks the lower limit of the size bins used in the assessment.

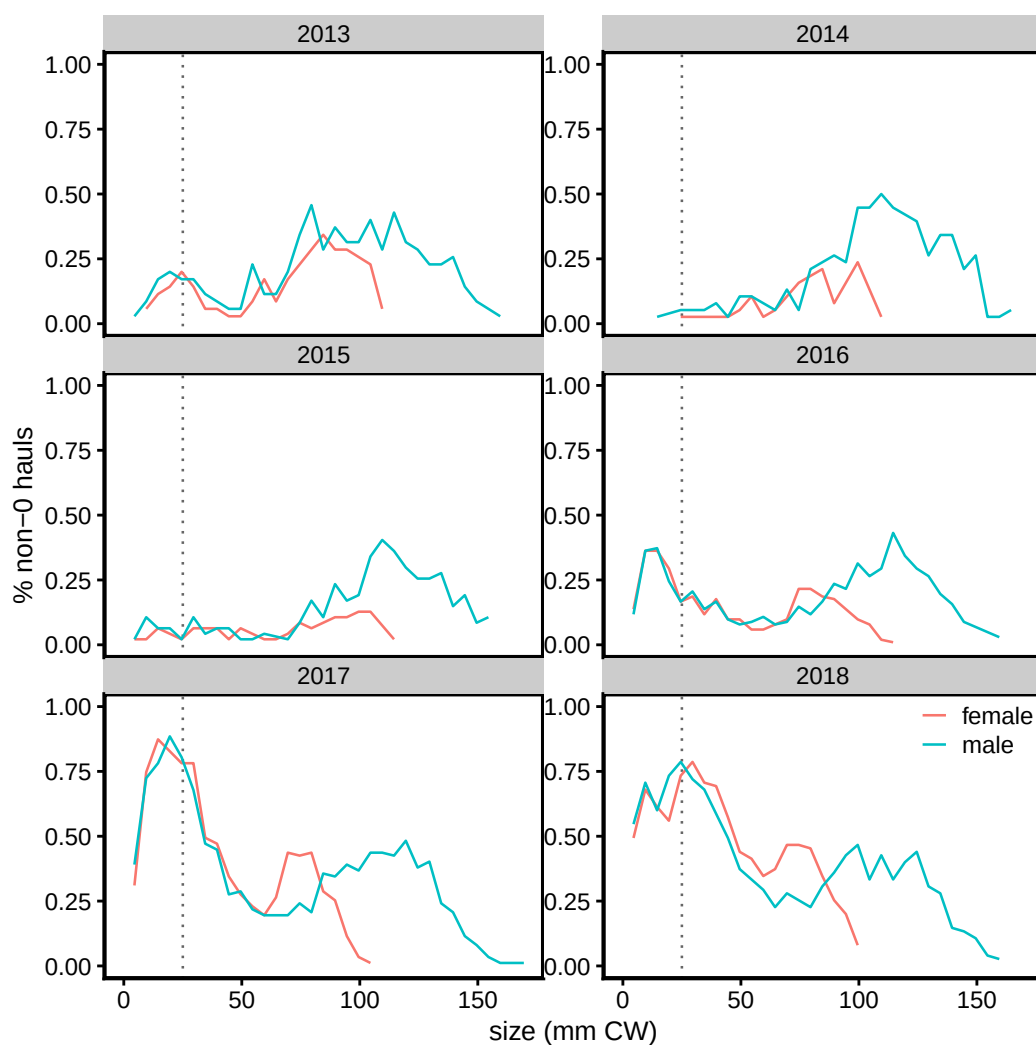


Figure 21. The percentage of non-zero catch hauls for the BSFRF gear in the SBS studies, by 5-mm size bin and sex. The dotted vertical line marks the lower limit of the size bins used in the assessment.

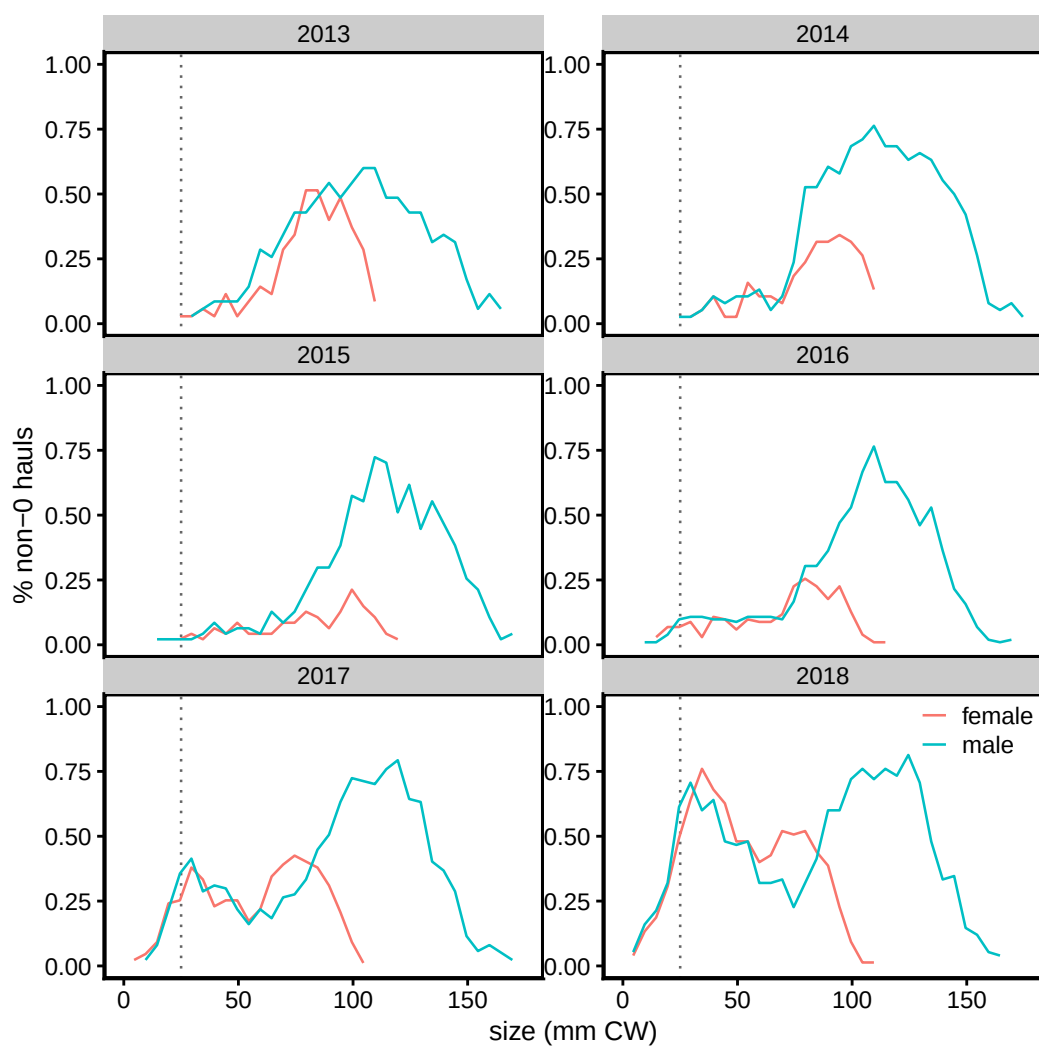


Figure 22. The percentage of non-zero catch hauls for the NMFS gear in the SBS studies, by 5-mm size bin and sex. The dotted vertical line marks the lower limit of the size bins used in the assessment.

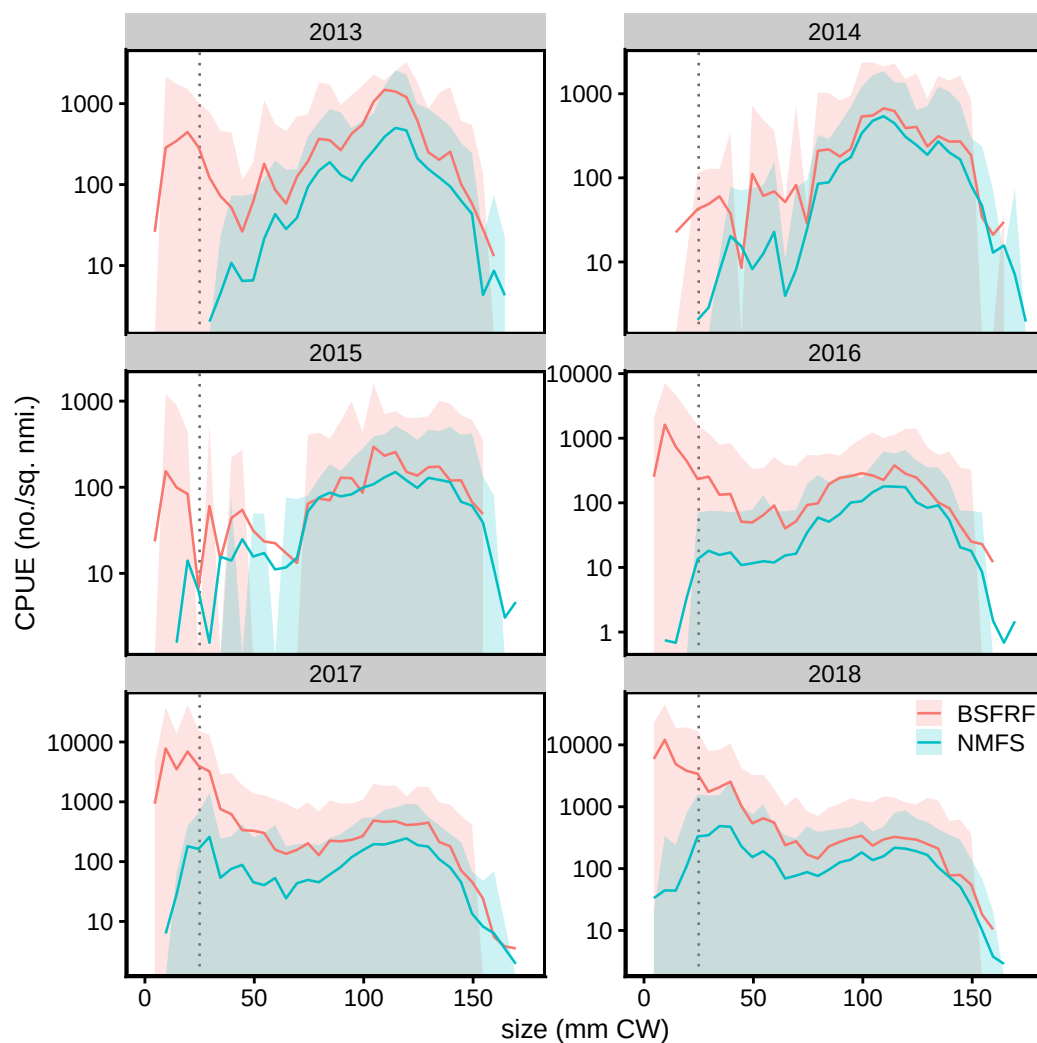


Figure 23. Male CPUE in the SBS studies, by 5-mm size bin and gear type. Solid lines: mean; shaded area: empirical 90% CI. The dotted vertical line marks the lower limit of the size bins used in the assessment. The y-axis is on a log scale to facilitate comparison across the range of values.

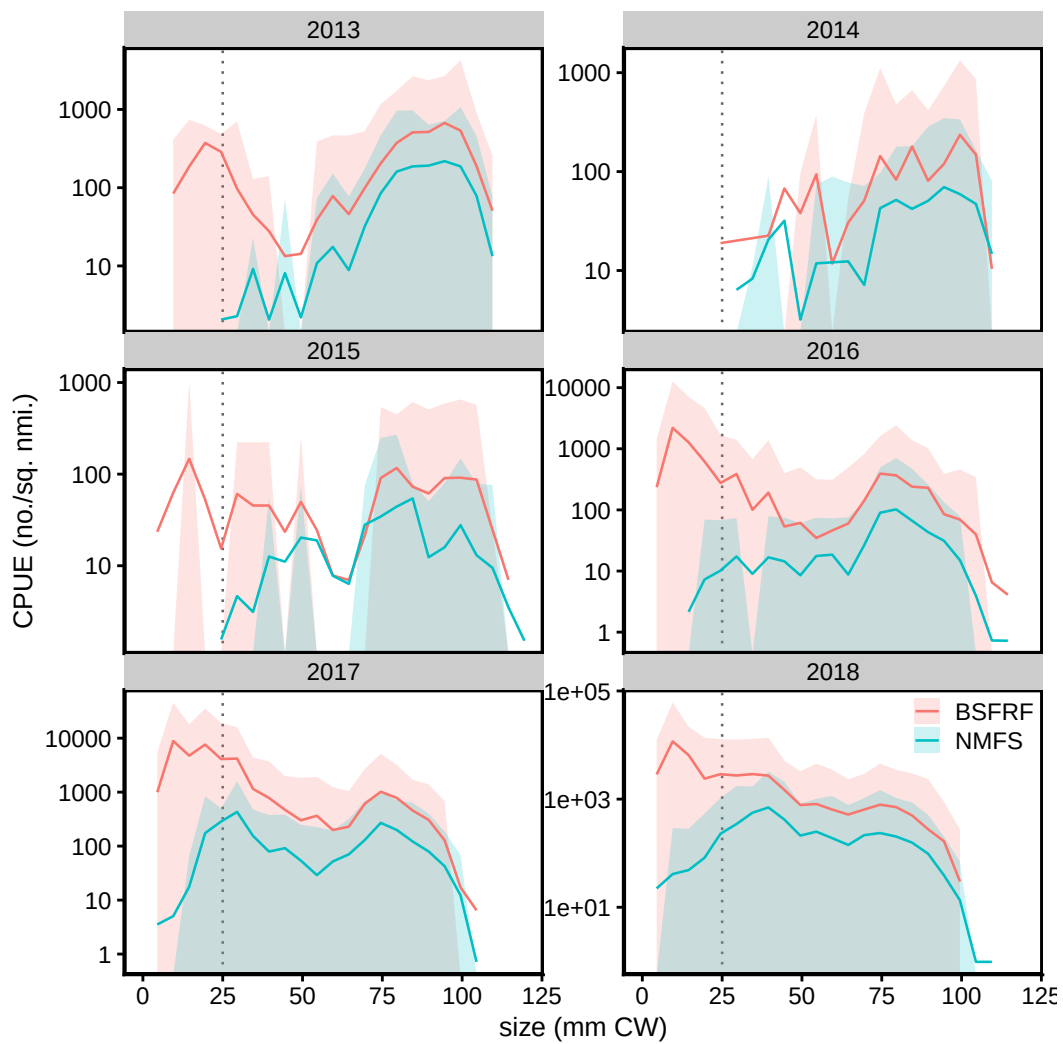


Figure 24. Female CPUE in the SBS studies, by 5-mm size bin and gear type. Solid lines: mean; shaded area: empirical 90% CI. The dotted vertical line marks the lower limit of the size bins used in the assessment. The y-axis is on a log scale to facilitate comparison across the range of values.

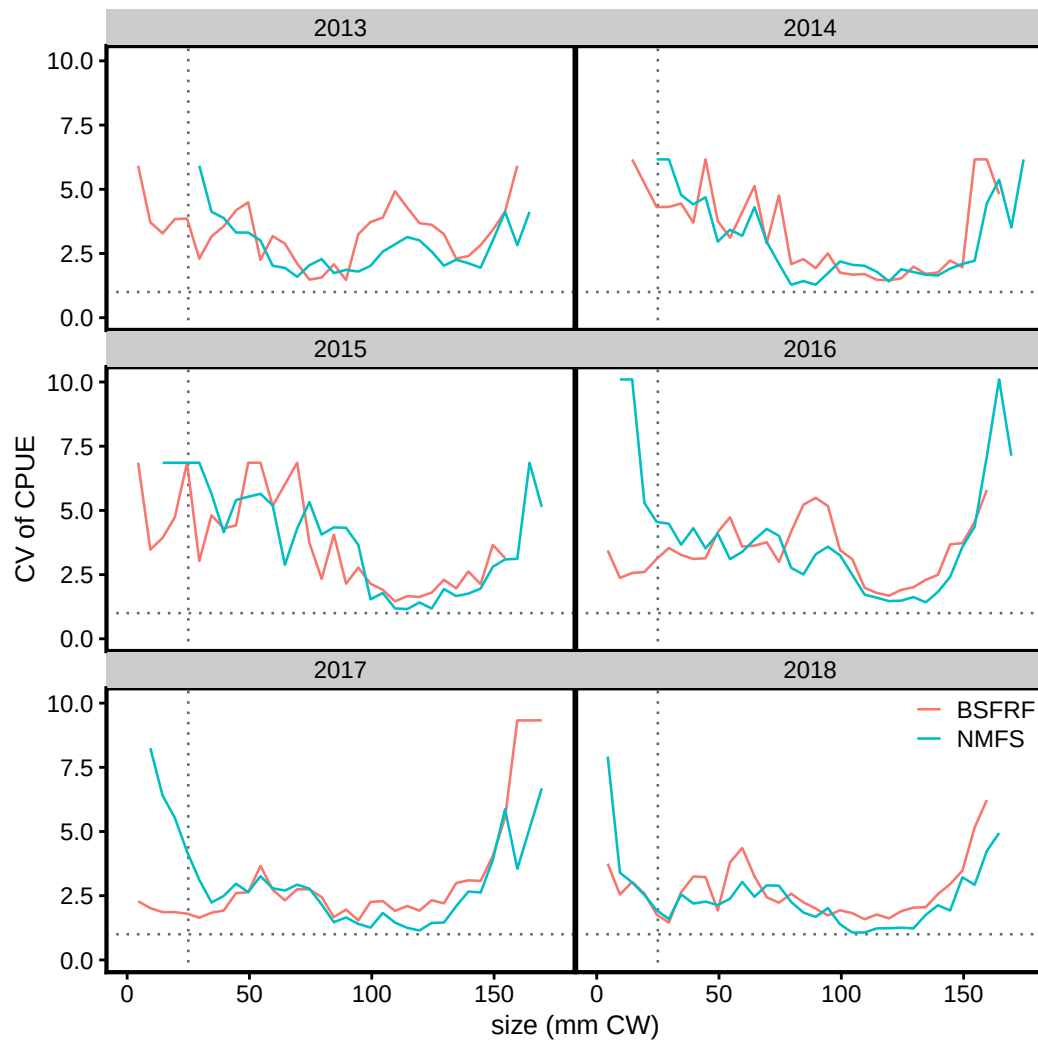


Figure 25. CVs for male CPUE in the SBS studies, by 5-mm size bin and gear type. The dotted vertical line marks the lower limit of the size bins used in the assessment. The horizontal line indicates a CV of 1.

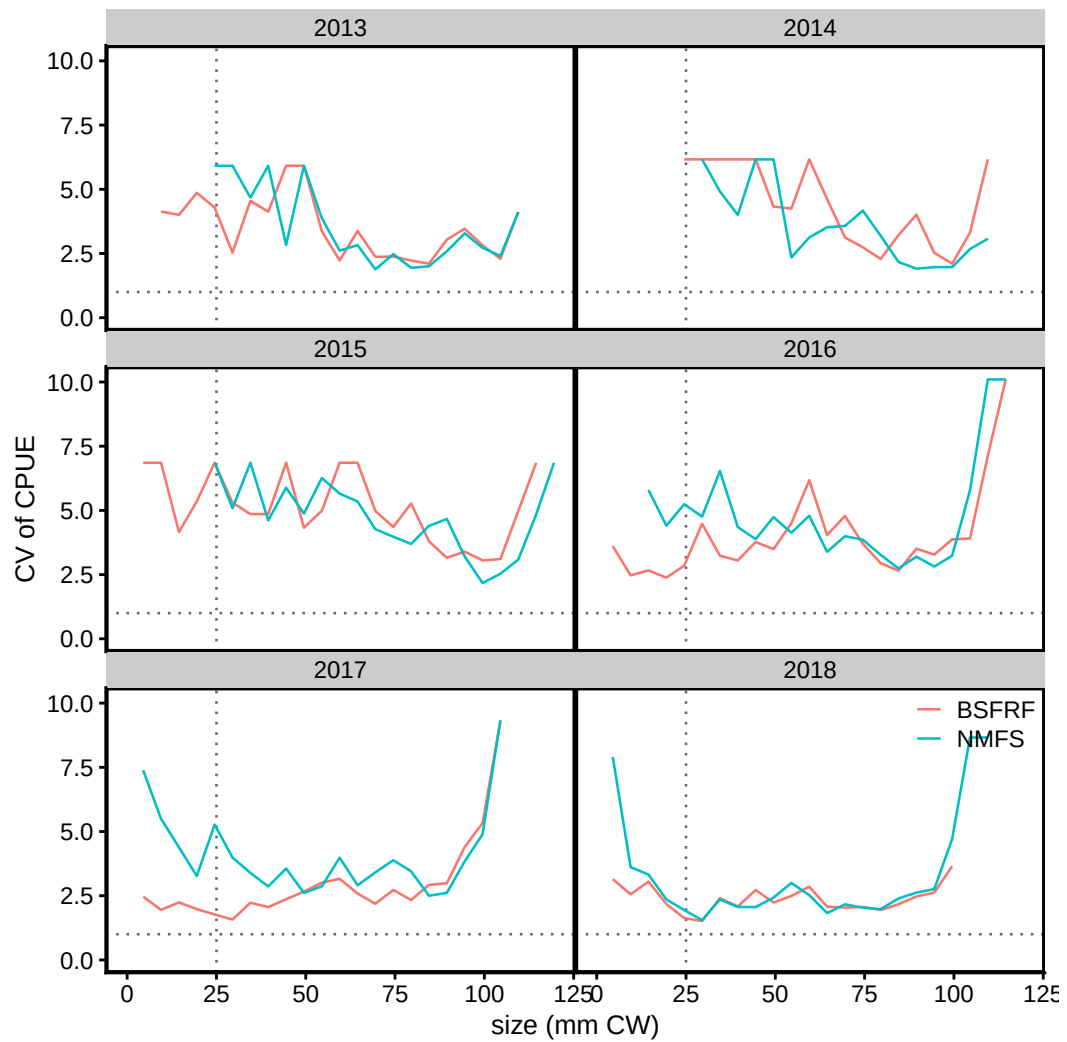


Figure 26. CVs for female CPUE in the SBS studies, by 5-mm size bin and gear type. The dotted vertical line marks the lower limit of the size bins used in the assessment. The horizontal line indicates a CV of 1.

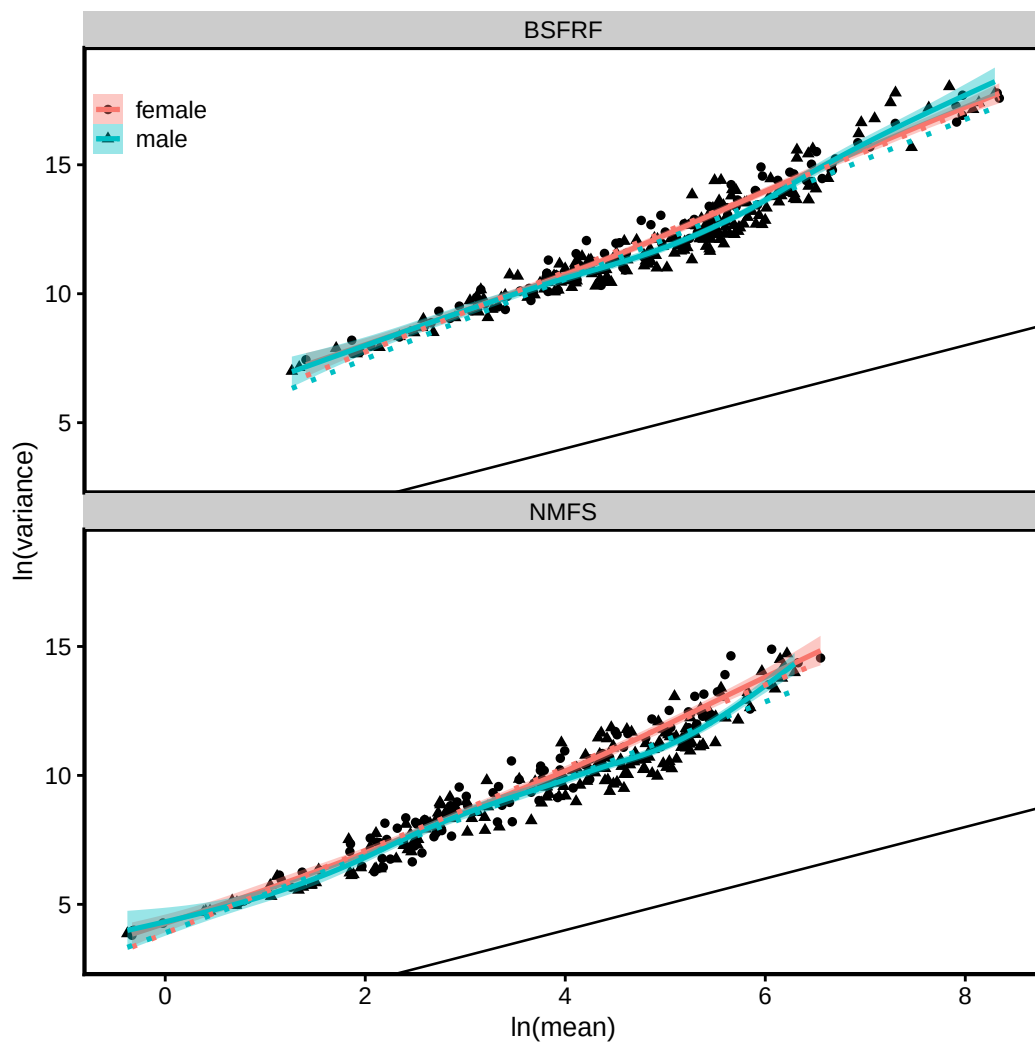


Figure 27. Variance in CPUE by 5 mm CW size bin plotted against mean CPUE, on the natural logarithm scale. Symbols: observed values (circles: females; triangles: males); colored shading/line: GAM smooth fits to observed values; dotted lines: linear regressions; solid black line: 1:1 line.

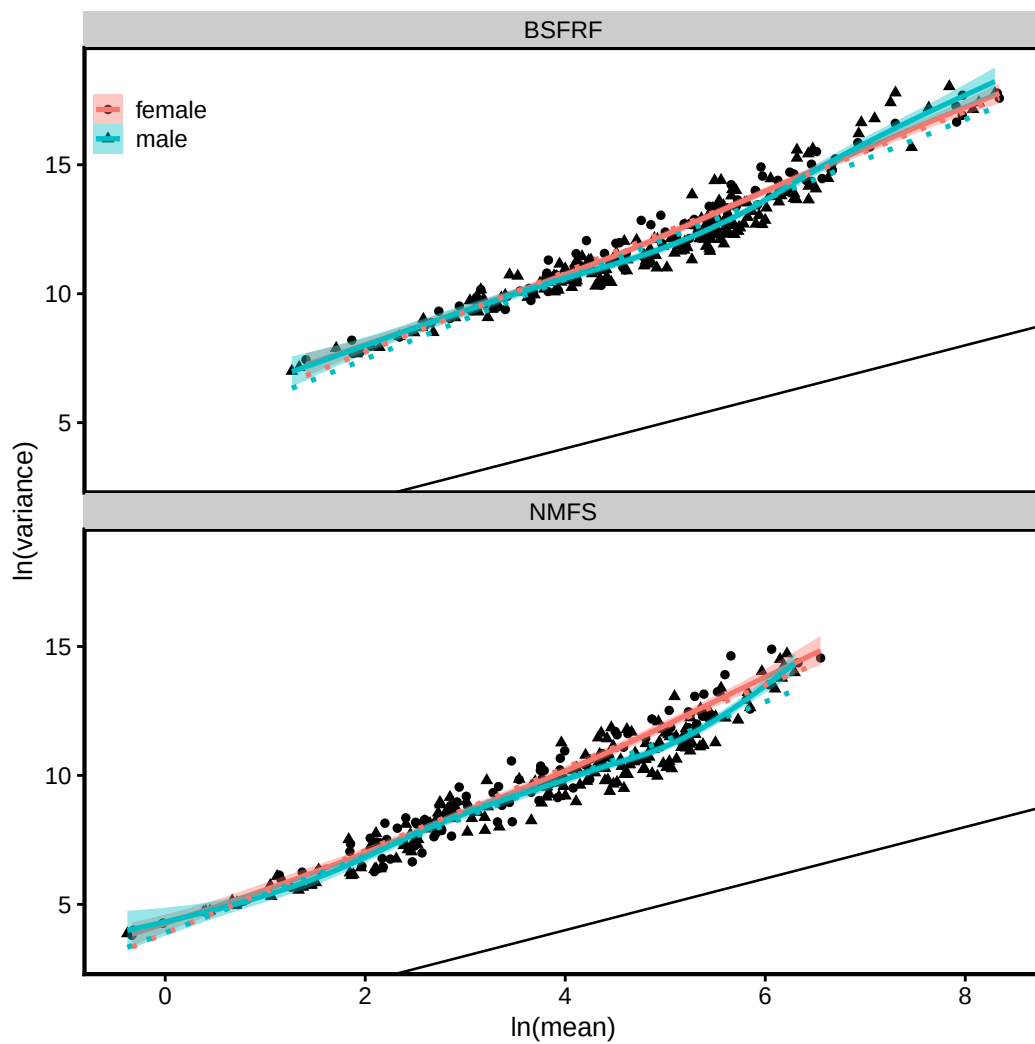


Figure 28. Variance in CPUE by 5 mm CW size bin plotted against mean CPUE, on the natural logarithm scale. Symbols: observed values (circles: BSFRF; triangles: NMFS); colored shading/line: GAM smooth fits to observed values; dotted lines: linear regressions; solid black line: 1:1 line.

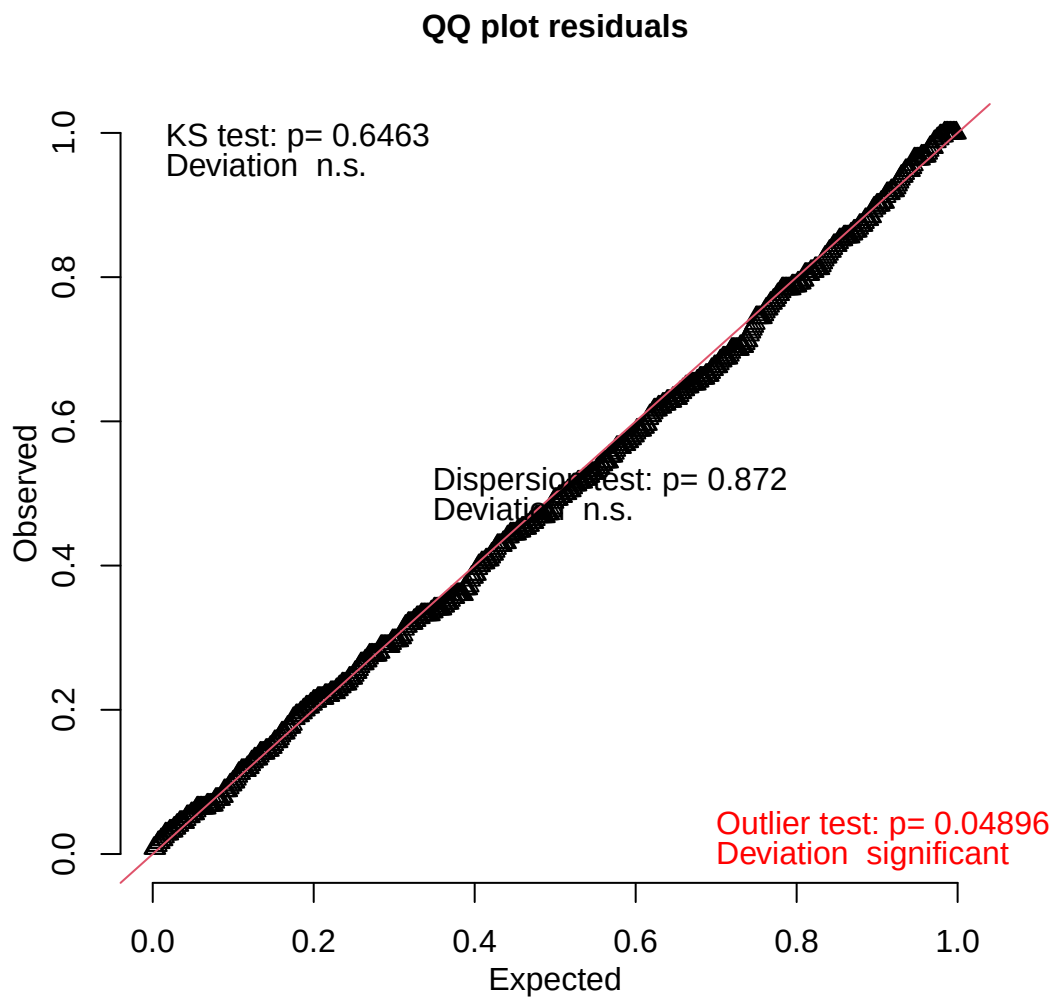


Figure 29. Quantile-quantile plot of scaled residuals ([Hartig 2022](#)) from the reduced model used to estimate the Tweedie power parameter for the observed CPUE data.

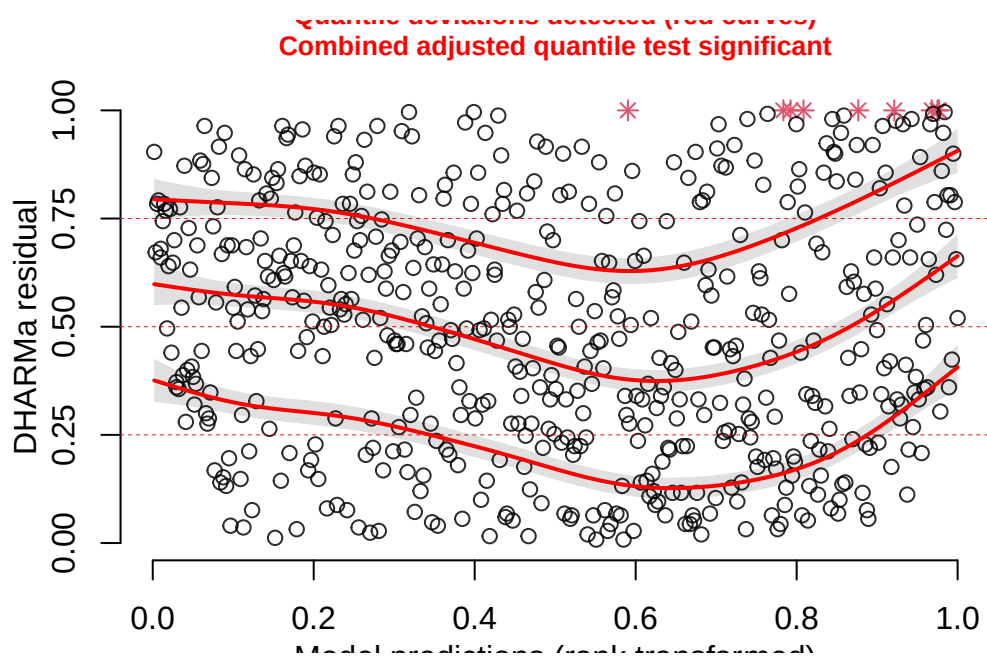


Figure 30. Scaled residuals (Hartig 2022) from the reduced model used to estimate the Tweedie power coefficient for the observed CPUE data.

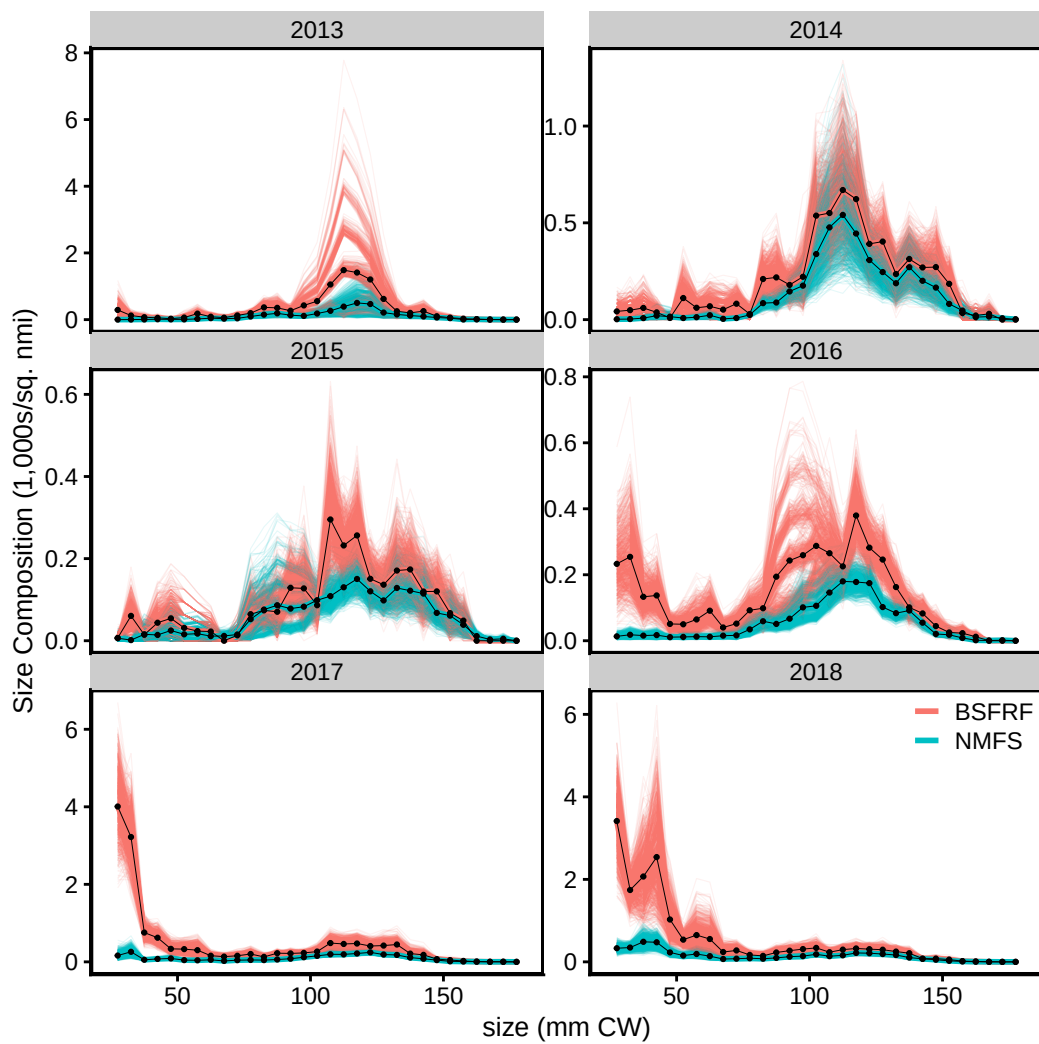


Figure 31. "Observed" (thick line, dots) and bootstrapped (thin lines) size compositions for males. Bootstrapping consisted of hierarchical resampling of paired hauls and measured crab within hauls 1000 times, with size compositions for both gear types calculated each time.

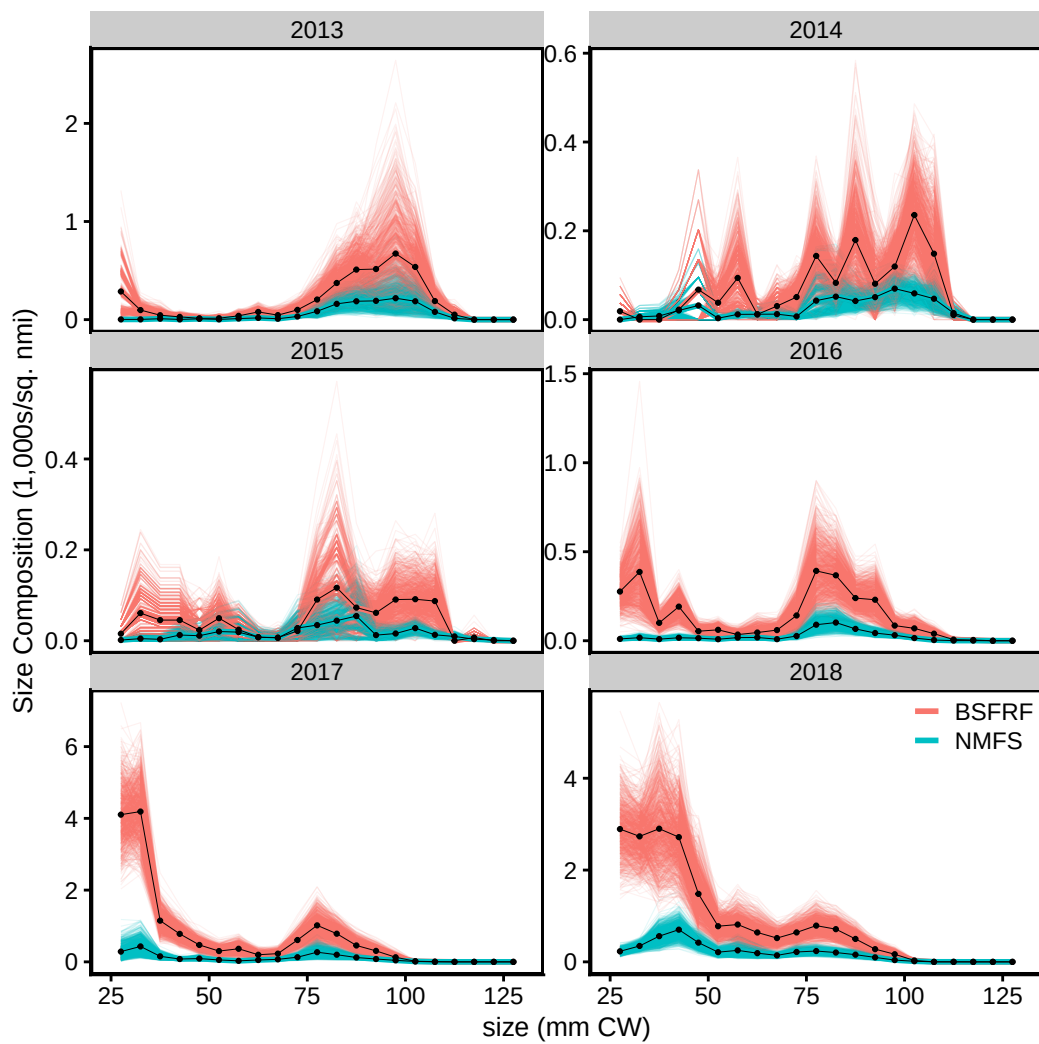


Figure 32. "Observed" (thick line, dots) and bootstrapped (thin lines) size compositions for females. Bootstrapping consisted of hierarchical resampling of paired hauls and measured crab within hauls 1000 times, with size compositions for both gear types calculated each time.

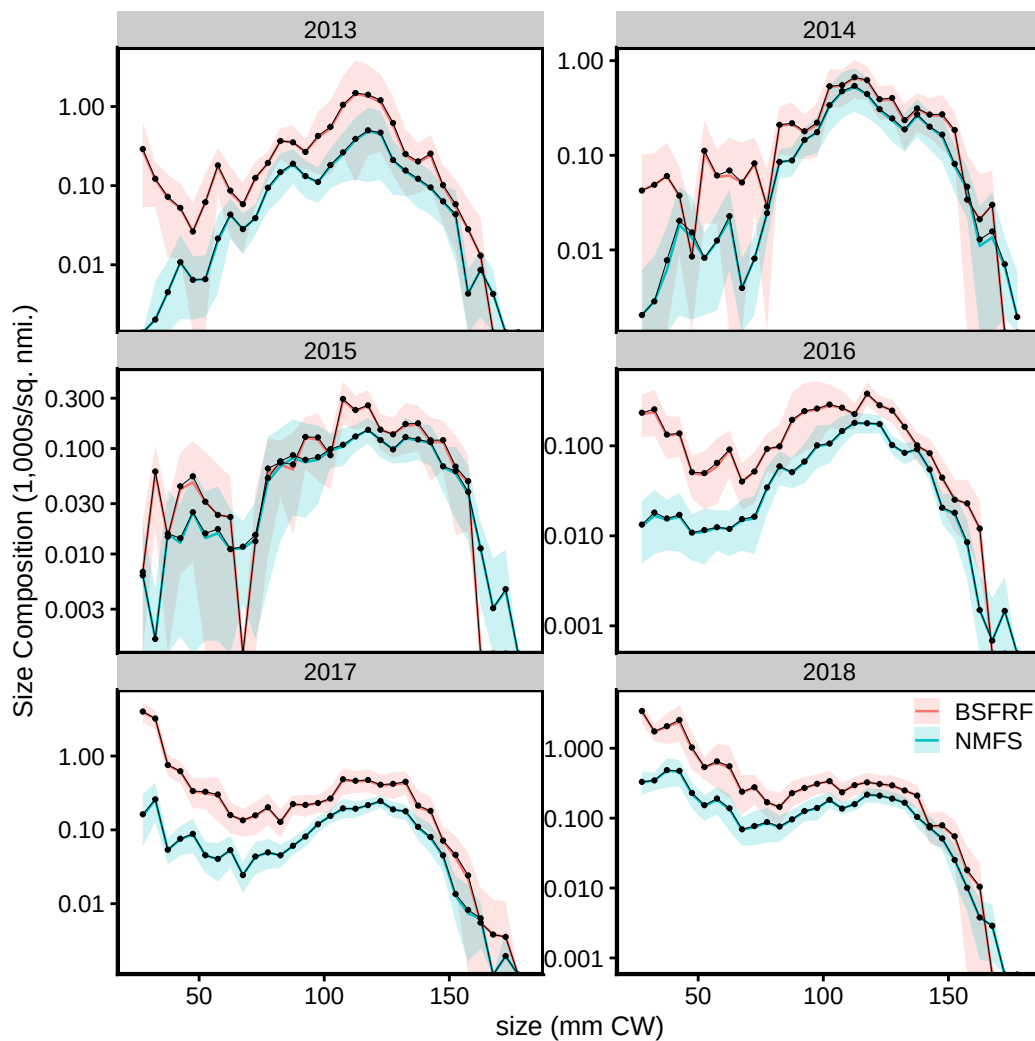


Figure 33. "Observed" (black line, dots) and bootstrapped median (lines) and 90 percent confidence intervals by gear type for male size compositions. Bootstrapping consisted of hierarchical resampling of paired hauls and measured crab within hauls 1000 times by study year, with size compositions for both gear types calculated each time.

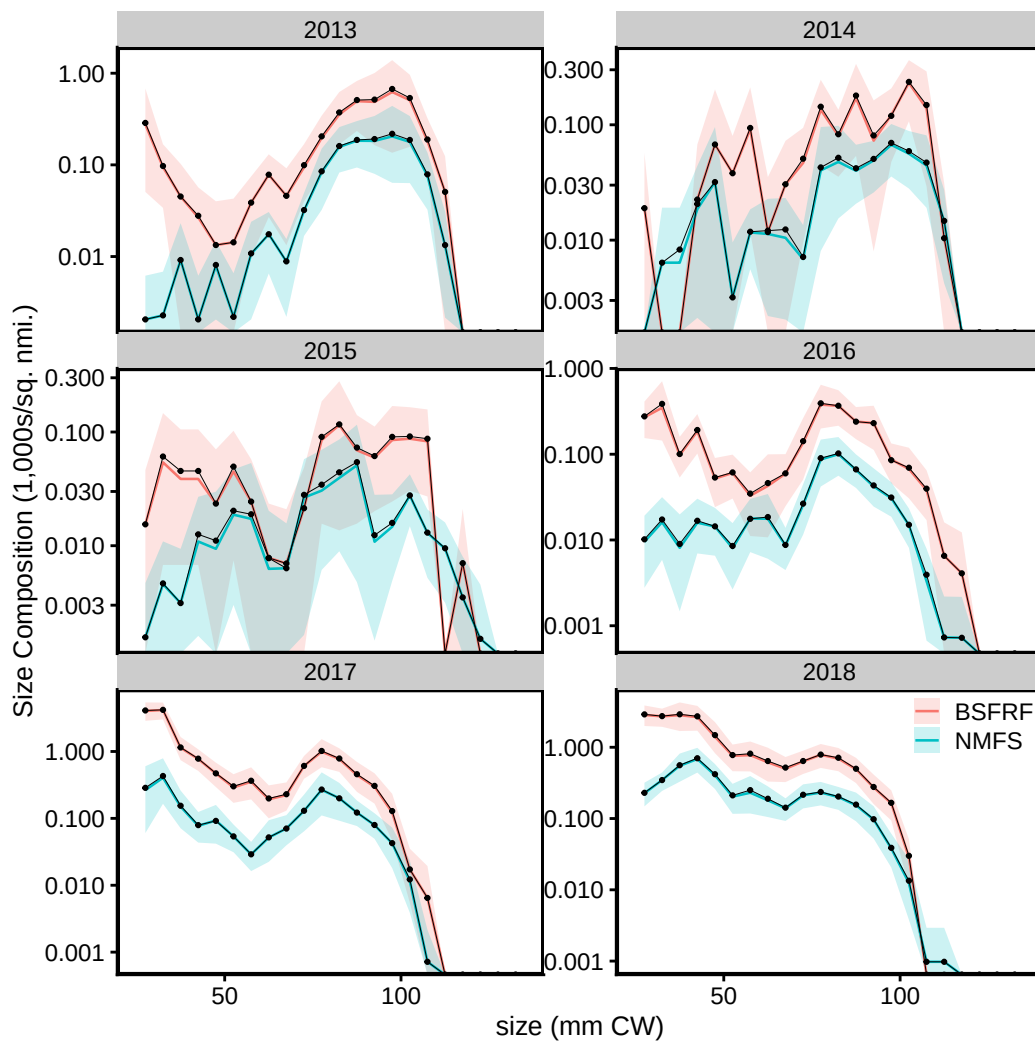


Figure 34. "Observed" (black line, dots) and bootstrapped median (lines) and 90 percent confidence intervals by gear type for female size compositions. Bootstrapping consisted of hierarchical resampling of paired hauls and measured crab within hauls 1000 times by study year, with size compositions for both gear types calculated each time.

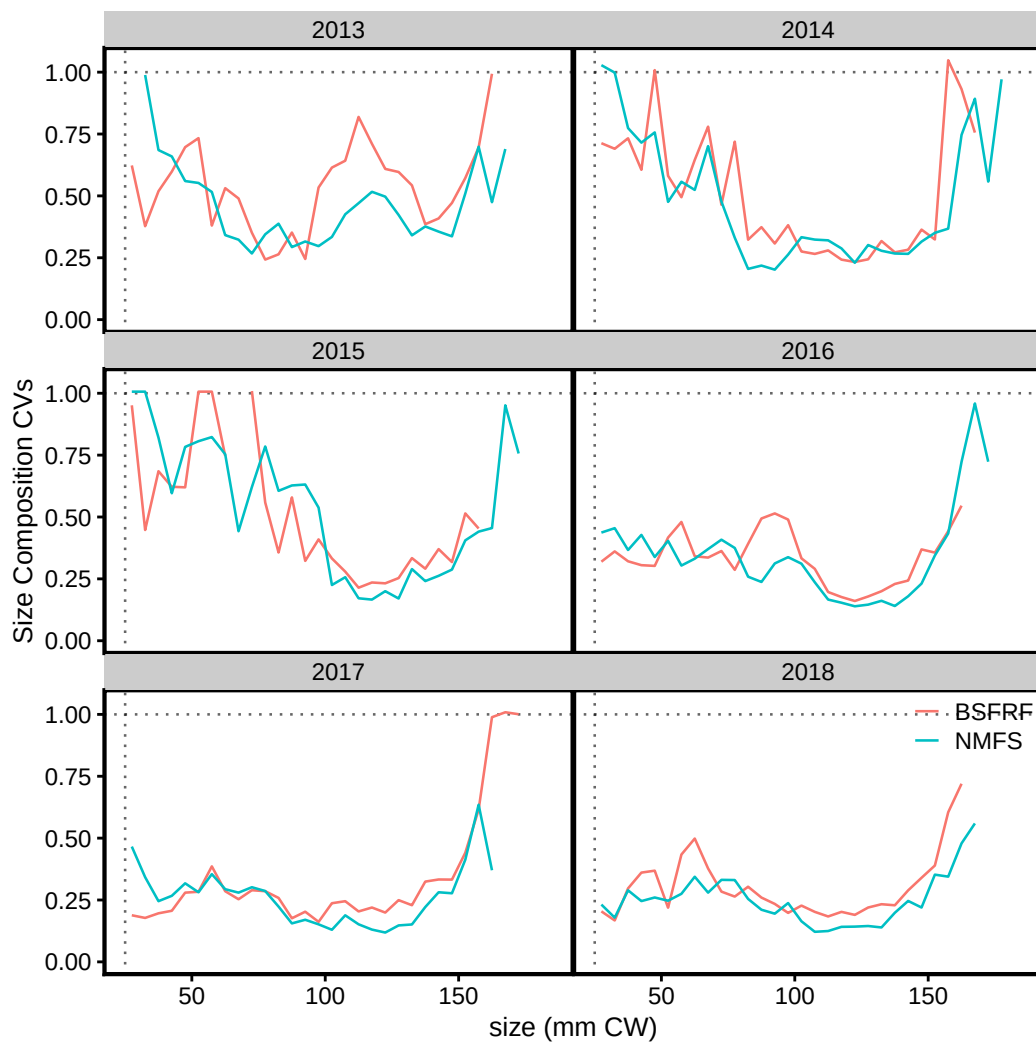


Figure 35. CVs from bootstrapped male size compositions. Bootstrapping consisted of hierarchical resampling of paired hauls and measured crab within hauls 1000 times by study year, with size compositions for both gear types calculated each time.

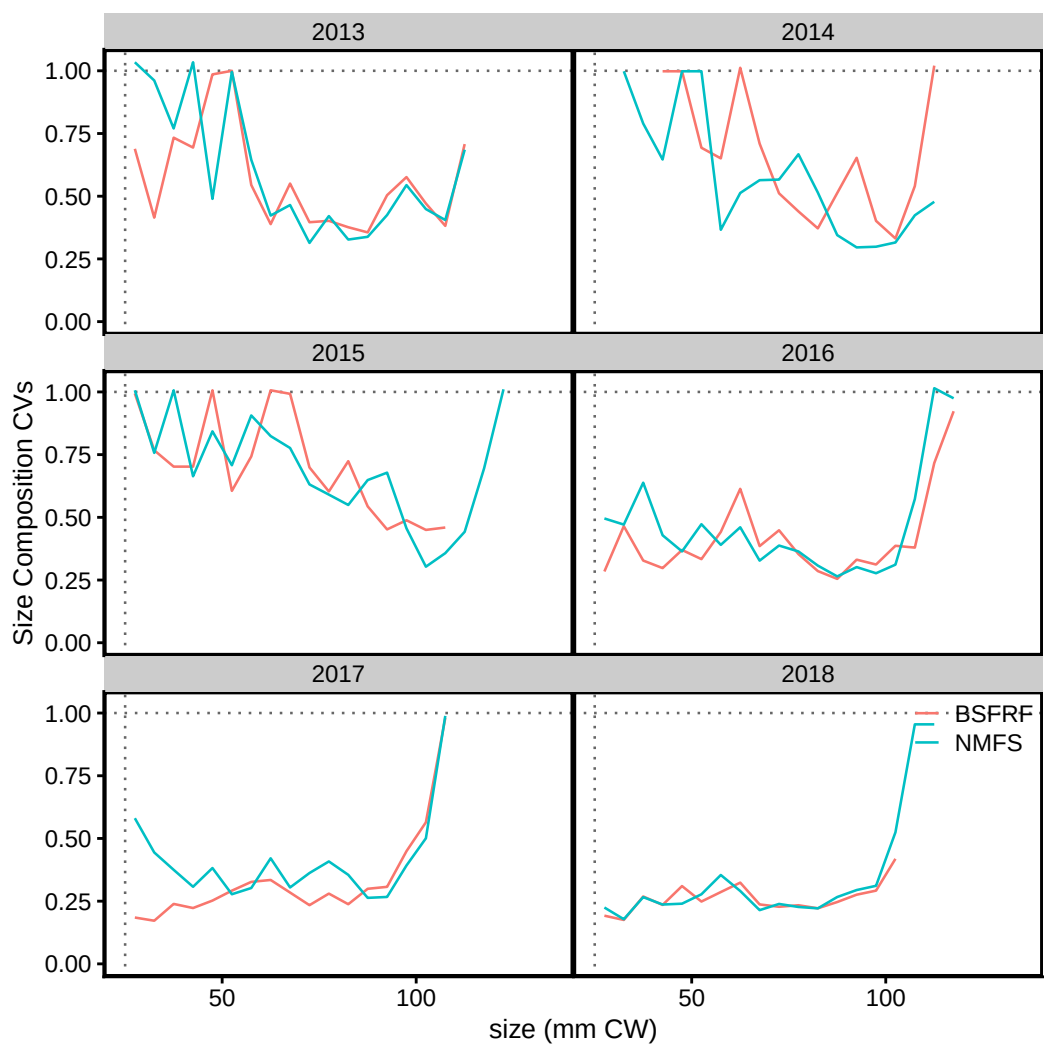


Figure 36. CVs from bootstrapped female size compositions. Bootstrapping consisted of hierarchical resampling of paired hauls and measured crab within hauls 1000 times by study year, with size compositions for both gear types calculated each time.

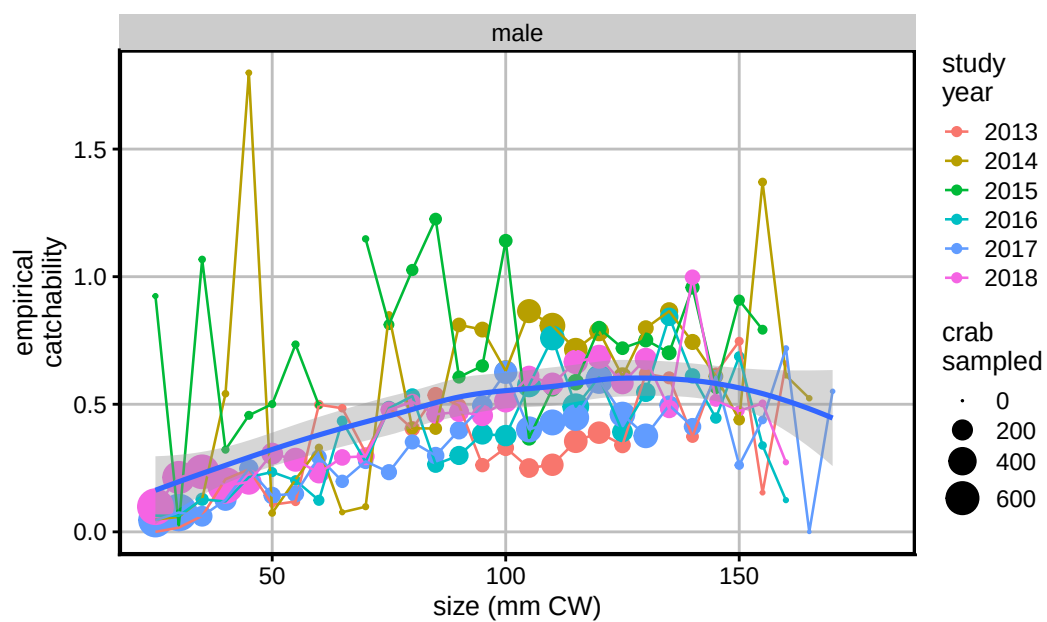


Figure 37. Comparison of annual empirical estimates of NMFS survey selectivity for male Tanner crab. An overall (unweighted) smoothed trend and simple uncertainty interval are indicated by the thick blue line and grey shading.

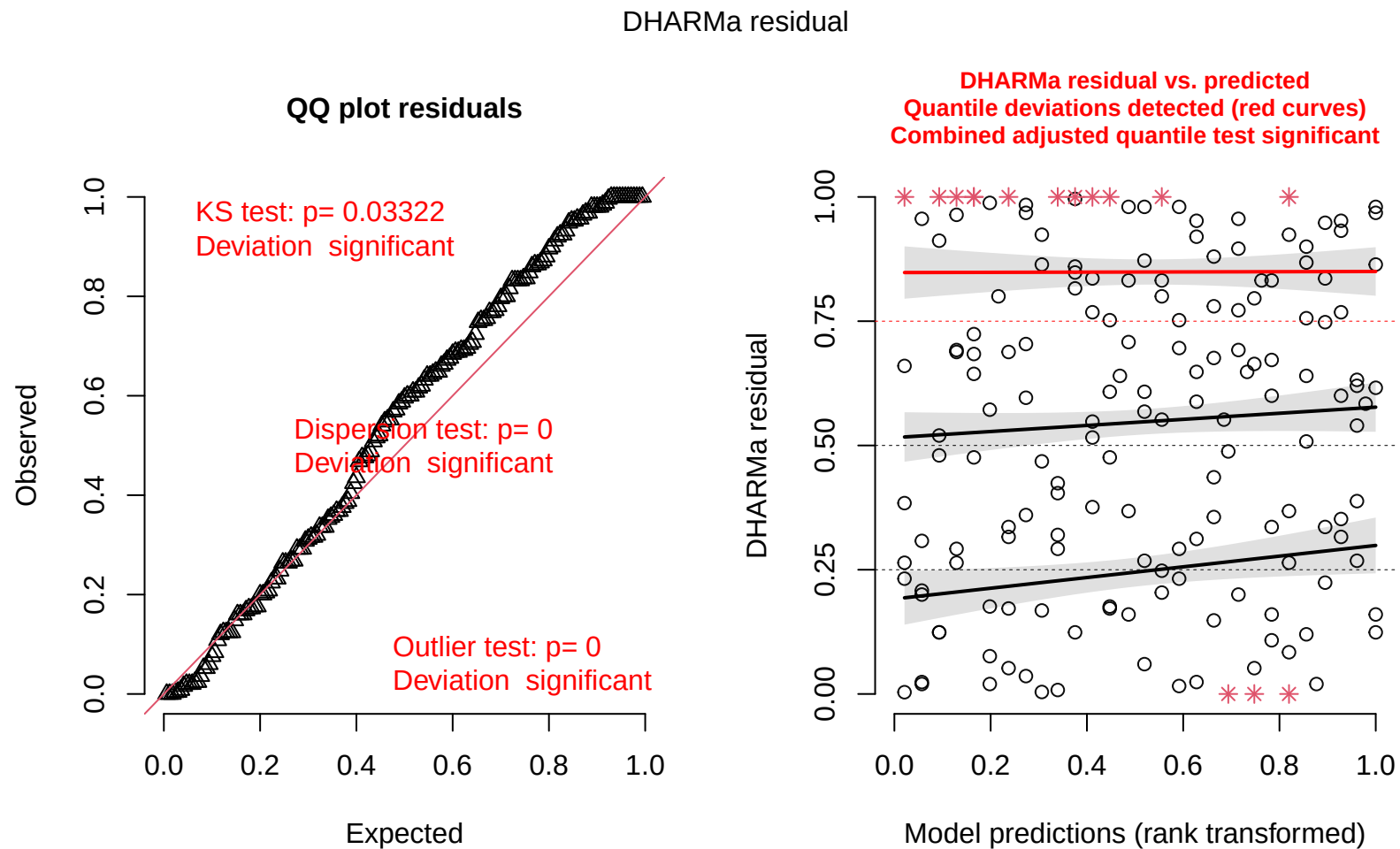


Figure 38. DHARMa residuals from Model 1 for males. Left: quantile-quantile plot for observed and expected residuals under the null hypothesis. Right: quantile deviations for residuals.

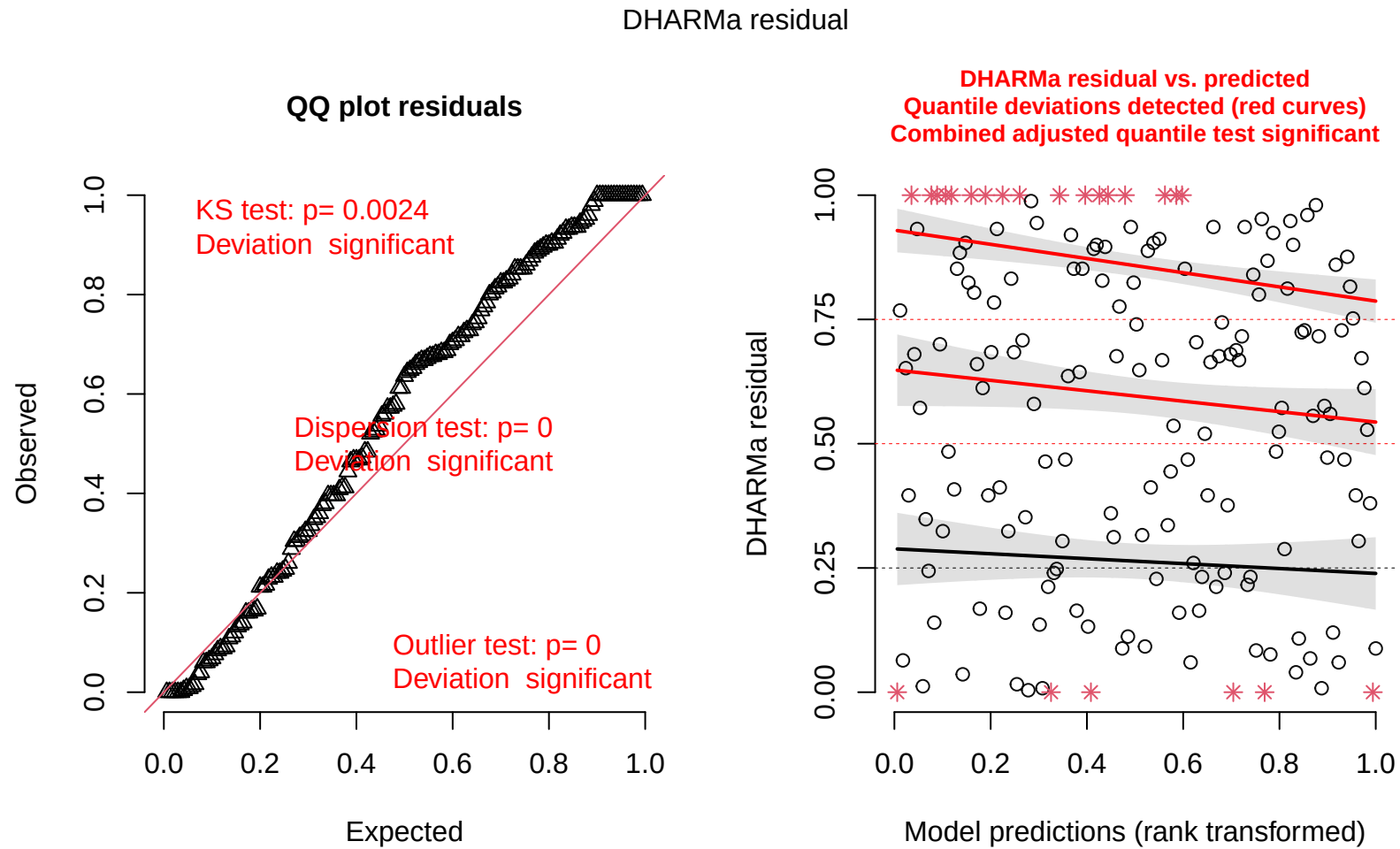


Figure 39. DHARMA residuals from Model 2 for males. Left: quantile-quantile plot for observed and expected residuals under null hypothesis. Right: quantile deviations for residuals.

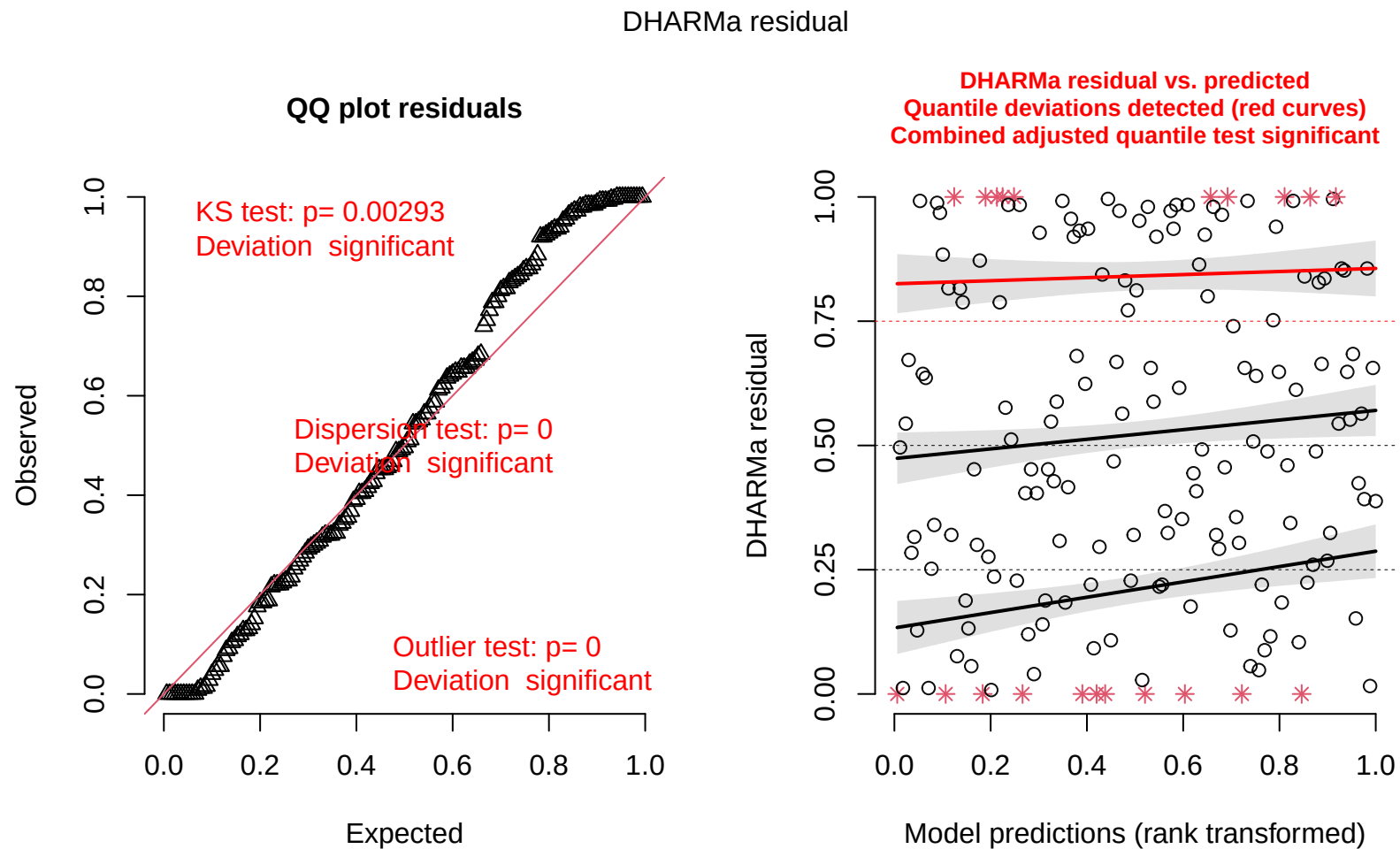


Figure 40. DHARMA residuals from Model 3 for males. Left: quantile-quantile plot for observed and expected residuals under the null hypothesis. Right: quantile deviations for residuals.

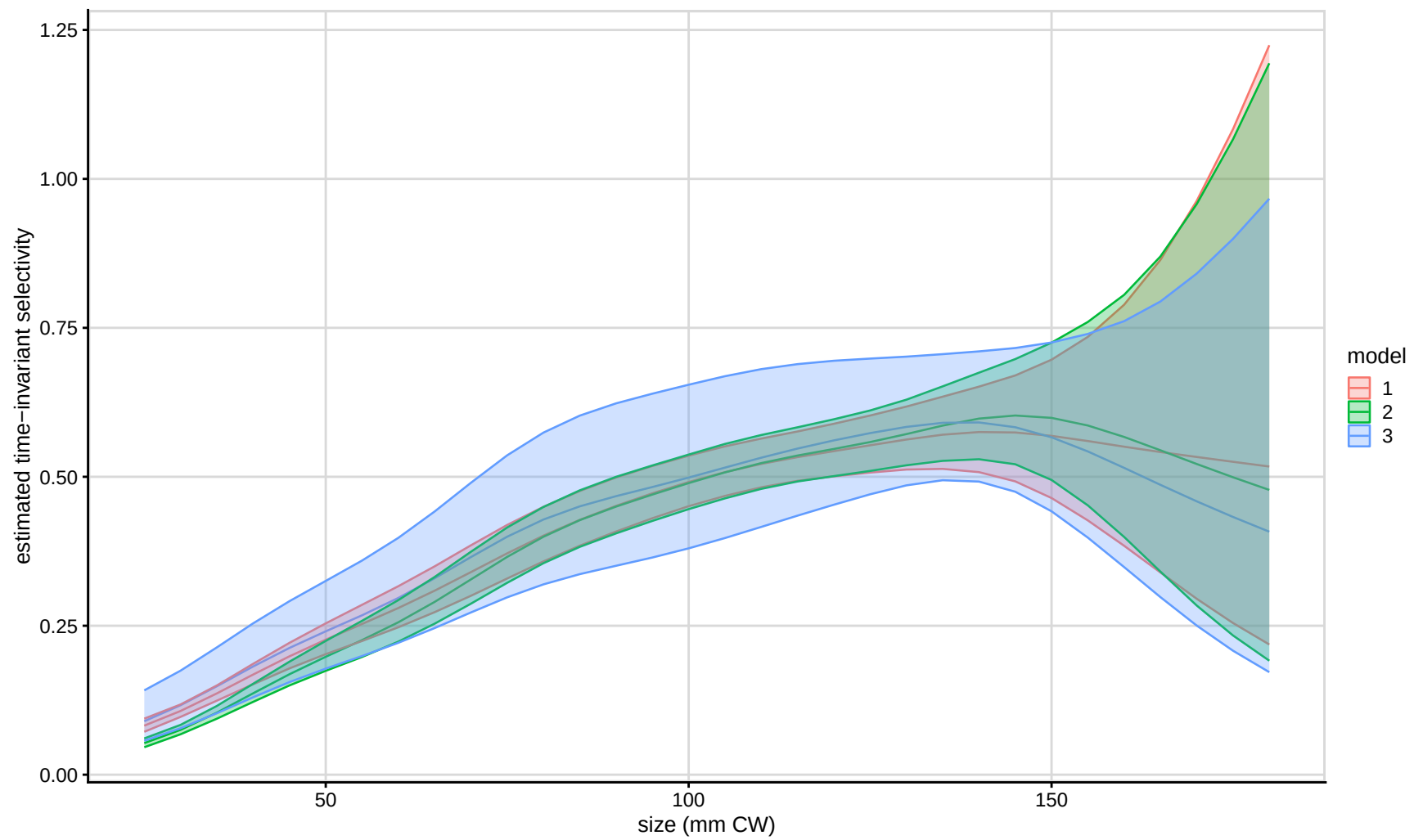


Figure 41. Comparison of the estimated time-invariant estimated selectivity, $S(z)$, for males from the three models.

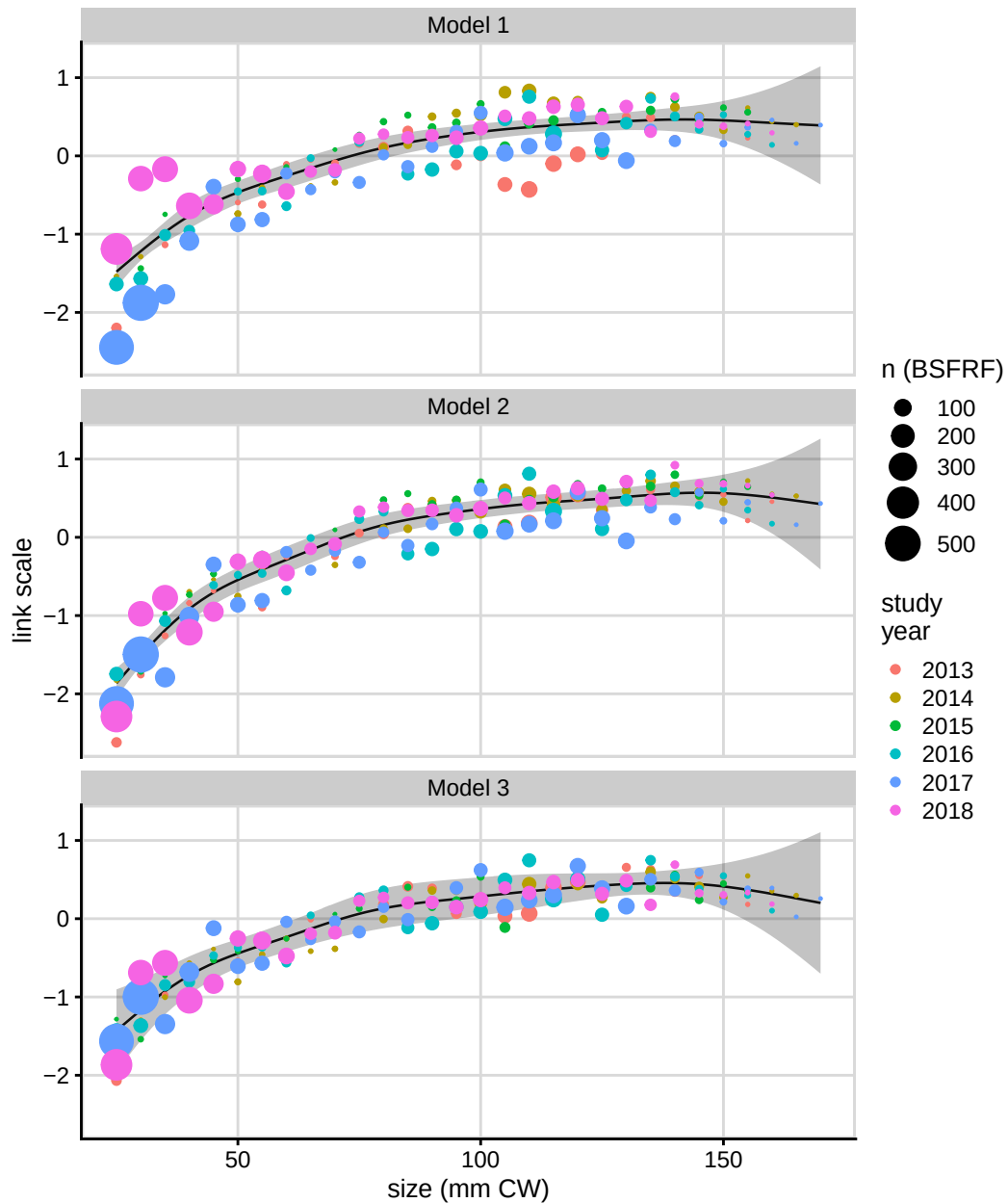


Figure 42. Partial residuals to the smooth term $s(z)$ (solid line; shading: 90% confidence interval) by size and year (the latter as a factor) on the link (natural logarithm) scale. The area of each residual is scaled to the number of males caught at size by the BSFRF gear. There is no term for “non-study” years.

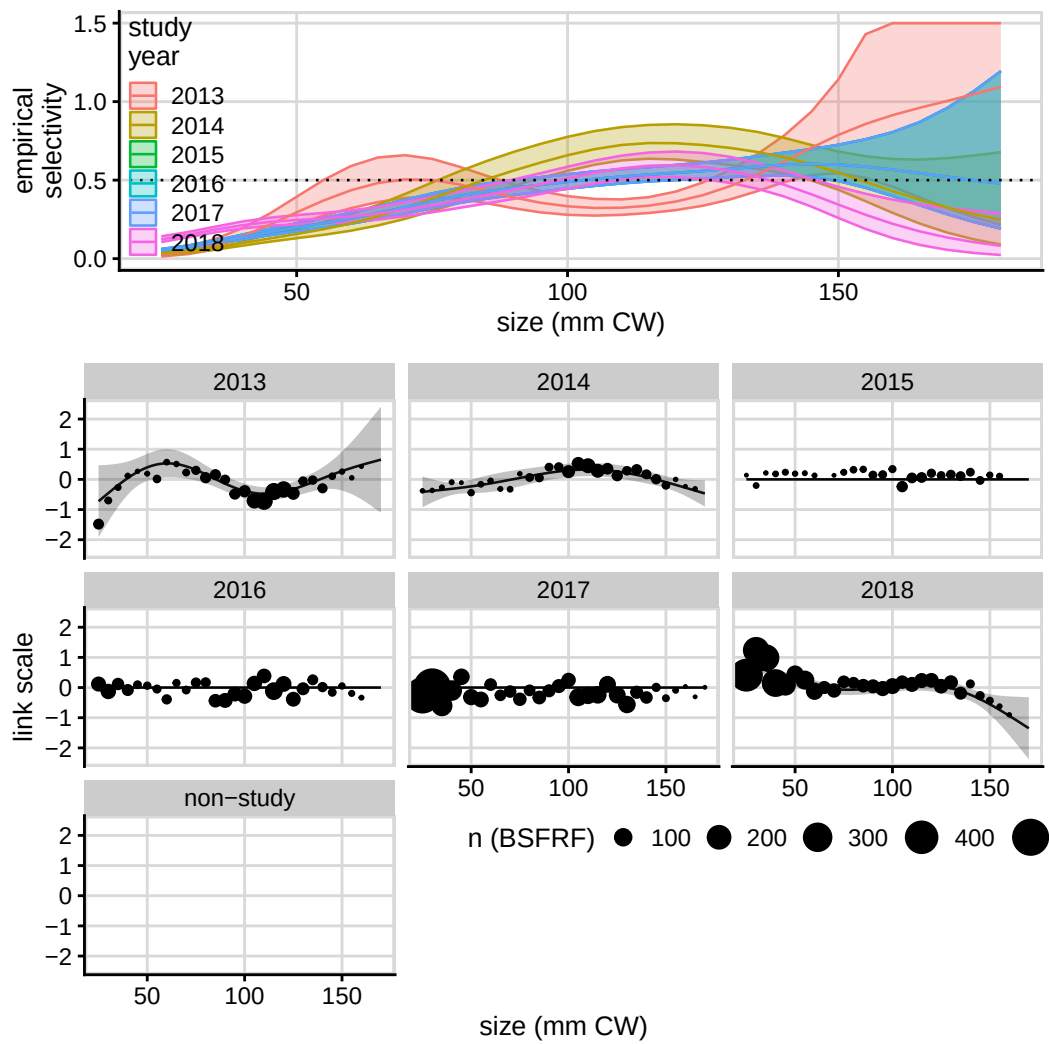


Figure 43. Estimated survey selectivity for males from Model 2. Upper plot: annual estimates by study year. Lower plot: factor smooth interaction terms $t_i(z, y, bs = \text{single } sz \text{ single})$ (solid line; shading: 90% confidence interval) by size and year (the latter as a factor) on the link (natural logarithm) scale. The area of each residual is scaled to the number of males caught at size by the BSFRF gear.

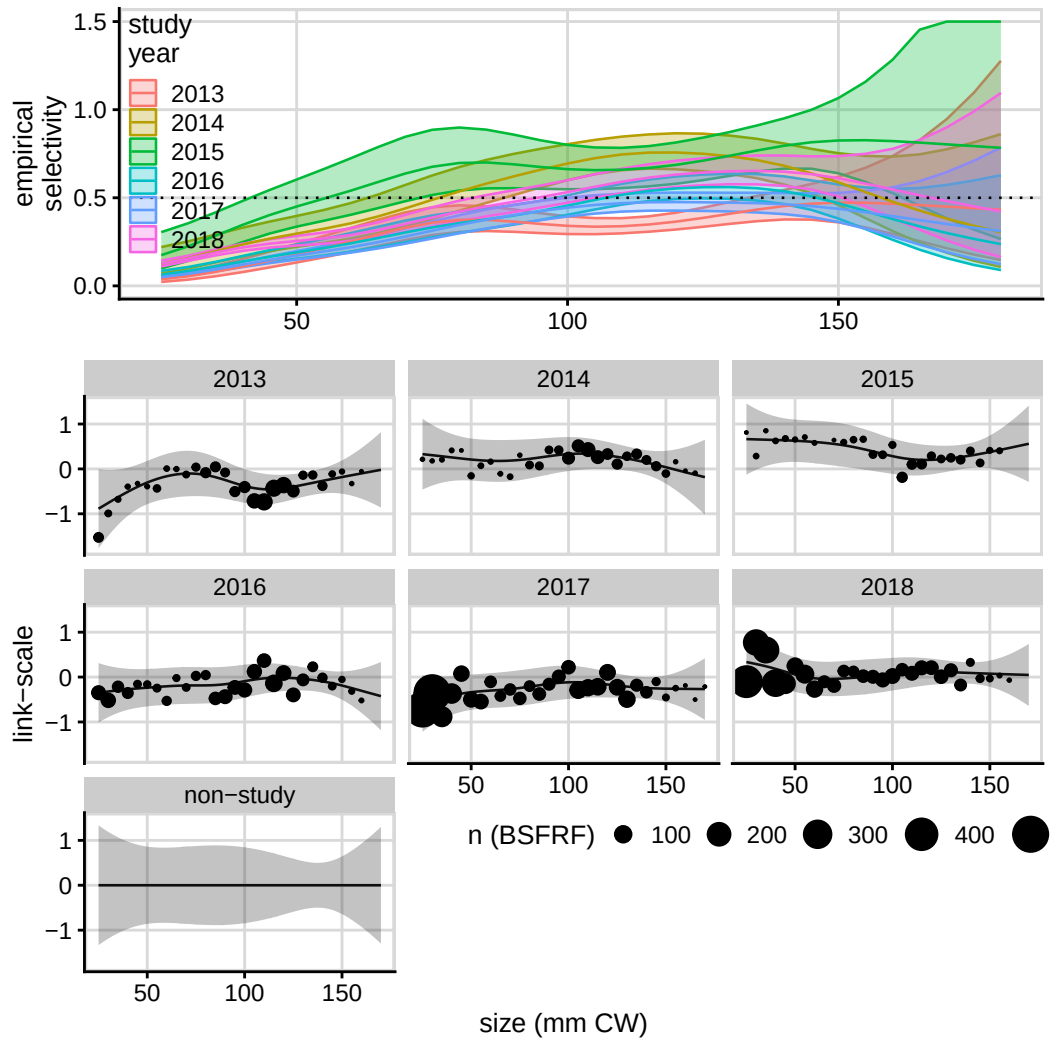


Figure 44. Estimated survey selectivity for males from Model 3. Upper plot: annual estimates by study year. Lower plot: random factor smooth interaction terms $s(z, y, bs = \text{fs})$ (solid line; shading: 90% confidence interval) by size and year (the latter as a factor) on the link (natural logarithm) scale. The area of each residual is scaled to the number of males caught at size by the BSFRF gear. 'non-study' denotes the link-scale term for any non-study year, reflecting the overall uncertainty associated with unobserved random effects.

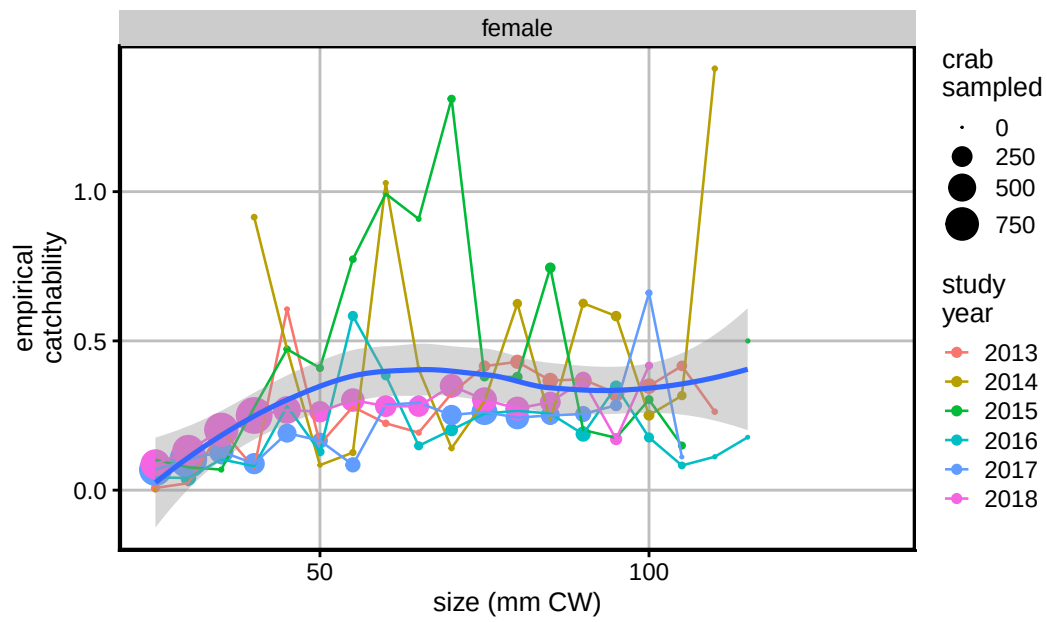


Figure 45. Comparison of annual empirical estimates of NMFS survey selectivity for female Tanner crab. An unweighted smoothed trend and simple uncertainty interval are indicated by the thick blue line and grey shading.

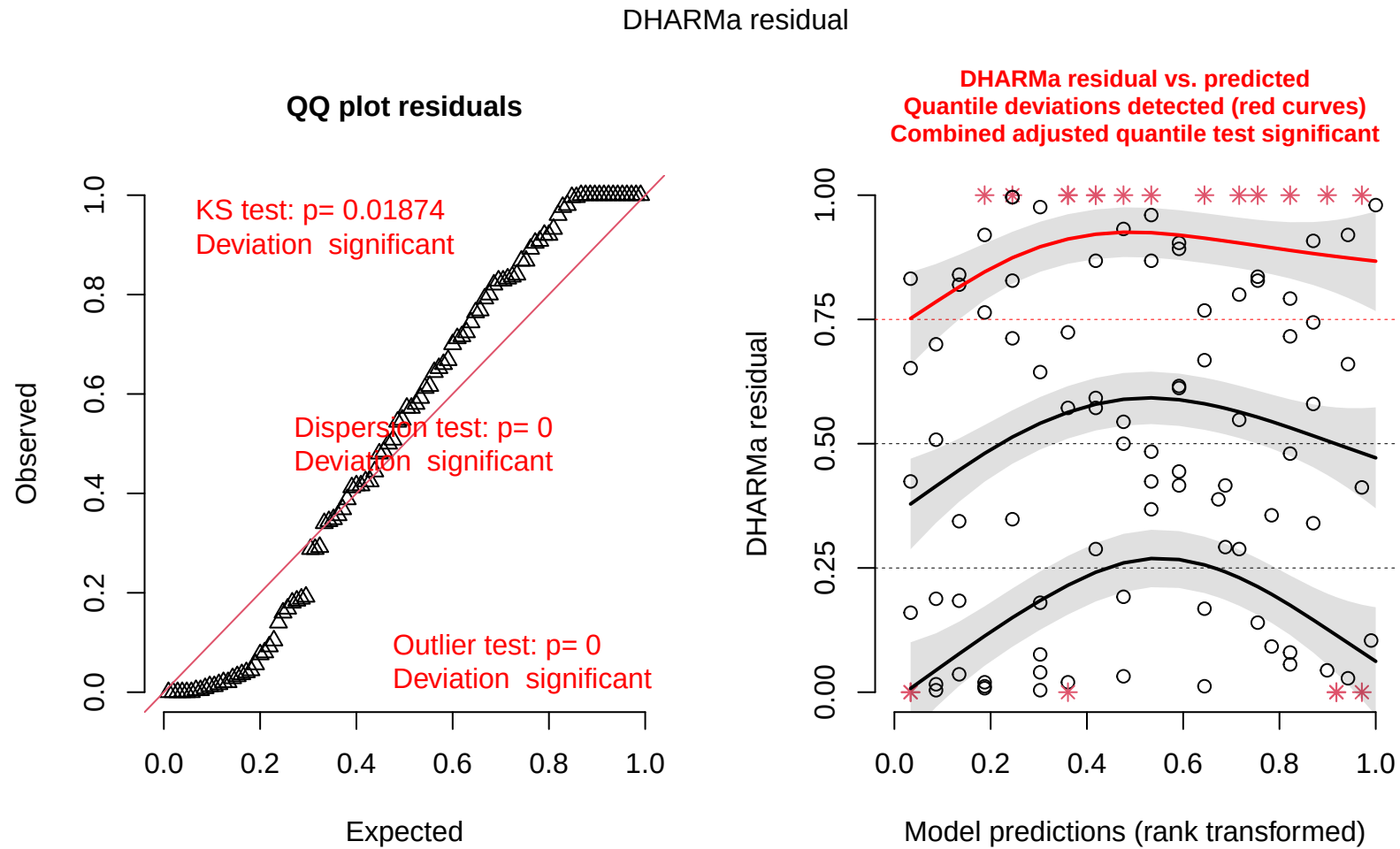


Figure 46. DHARMA residuals from Model 1 for females. Left: quantile-quantile plot for observed and expected residuals under the null hypothesis. Right: quantile deviations for residuals.

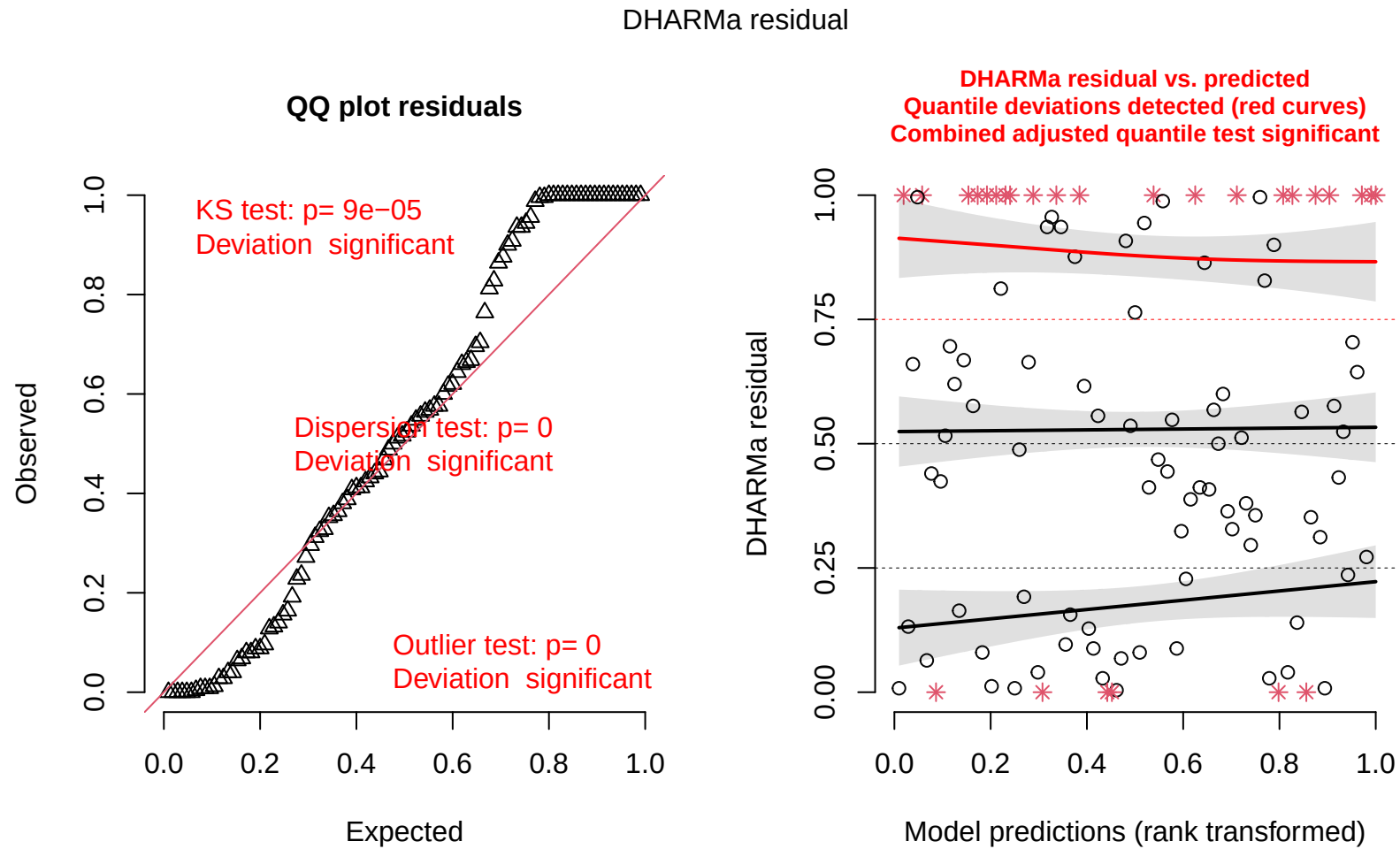


Figure 47. DHARMa residuals from Model 2 for females. Left: quantile-quantile plot for observed and expected residuals under null hypothesis. Right: quantile deviations for residuals.

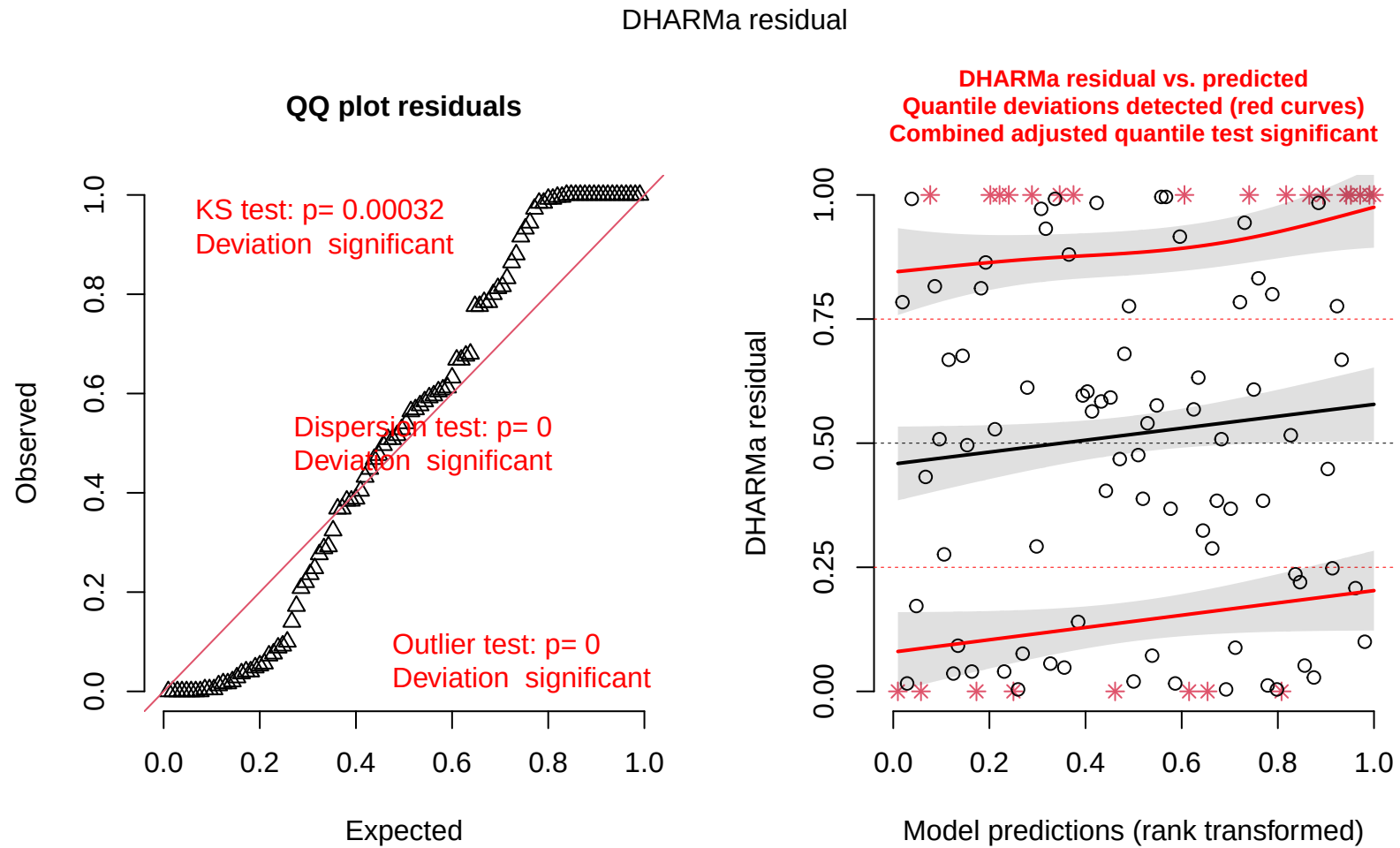


Figure 48. DHARMa residuals from Model 3 for females. Left: quantile-quantile plot for observed and expected residuals under the null hypothesis. Right: quantile deviations for residuals.

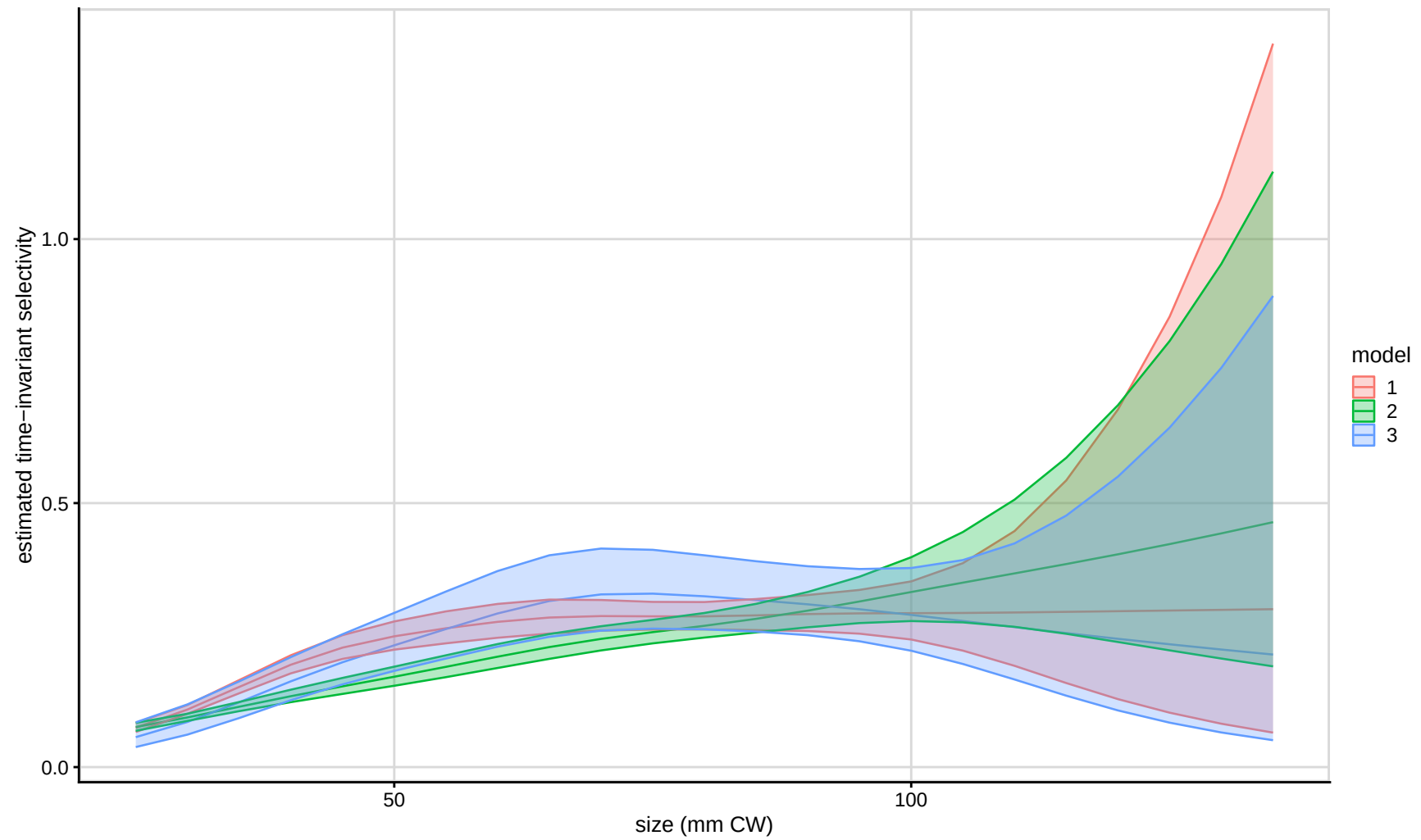


Figure 49. Comparison of the estimated time-invariant estimated selectivity, $S(z)$, for females from the three models.

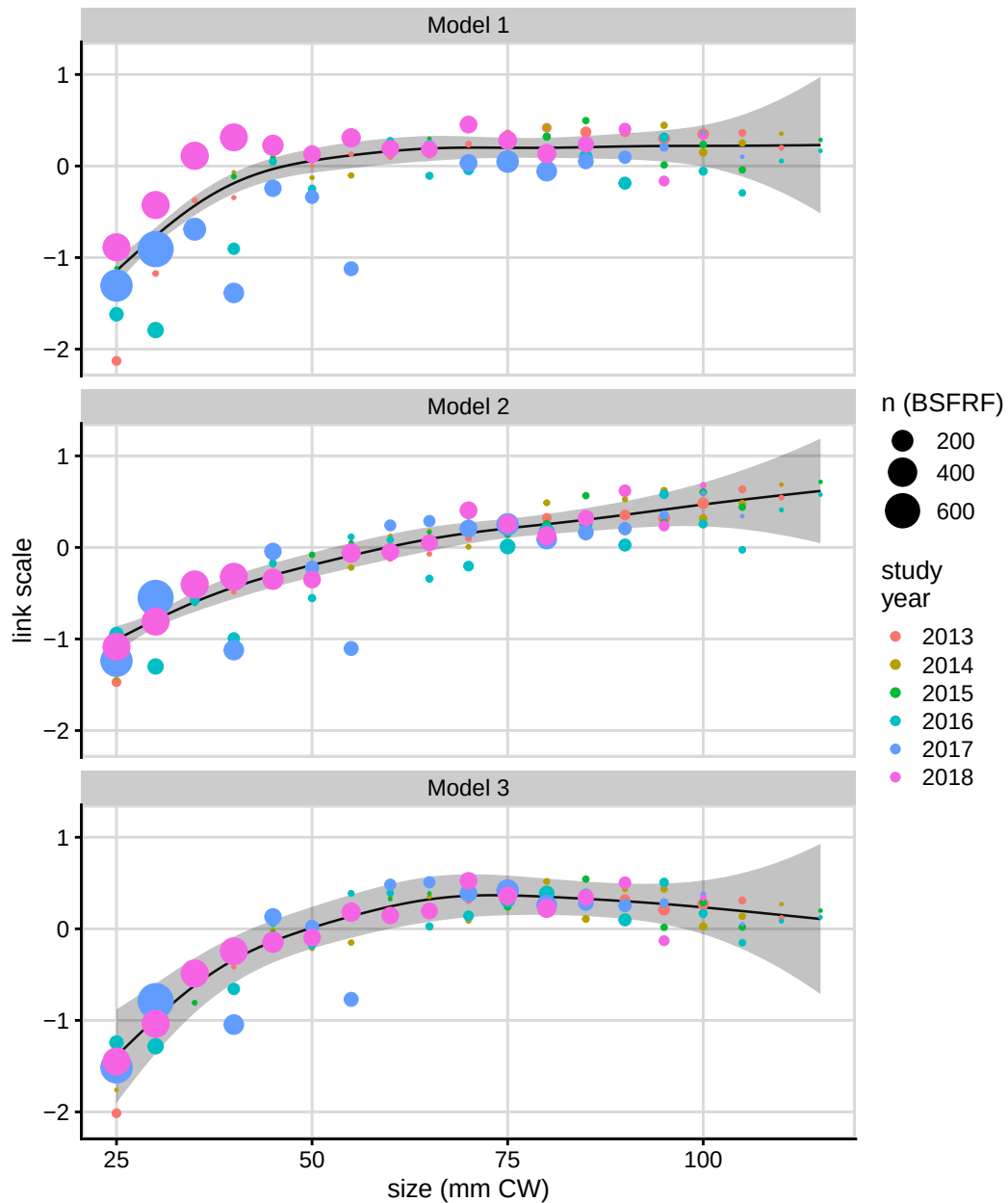


Figure 50. Partial residuals to the smooth term $s(z)$ (solid line; shading: 90% confidence interval) by size and year (the latter as a factor) on the link (natural logarithm) scale. The area of each residual is scaled to the number of females caught at size by the BSFRF gear. There is no term for "non-study" years.

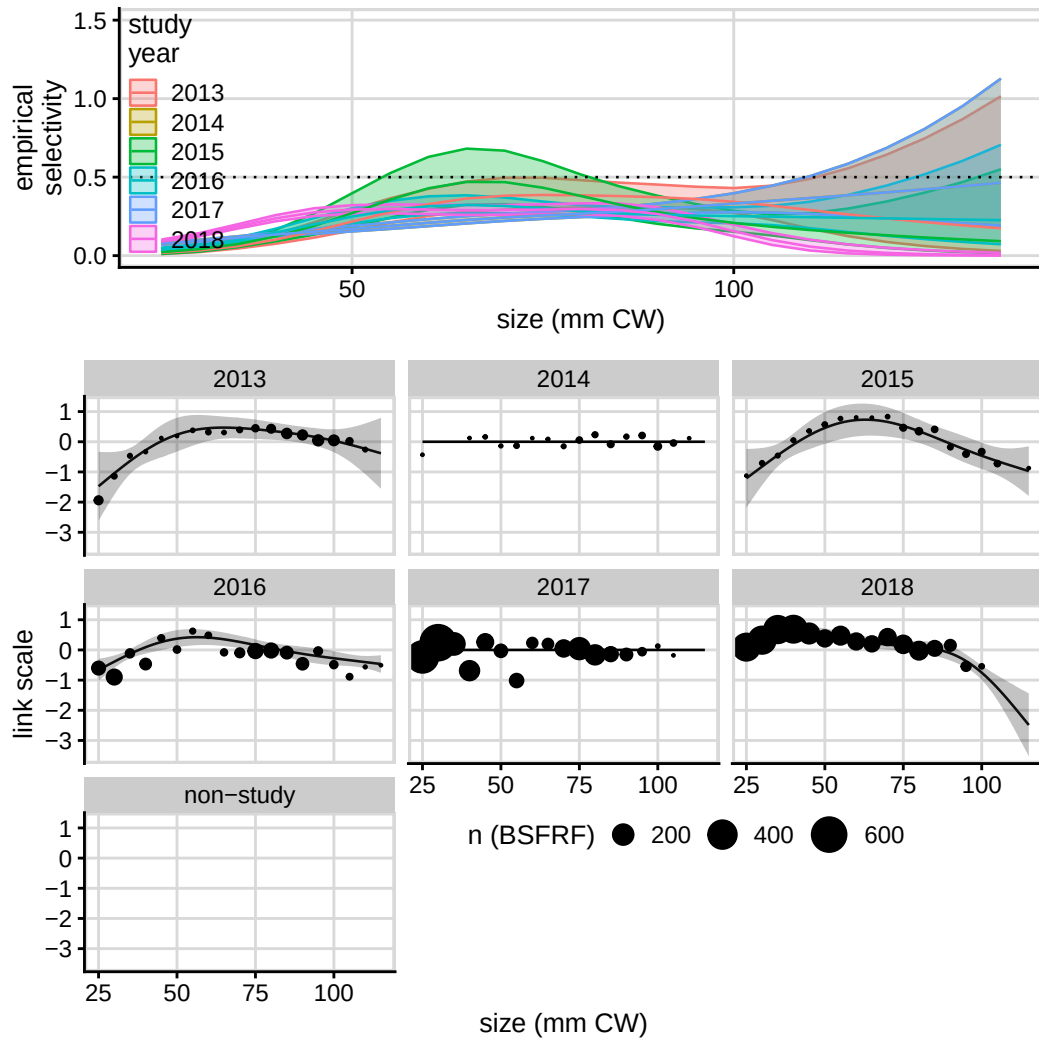


Figure 51. Estimated survey selectivity for females from Model 2. Upper plot: annual estimates by study year. Lower plot: factor smooth interaction terms $\text{ti}(z, y, \text{bs} = \text{single } sz \text{ single})$ (solid line; shading: 90% confidence interval) by size and year (the latter as a factor) on the link (natural logarithm) scale. The area of each residual is scaled to the number of females caught at size by the BSFRF gear.

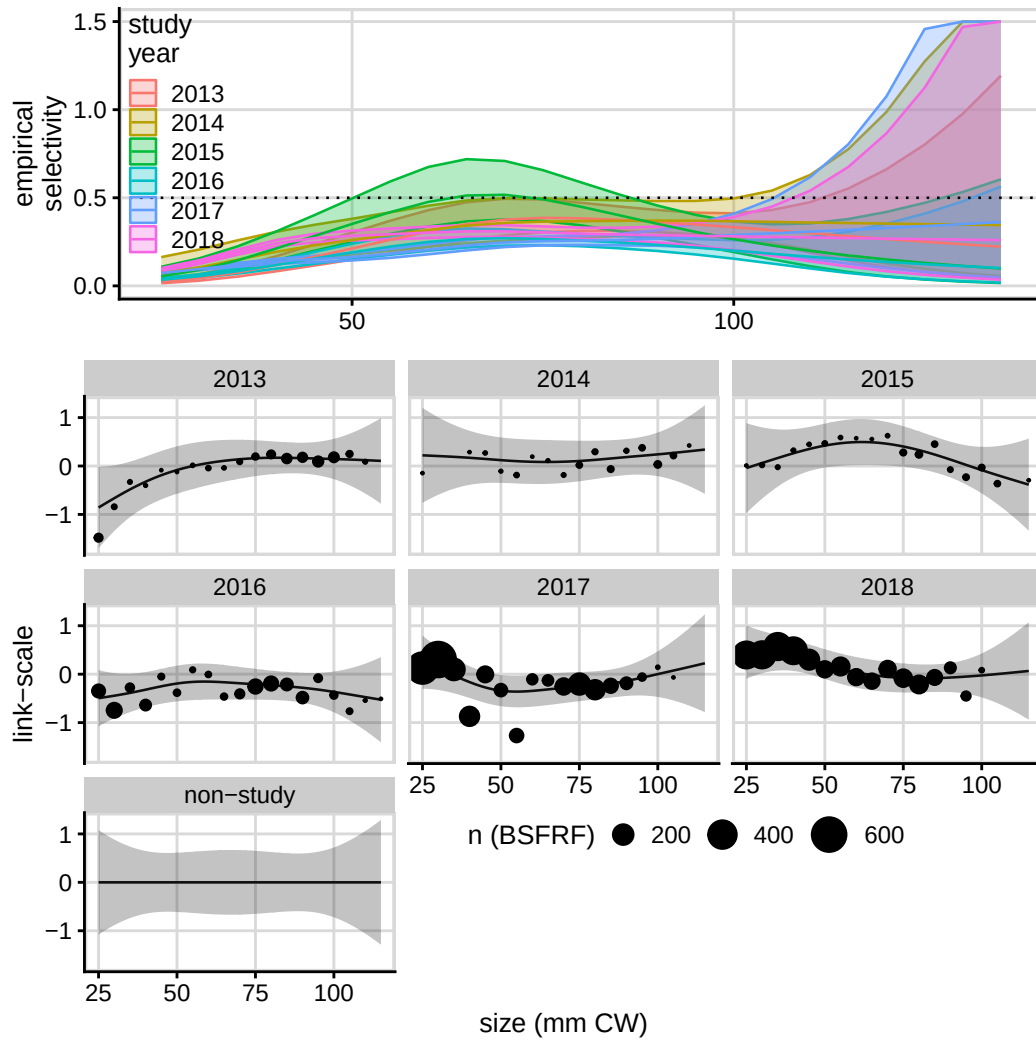


Figure 52. Estimated survey selectivity for females from Model 3. Upper plot: annual estimates by study year. Lower plot: random factor smooth interaction terms $s(z, y, bs = \text{fs})$ (solid line; shading: 90% confidence interval) by size and year (the latter as a factor) on the link (natural logarithm) scale. The area of each residual is scaled to the number of females caught at size by the BSFRF gear. 'non-study' denotes the link-scale term for any non-study year, reflecting the overall uncertainty associated with unobserved random effects.

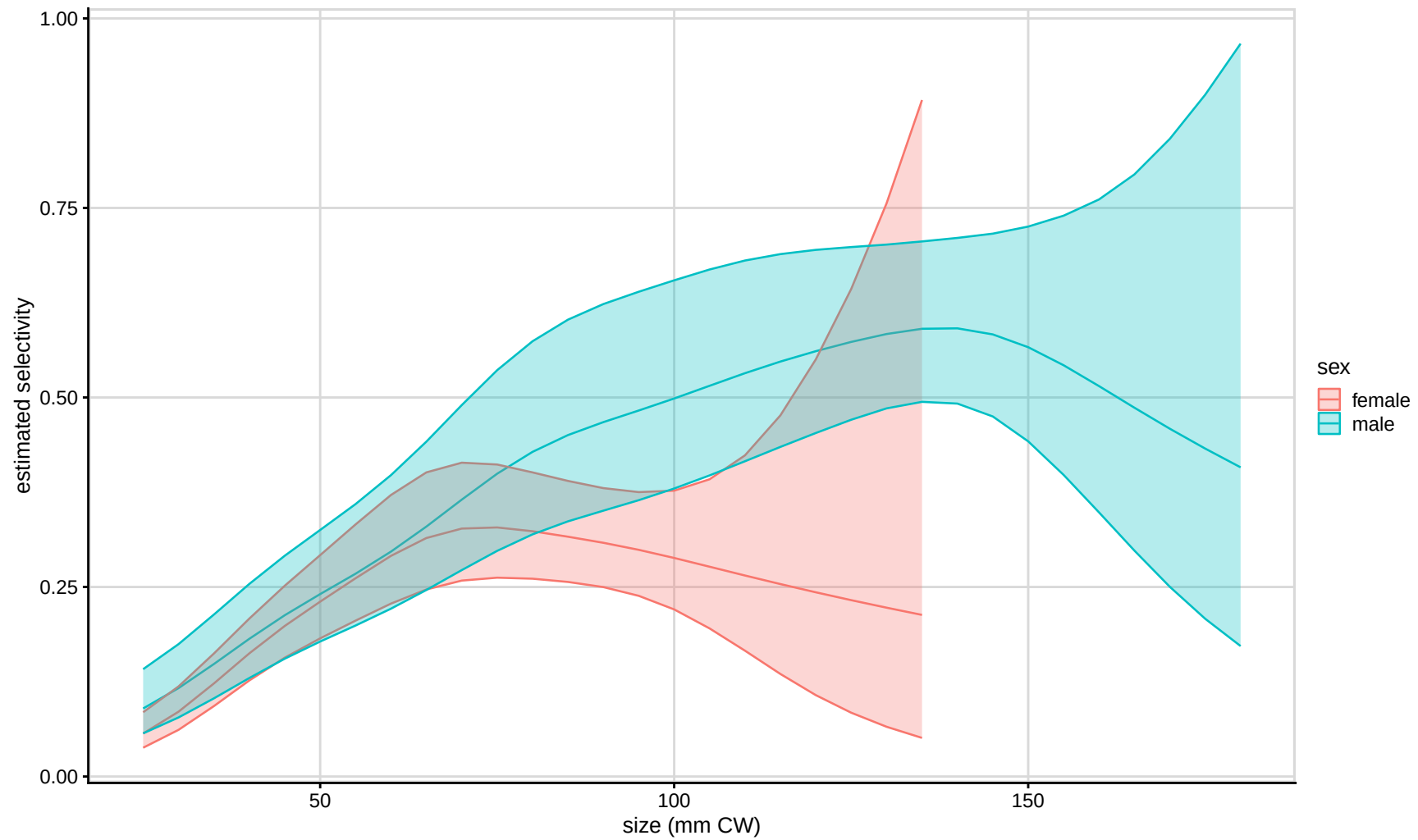


Figure 53. Estimated selectivity for arbitrary years, $S'(z)$, from the model regarded as the most satisfactory for each sex (Model 3, in both cases).

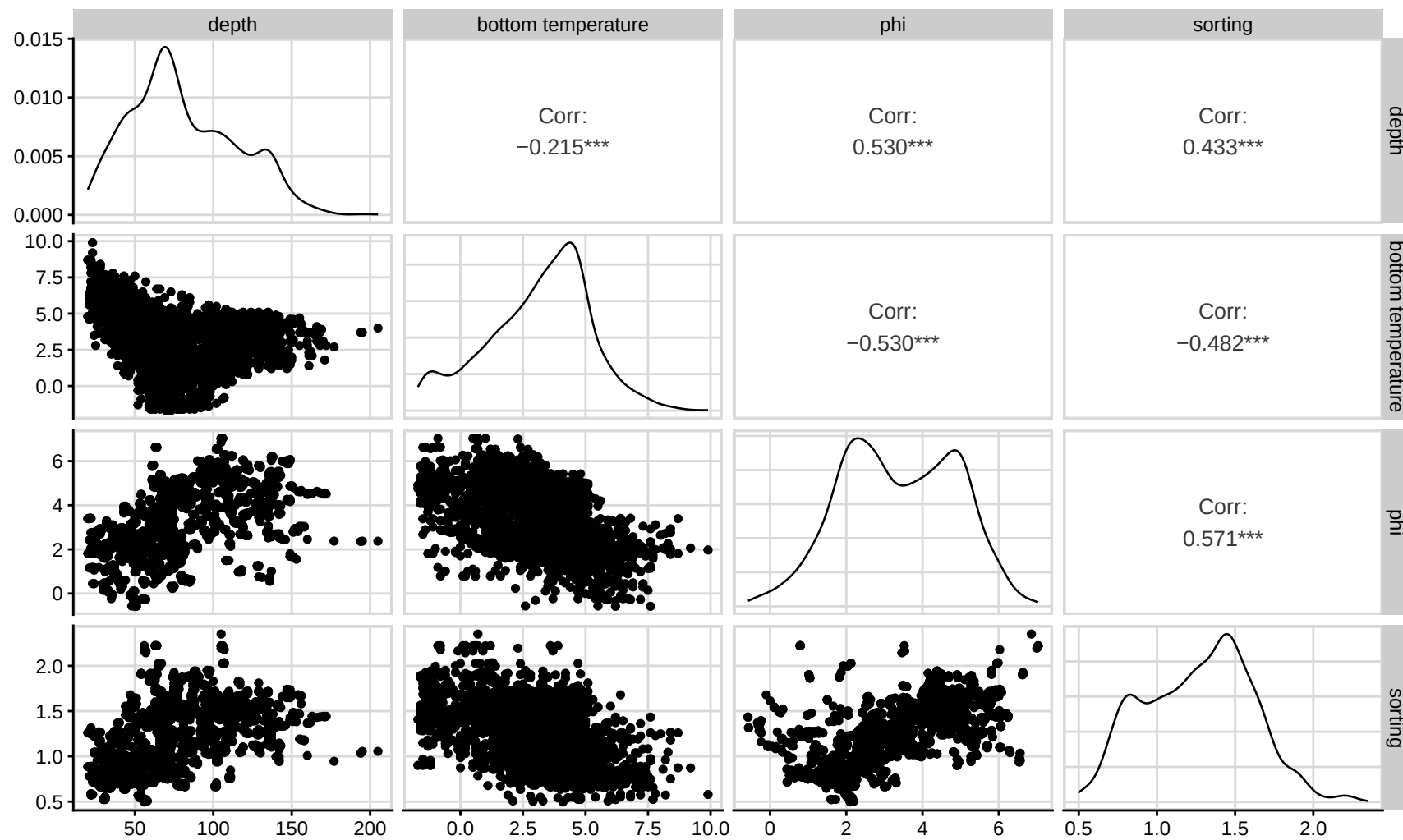


Figure 54. Pairs plot showing correlations among depth, bottom temperature, and interpolated mean grain size ('phi', in phi units) and sorting coefficient sediment characteristics at NMFS SBS haul locations (2013-2018).

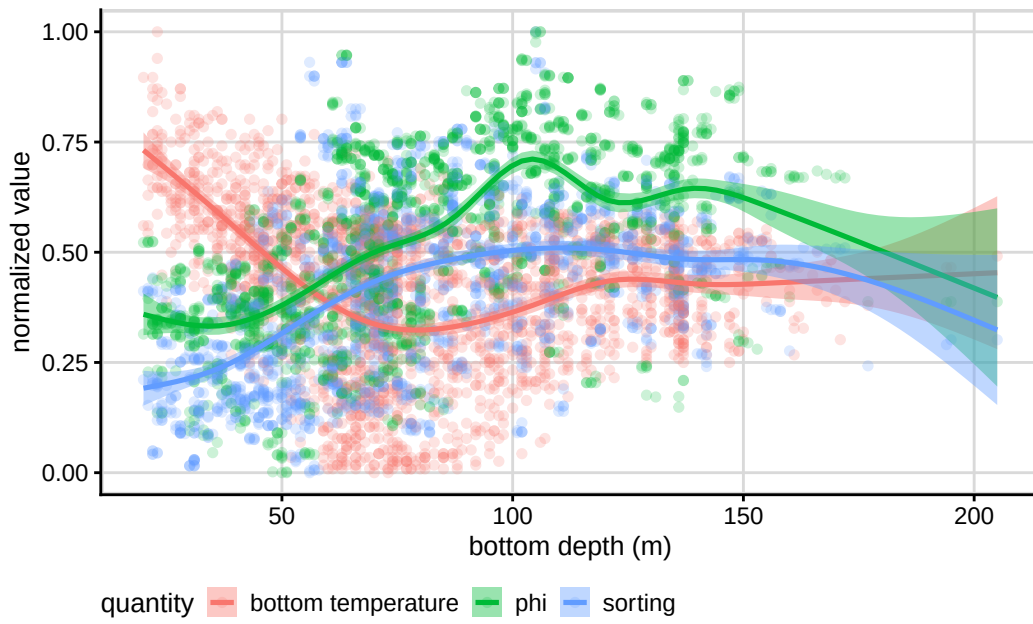


Figure 55. Relative distributions of bottom temperature, interpolated mean grain size ('phi', in phi units), and sorting coefficient with haul depth at NMFS SBS haul locations (2013-2018).

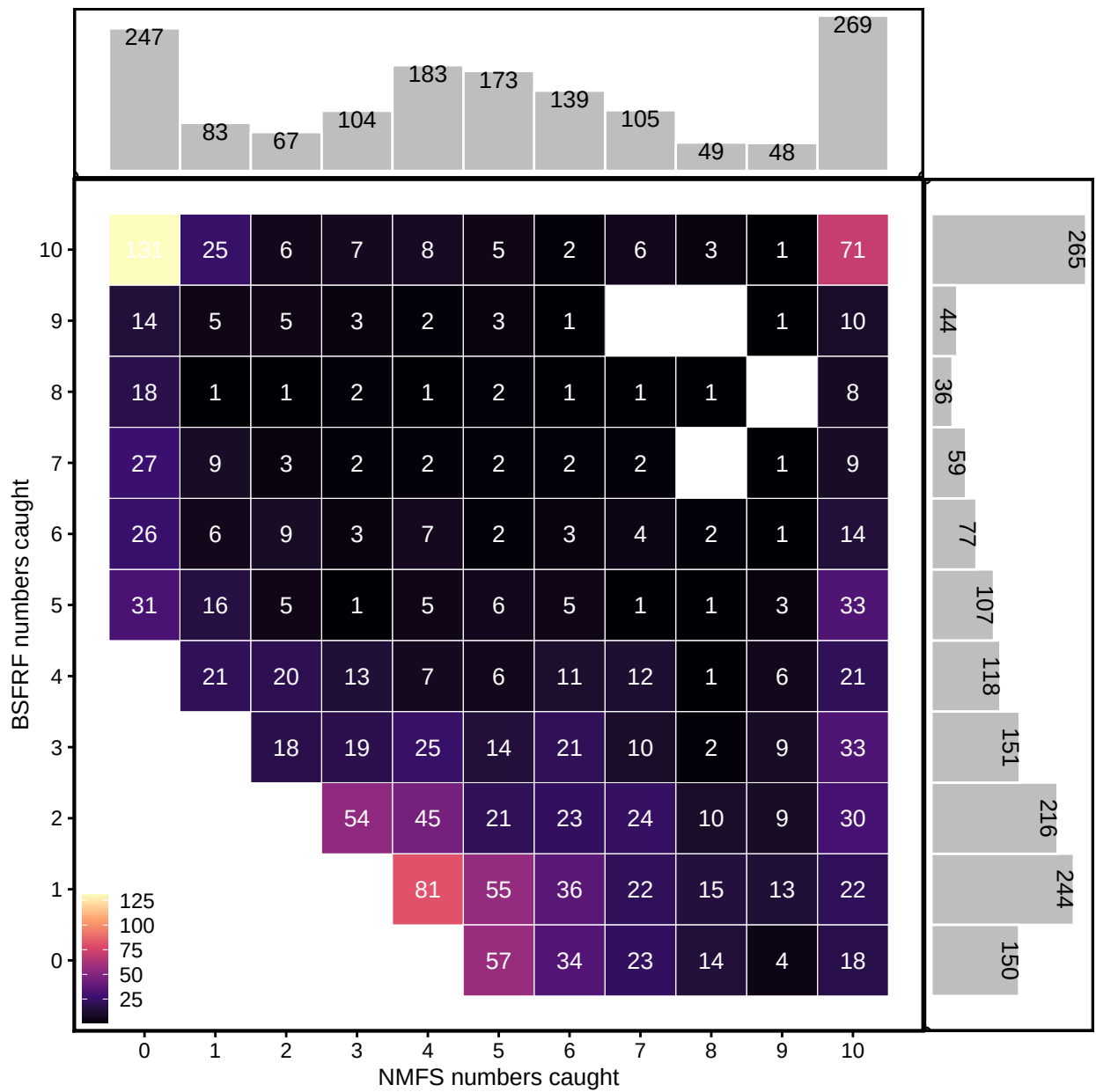


Figure 56. Joint histogram of numbers of males caught per size bin per SBS haul. Catch numbers > 10 have been aggregated into the number = 10 bin. Any size bin/haul with less than 5 males caught between both gear types have been excluded.

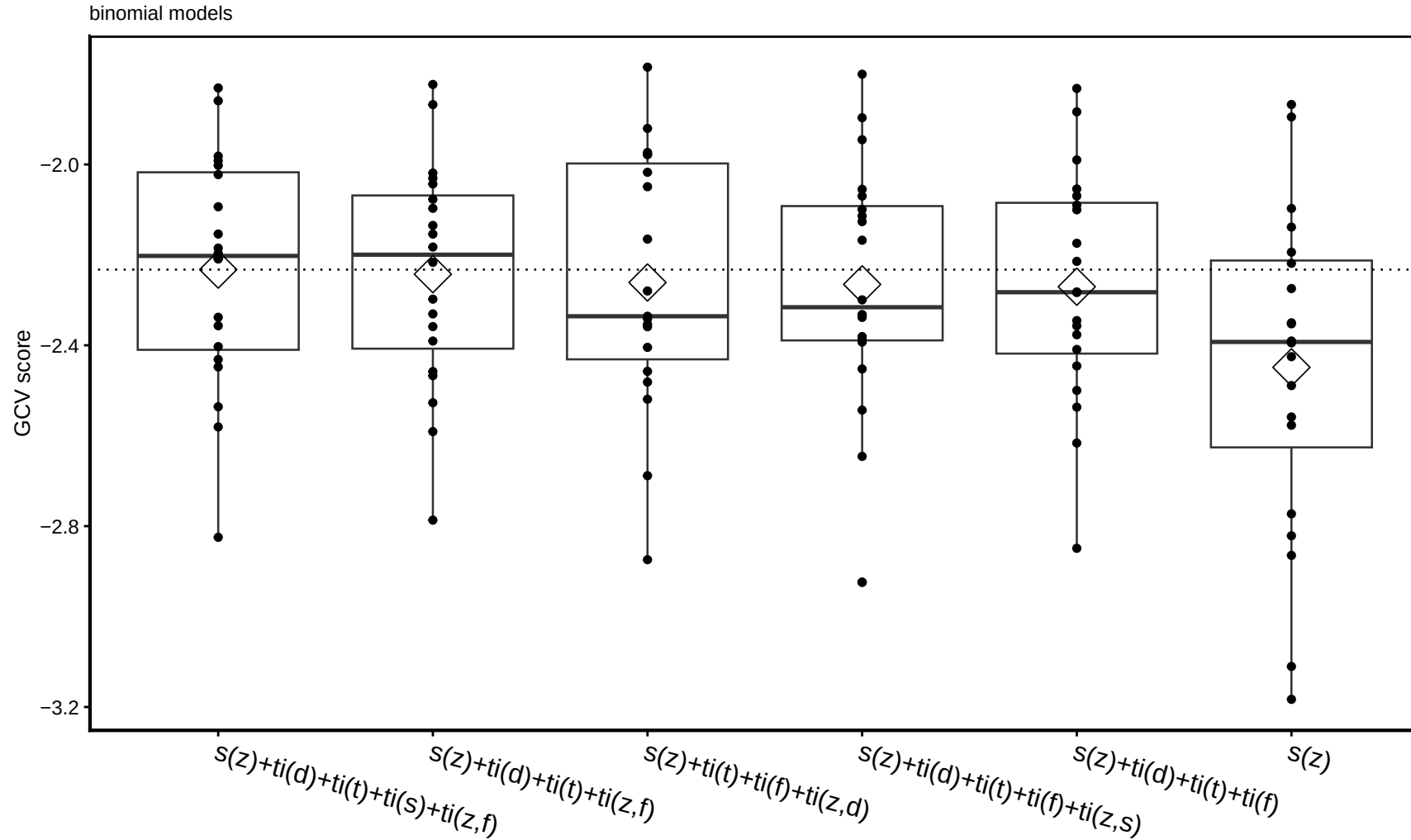


Figure 57. Scores from the top-ranking binomial models, plus the base model, for males. Diamond: mean score; points: fold-specific scores; box: box-and-whiskers plot indicating median and first and third quantiles; dotted line indicate the best mean score. The model with smooth terms ' $s(z)+ti(d)+ti(t)+ti(s)+ti(z,f)$ ' was considered to be the best, based on selecting the simplest 'equivalent' model after a comparison of the mean and range of the scores across the models.

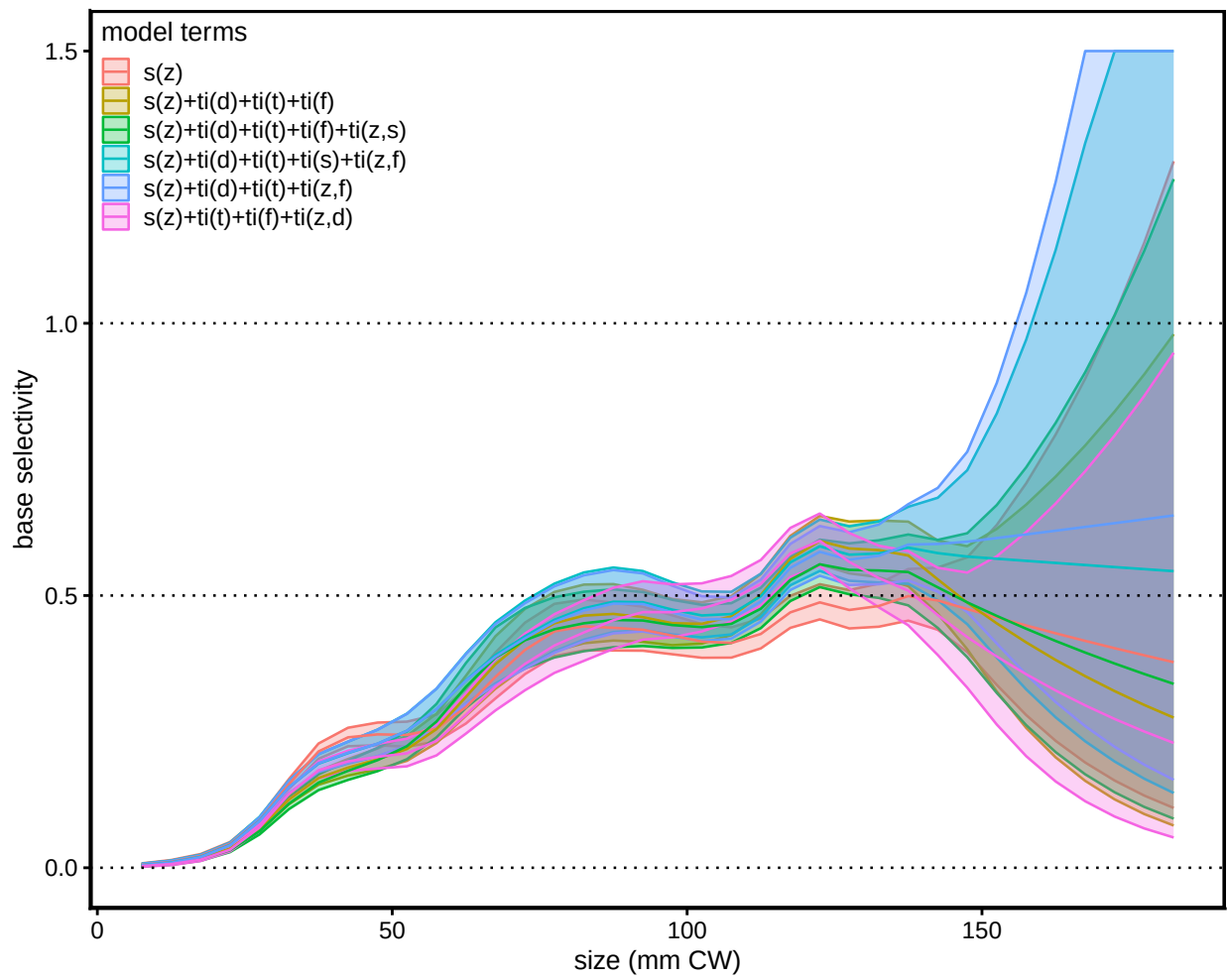


Figure 58. Base selectivity (no covariate effects) estimated from the top-ranking binomial models for males.

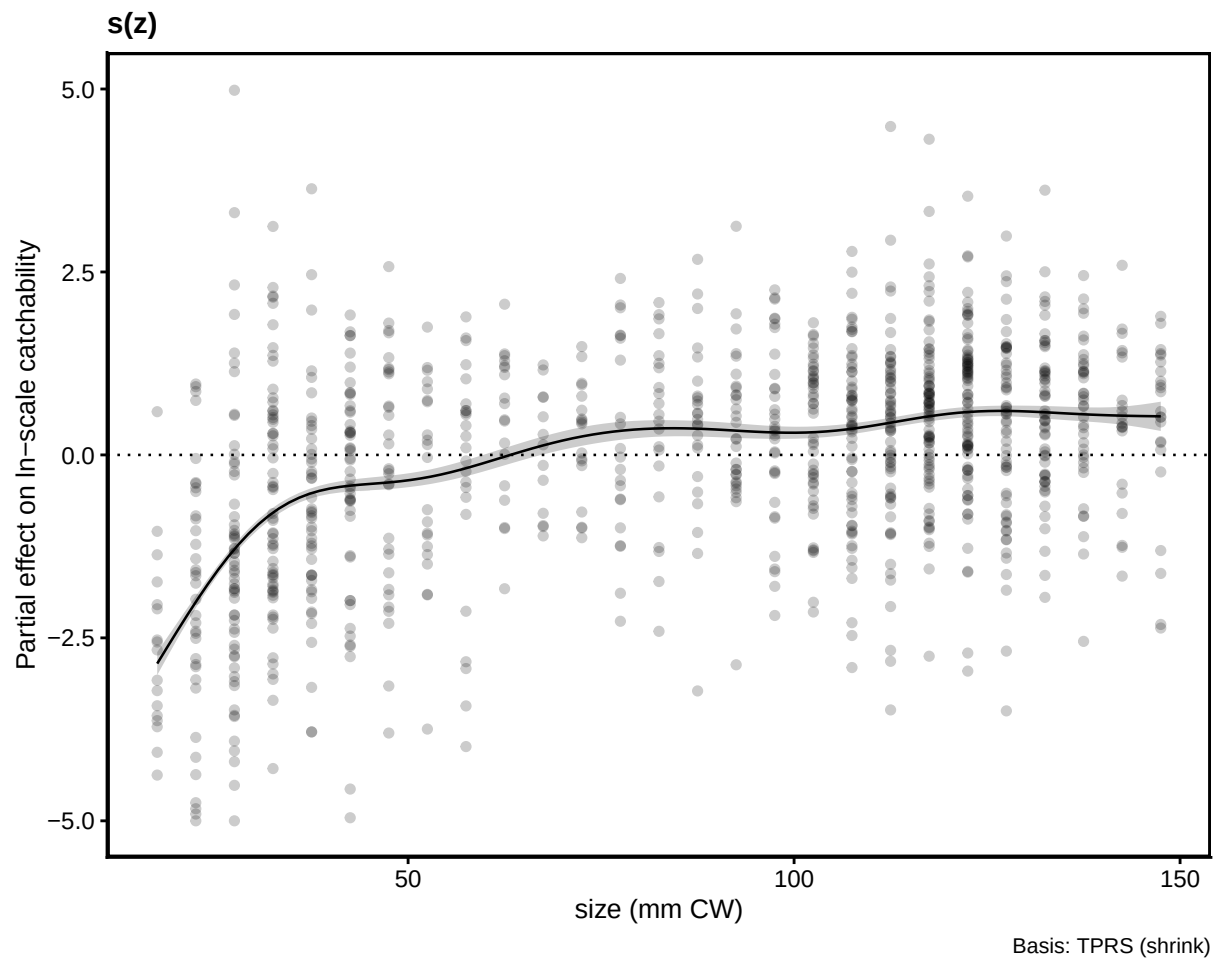


Figure 59. Partial effect of size (mm CW) on ln-scale catchability. Partial residuals for the response 'p' are shown on the link scale of the model, based on its binomial error distribution and logit link. Residuals beyond y-axis bounds are plotted at the bound.

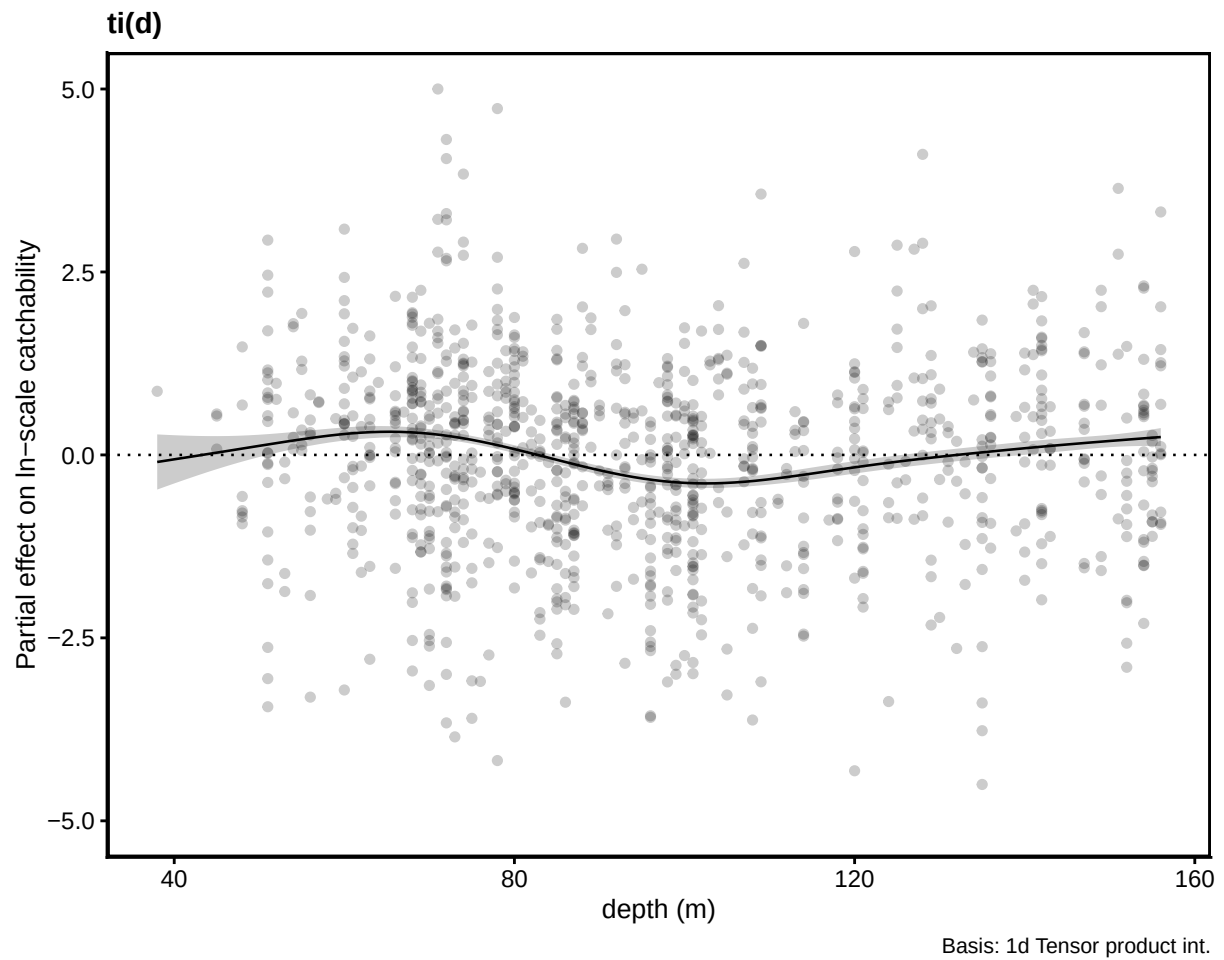


Figure 60. Partial effect of depth (m) on ln-scale catchability. Partial residuals for the response 'p' are shown on the link scale of the model, based on its binomial error distribution and logit link. Residuals beyond y-axis bounds are plotted at the bound.

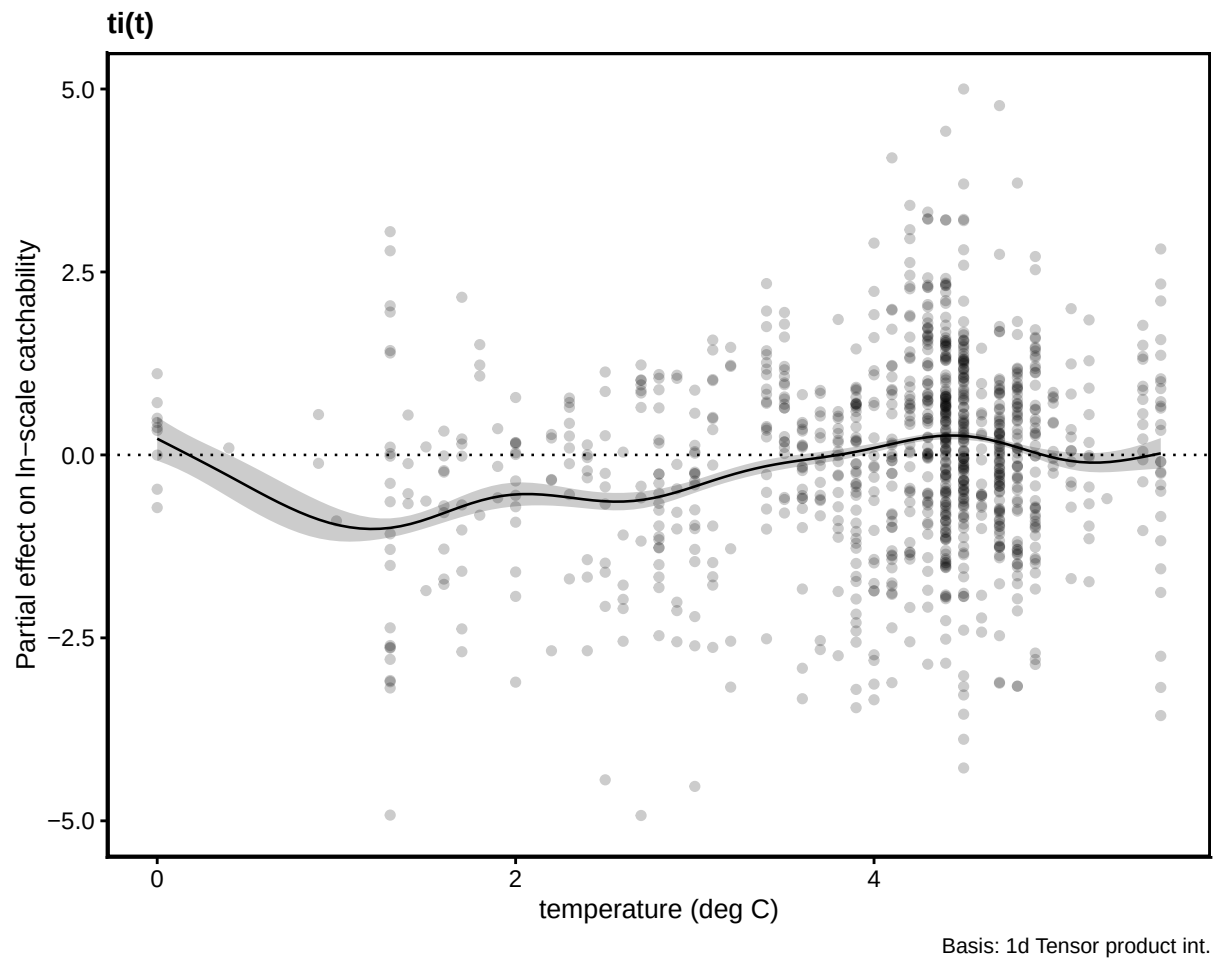


Figure 61. Partial effect of temperature (deg C) on ln-scale catchability. Partial residuals for the response 'p' are shown on the link scale of the model, based on its binomial error distribution and logit link. Residuals beyond y-axis bounds are plotted at the bound.

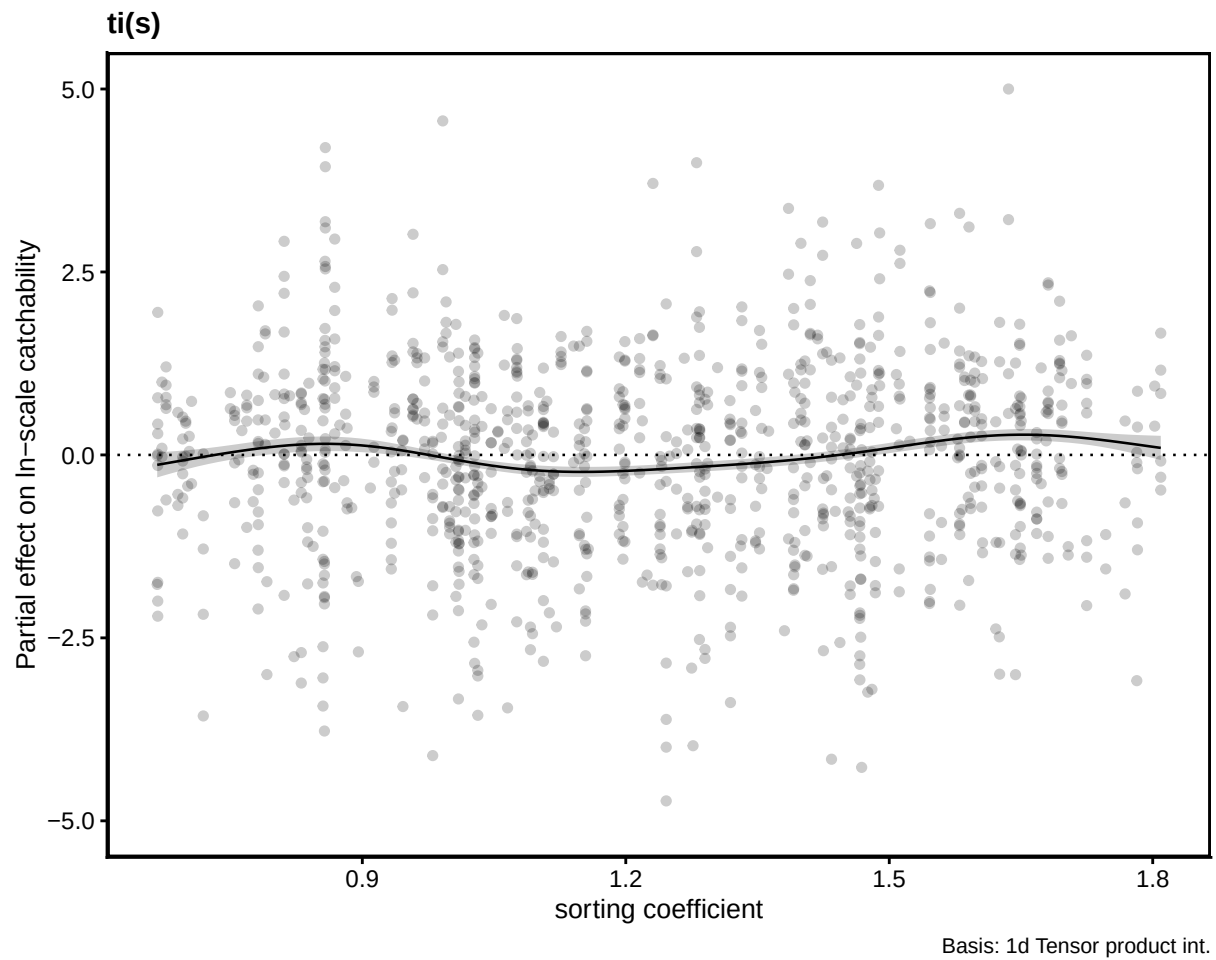


Figure 62. Partial effect of sorting coefficient on ln-scale catchability. Partial residuals for the response 'p' are shown on the link scale of the model, based on its binomial error distribution and logit link. Residuals beyond y-axis bounds are plotted at the bound.

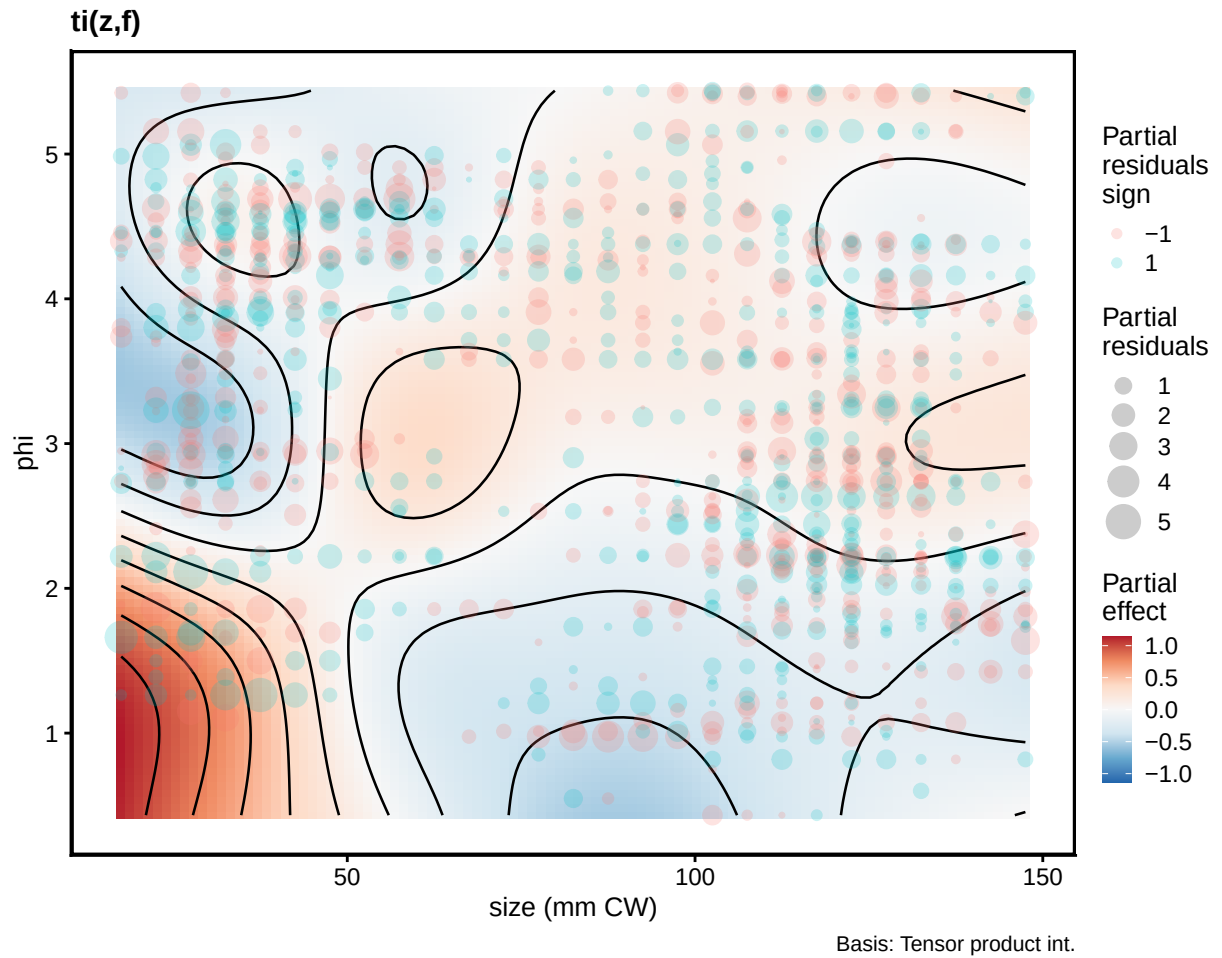


Figure 63. Partial effects of size (mm CW) and phi on ln-scale catchability. Partial residuals for the response 'p' are shown on the link scale of the model, based on its binomial error distribution and logit link. Residuals beyond y-axis bounds are plotted at the bound.

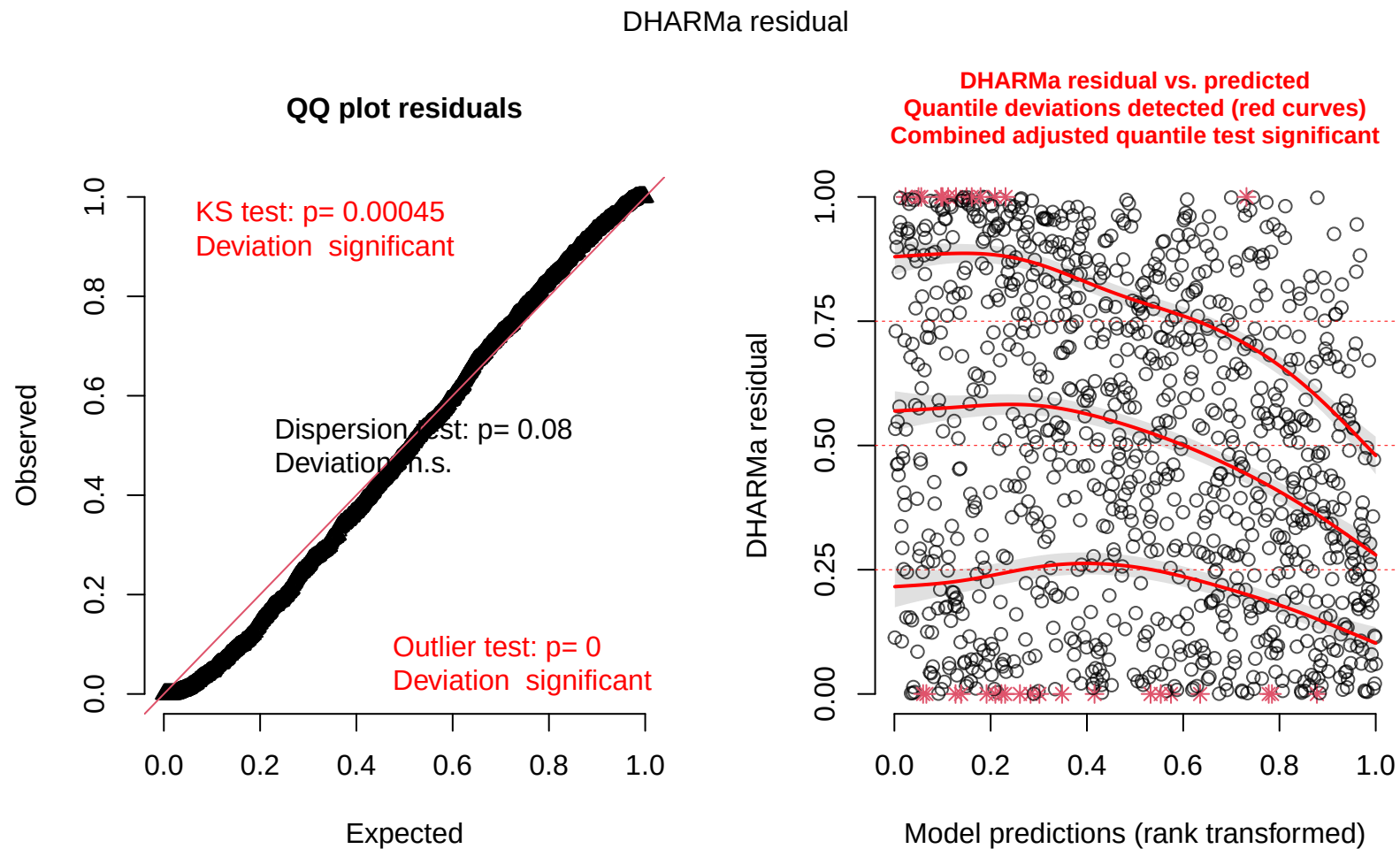


Figure 64. DHARMa residuals from the 'best' binomial model for males. Left: quantile-quantile plot for observed and expected residuals under the null hypothesis. Right: quantile deviations for residuals.

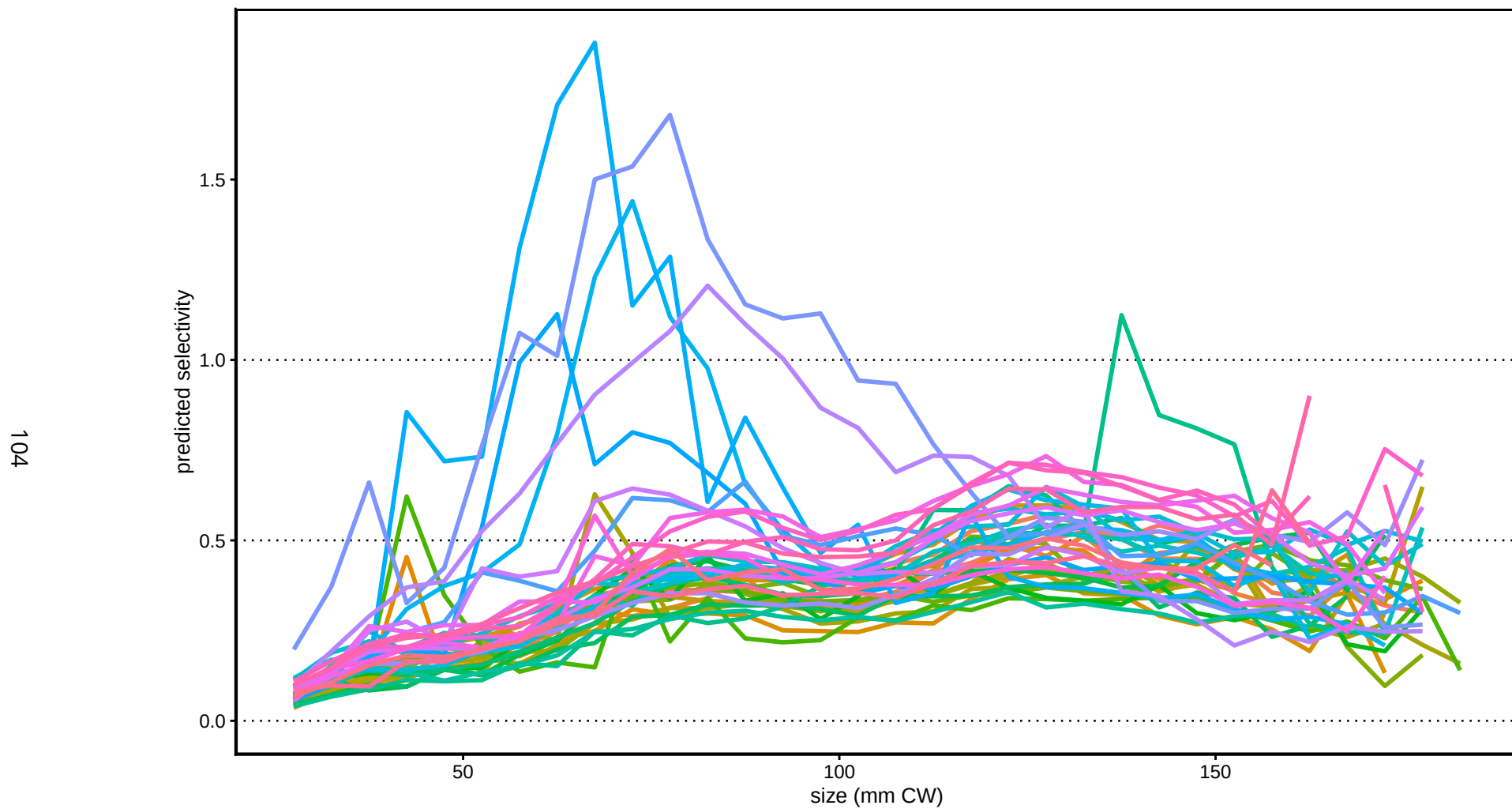


Figure 65. Haul-specific and annual mean catchability curves for males, weighted by size-specific CPUE, estimated using the best binomial model with covariates.

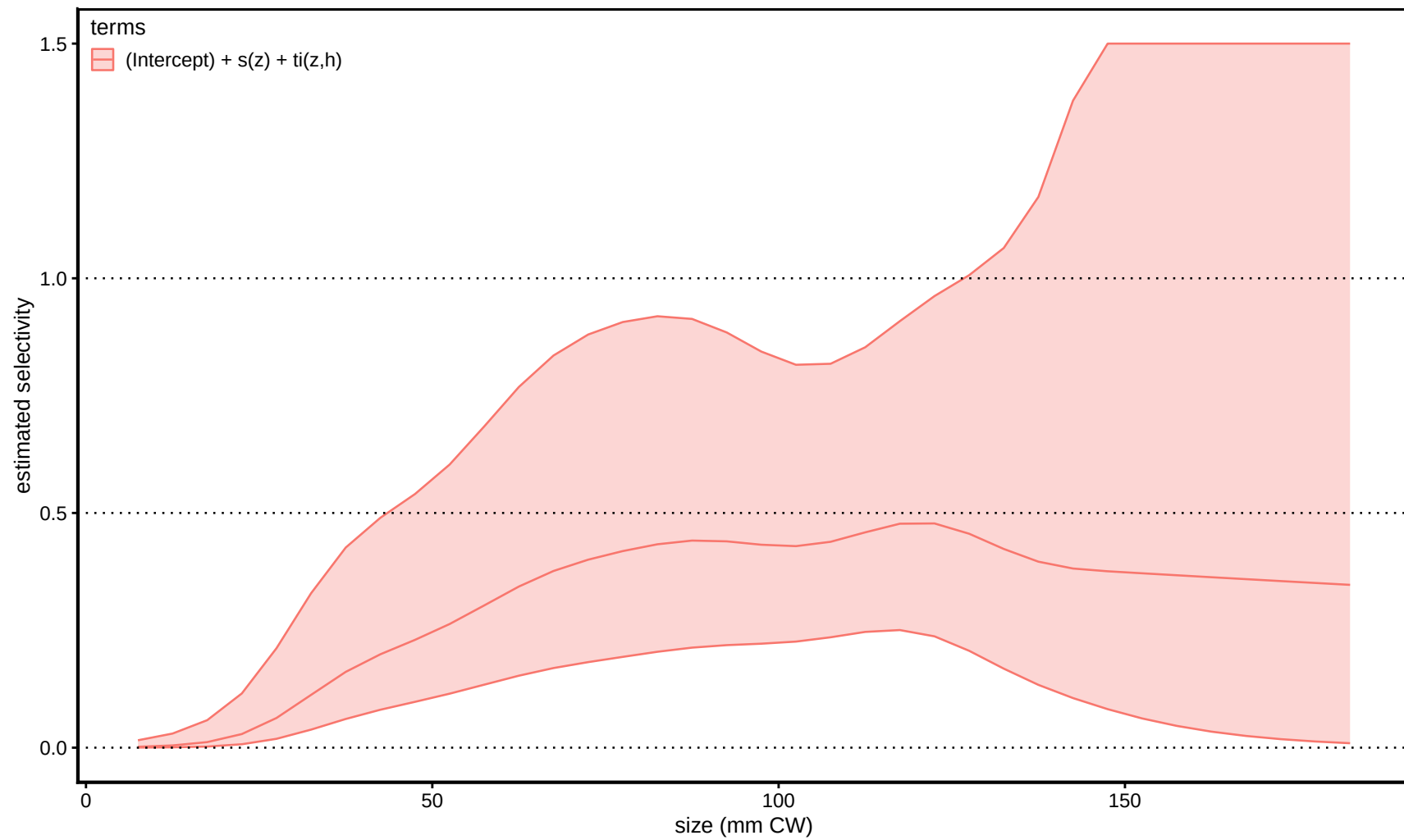


Figure 66. Estimated haul-level catchability for males based on the random effects binomial model (no covariates).

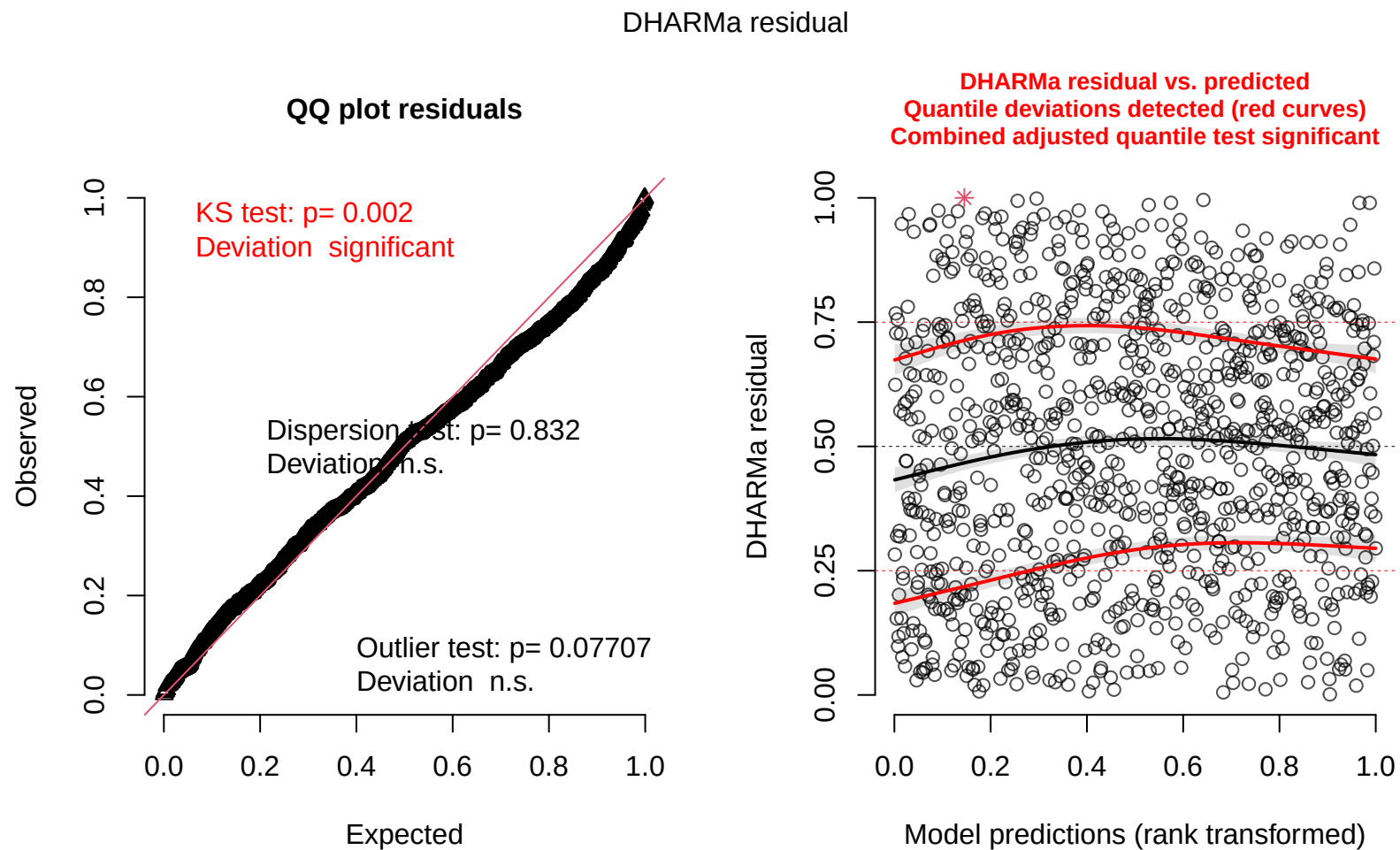


Figure 67. DHARMA residuals from the random effects binomial model for males. Left: quantile-quantile plot for observed and expected residuals under the null hypothesis. Right: quantile deviations for residuals.

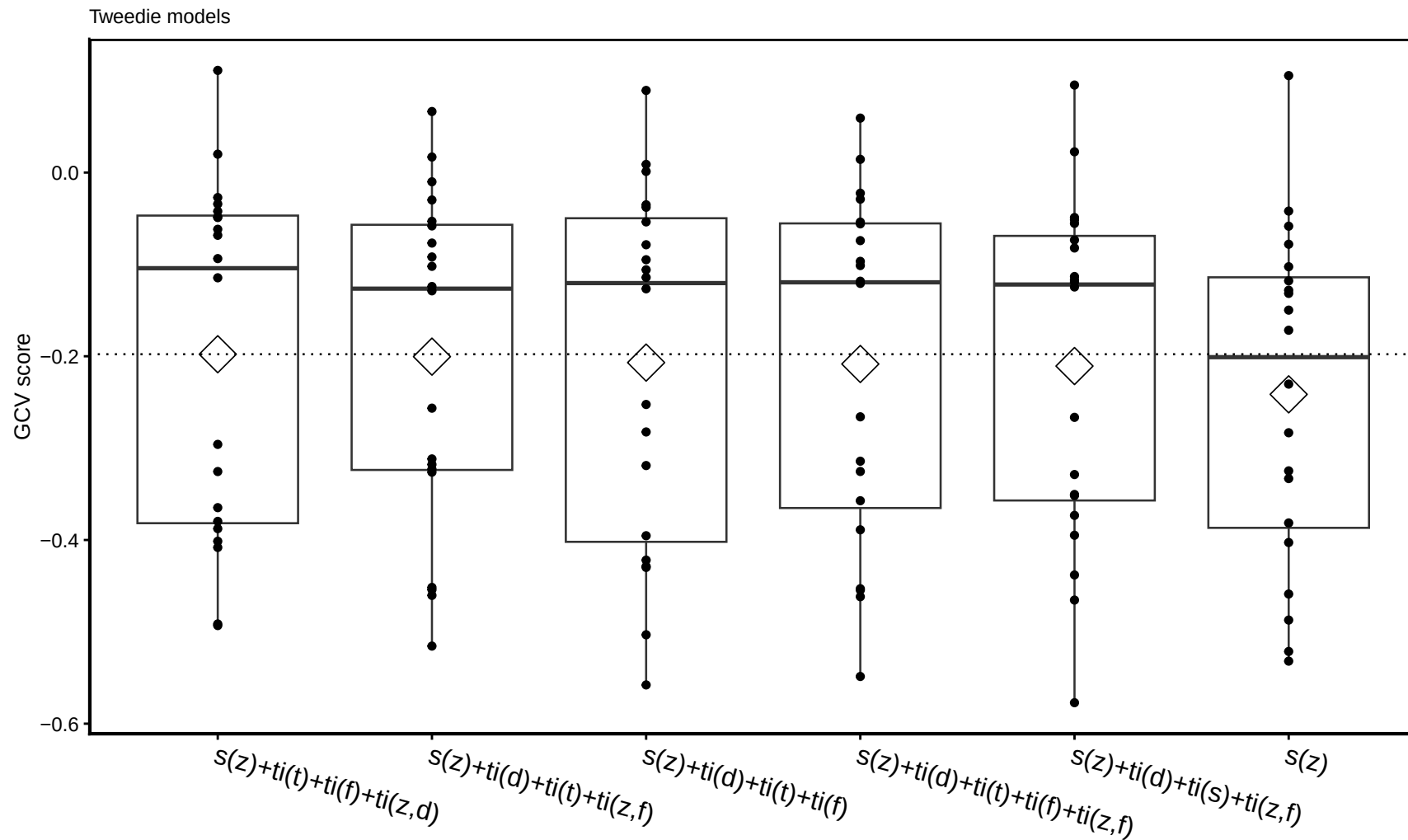


Figure 68. Scores from the top-ranking Tweedie models for males, plus the base model. Diamond: mean score; points: fold-specific scores; box: box-and-whiskers plot indicating median and first and third; dotted line indicate the best mean score. The model with smooth terms ' $s(z)+ti(t)+ti(f)+ti(z,d)$ ' was considered to be the best, based on selecting the simplest 'equivalent' model after a comparison of the mean and range of the scores across the models.

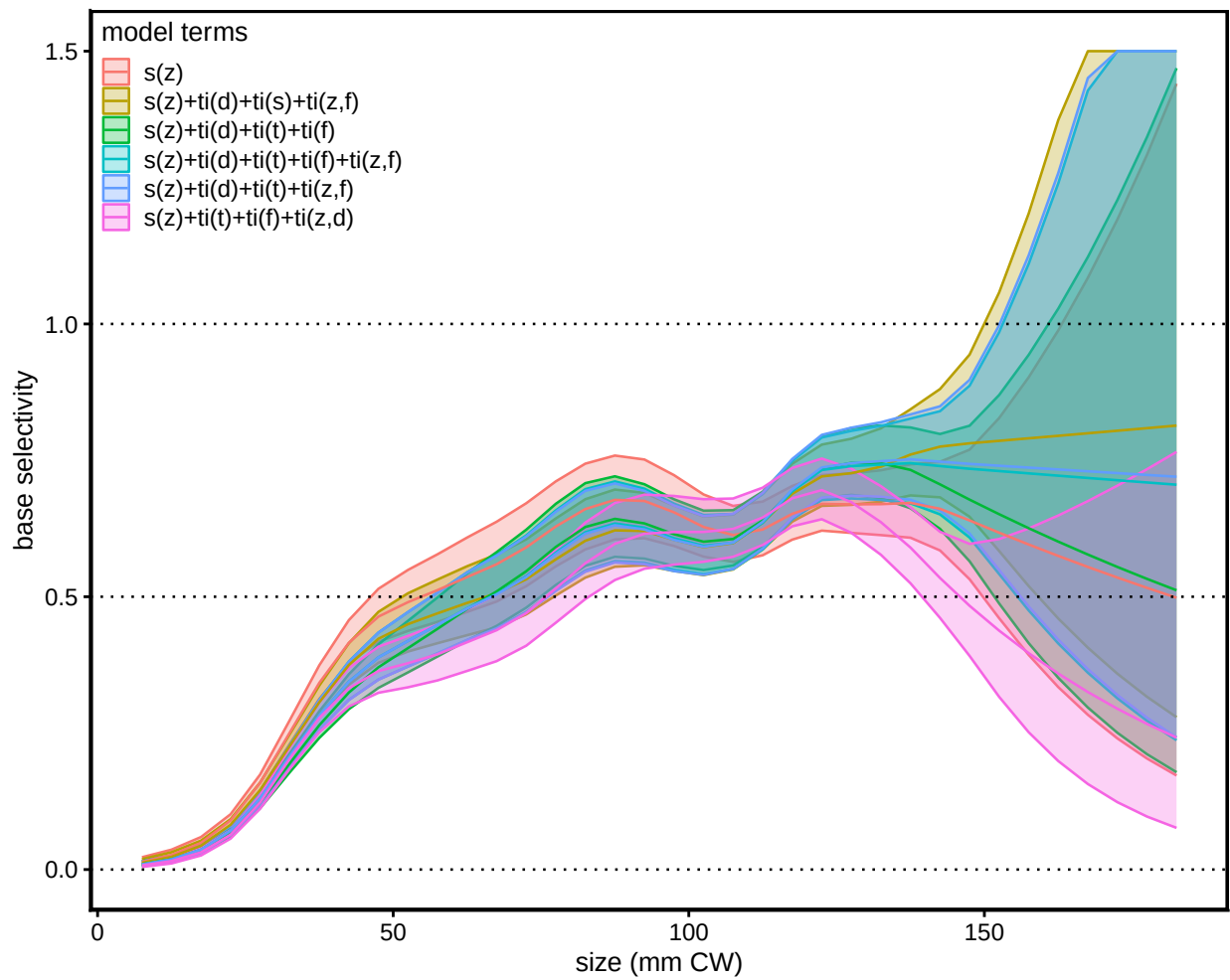


Figure 69. Base selectivity (no covariate effects) estimated from the top-ranking Tweedie models for males.

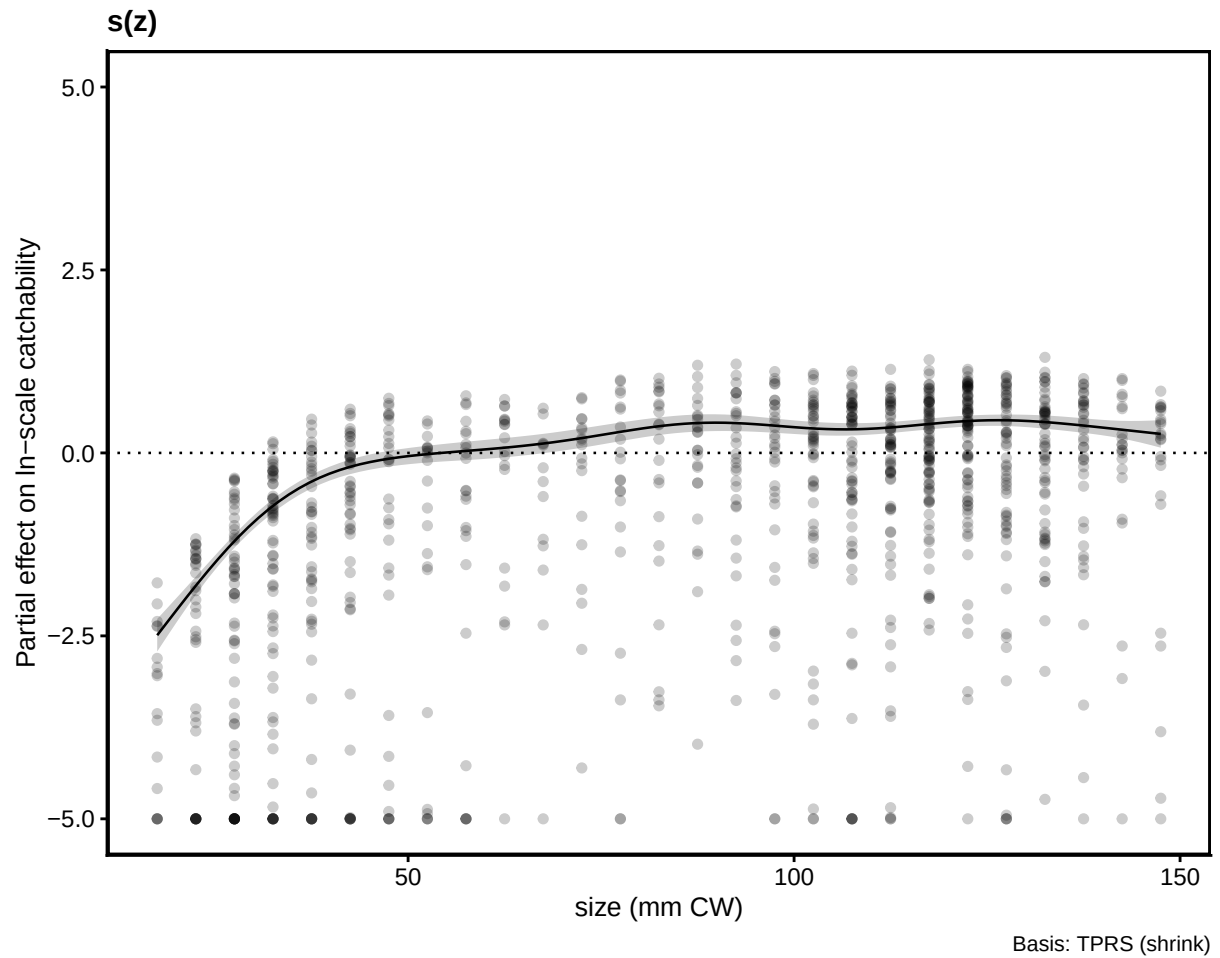


Figure 70. Partial effect of size (mm CW) on ln-scale catchability. Partial residuals for the response 'obsR' are shown on the link scale of the model, based on its Tweedie($p=1.99$) error distribution and log link. Residuals beyond y-axis bounds are plotted at the bound.

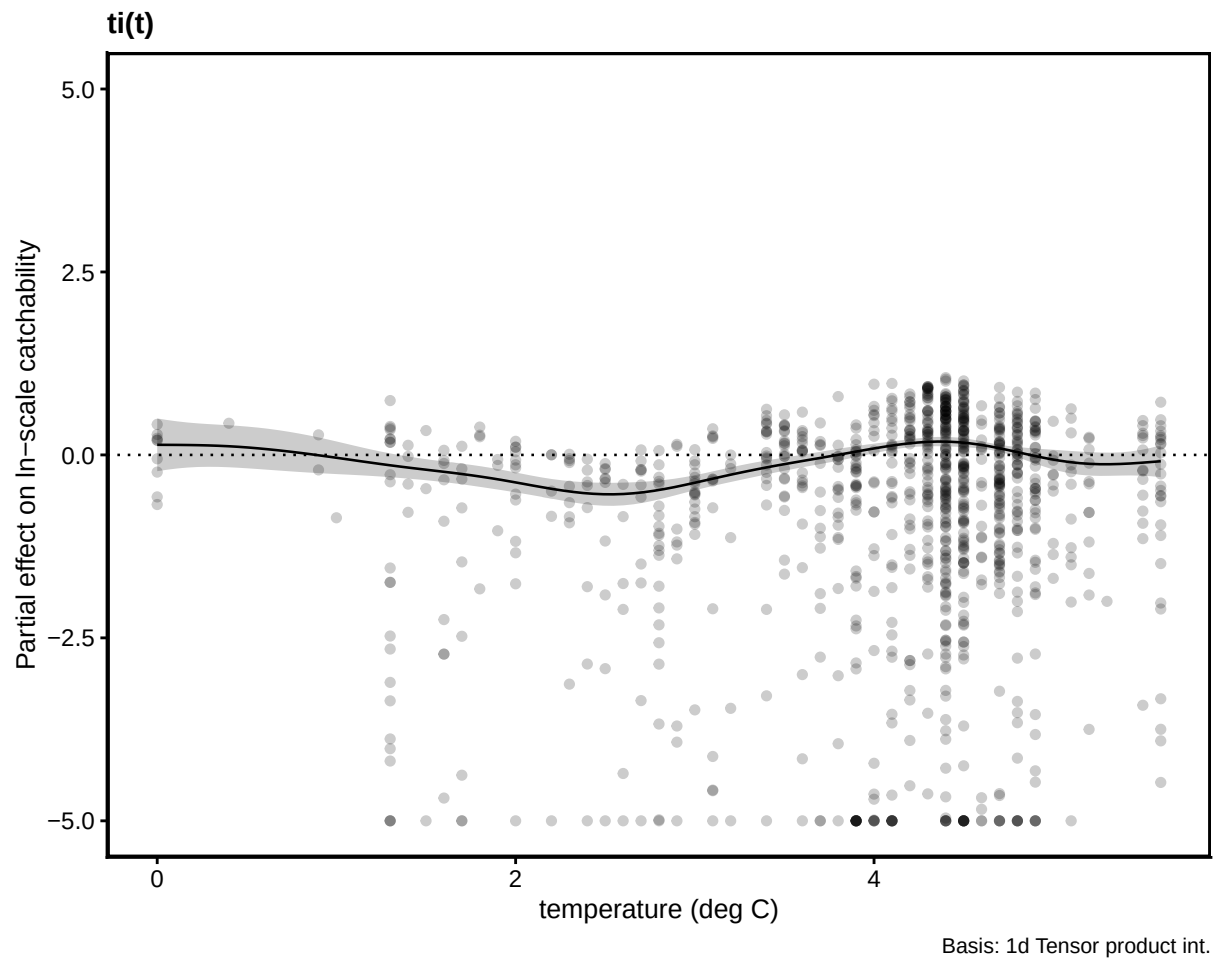


Figure 71. Partial effect of temperature (deg C) on ln-scale catchability. Partial residuals for the response 'obsR' are shown on the link scale of the model, based on its Tweedie($p=1.99$) error distribution and log link. Residuals beyond y-axis bounds are plotted at the bound.

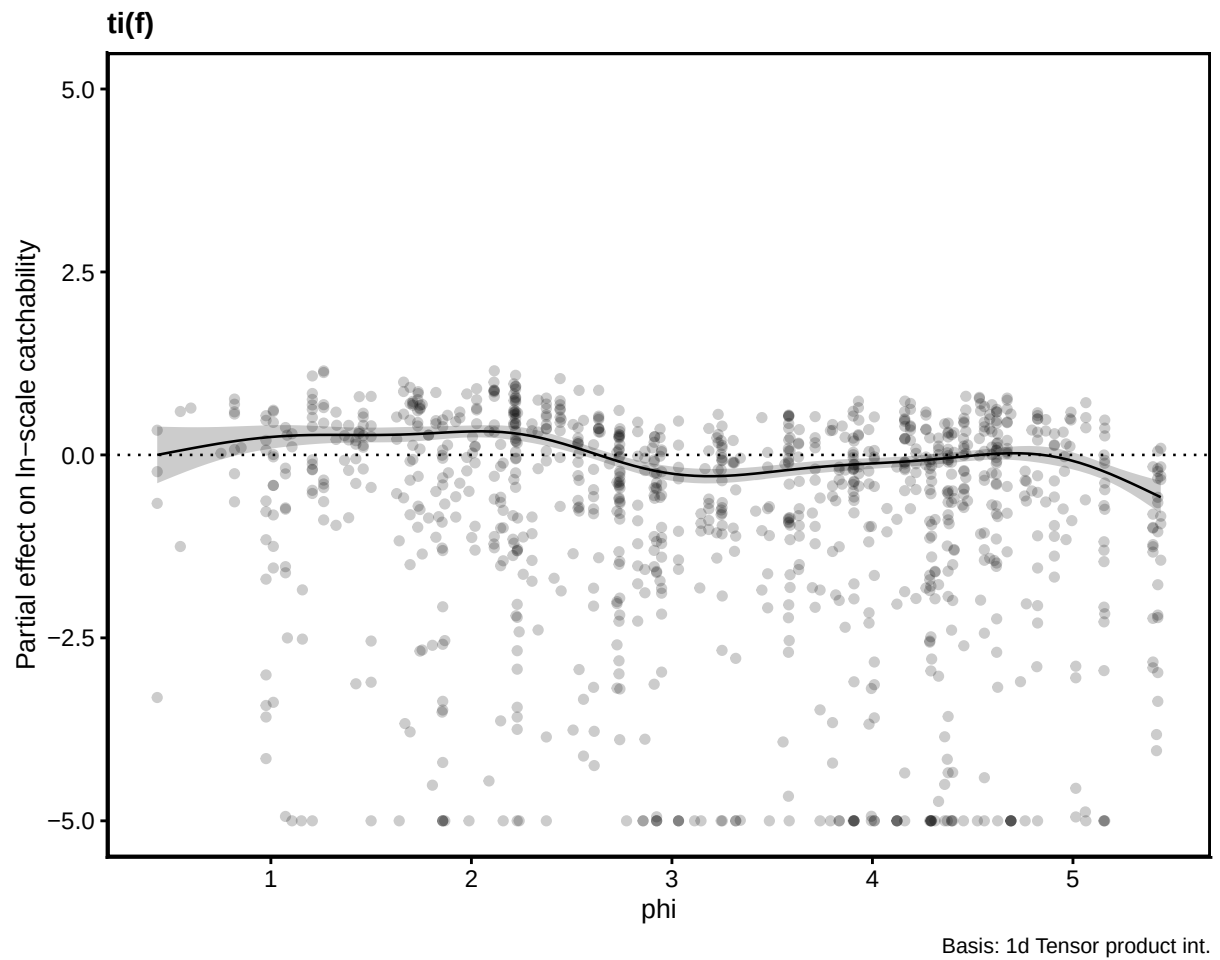


Figure 72. Partial effect of ϕ on ln-scale catchability. Partial residuals for the response 'obsR' are shown on the link scale of the model, based on its Tweedie($p=1.99$) error distribution and log link. Residuals beyond y-axis bounds are plotted at the bound.

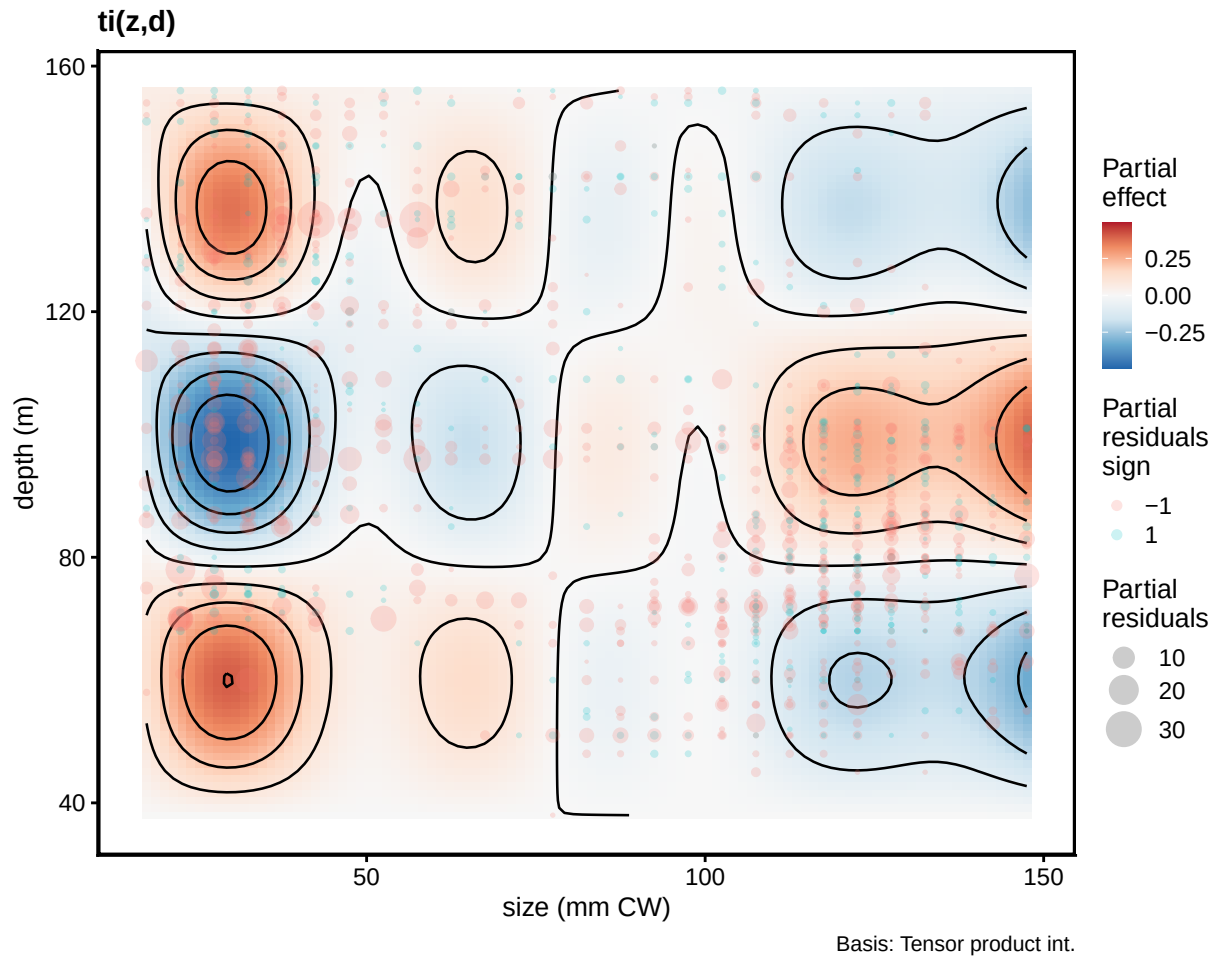


Figure 73. Partial effects of size (mm CW) and depth (m) on In-scale catchability. Partial residuals for the response 'obsR' are shown on the link scale of the model, based on its Tweedie($p=1.99$) error distribution and log link. Residuals beyond y-axis bounds are plotted at the bound.

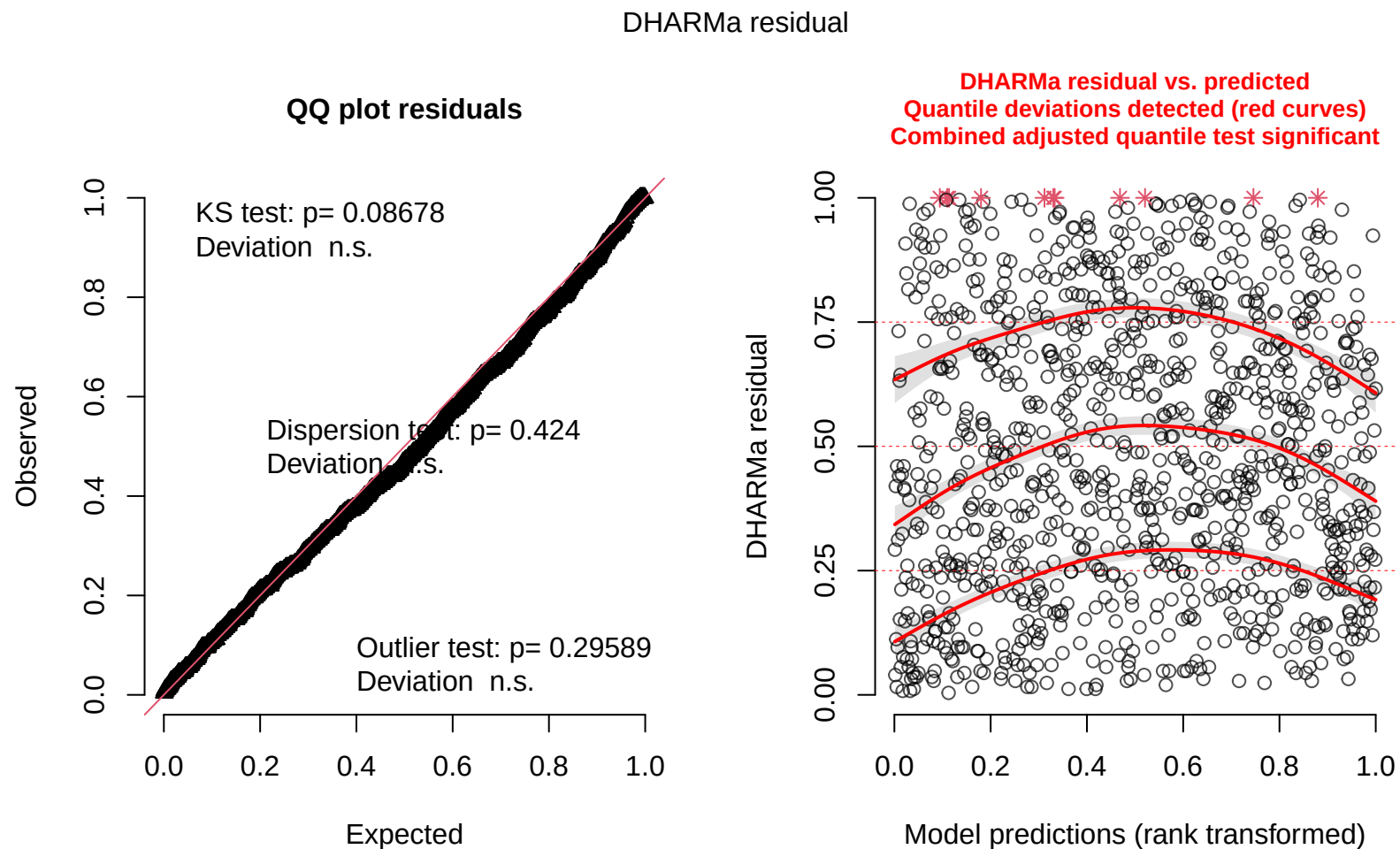


Figure 74. DHARMa residuals from the 'best'Tweedie model for males. Left: quantile-quantile plot for observed and expected residuals under the null hypothesis. Right: quantile deviations for residuals.

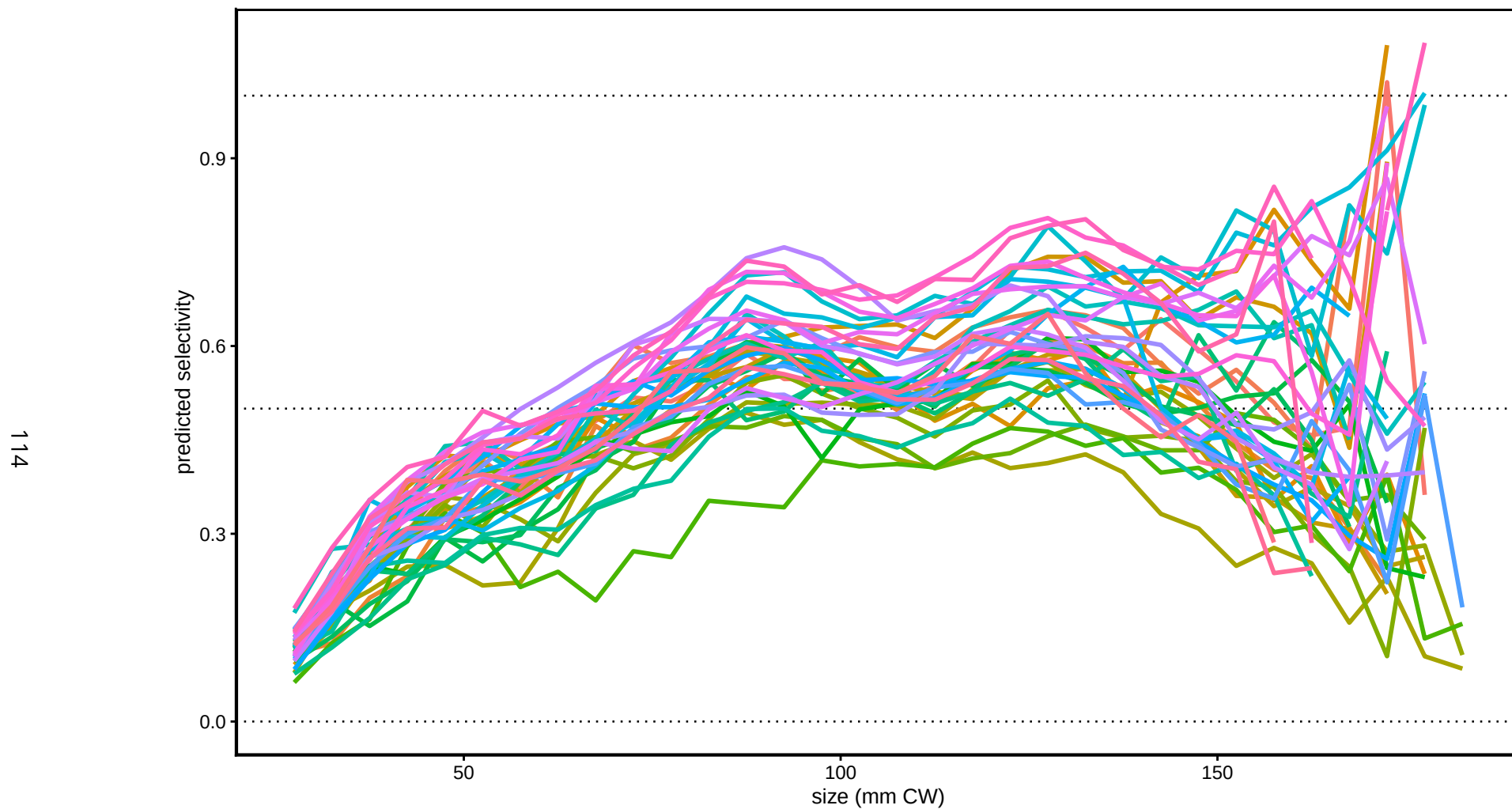


Figure 75. Haul-specific and annual mean catchability curves for males, weighted by size-specific CPUE, estimated using the best Tweedie model with covariates.

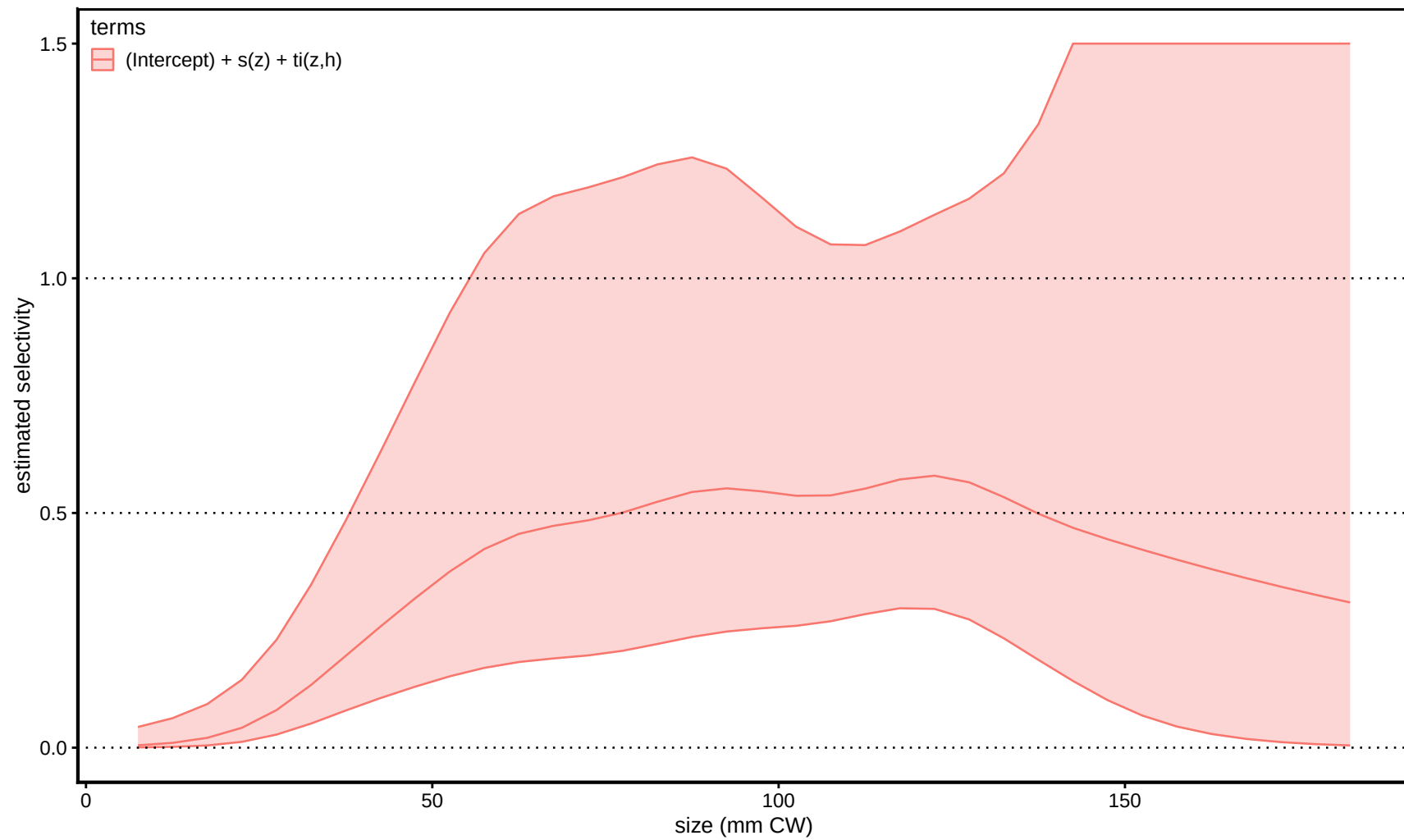


Figure 76. Estimated haul-level catchability for males based on the random effects Tweedie model (no covariates).

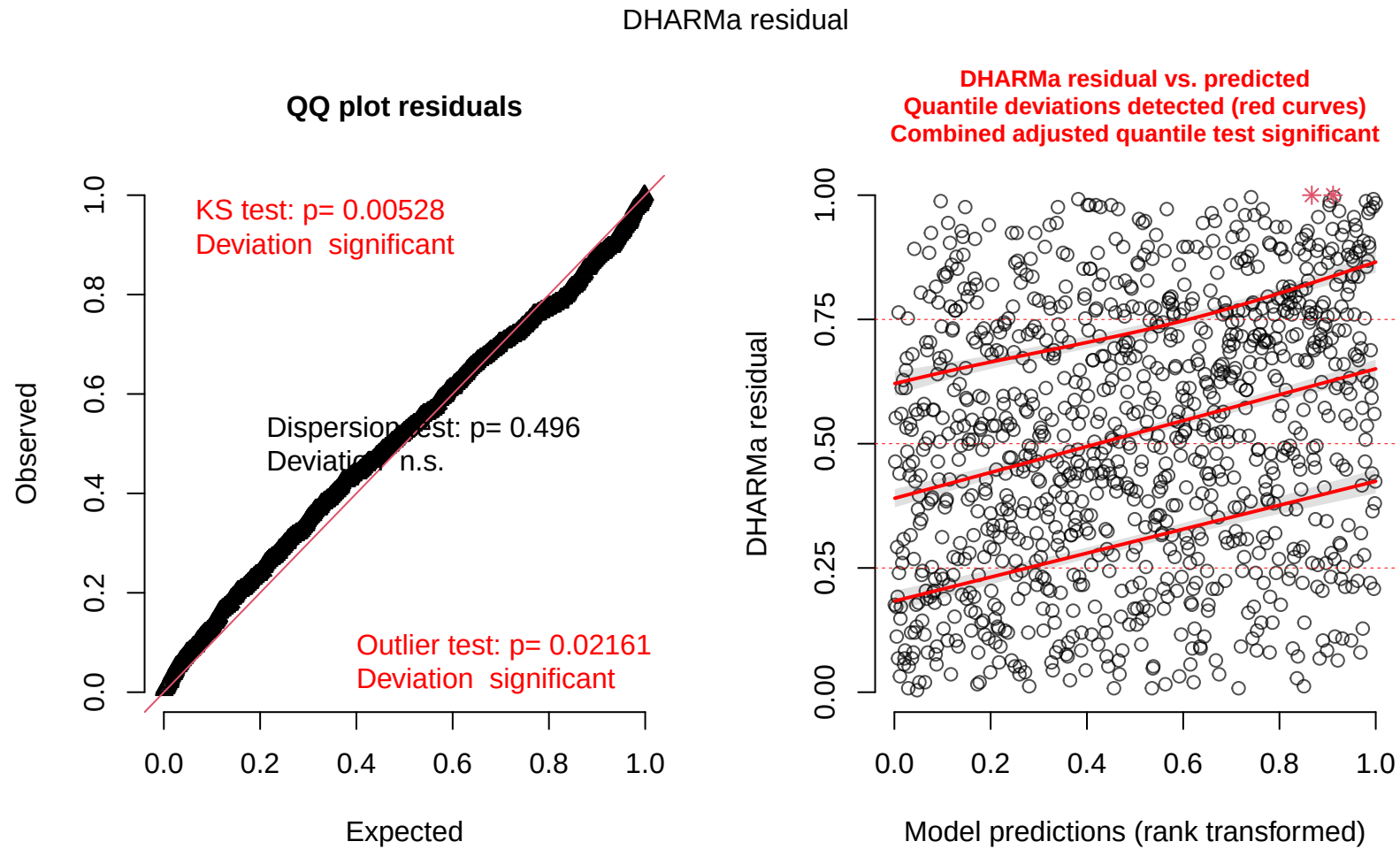


Figure 77. DHARMa residuals from the random effects Tweedie model for males. Left: quantile-quantile plot for observed and expected residuals under the null hypothesis. Right: quantile deviations for residuals.